

The Dark Age of the Cosmos

McGill Radio Lab

2025



170 MYA



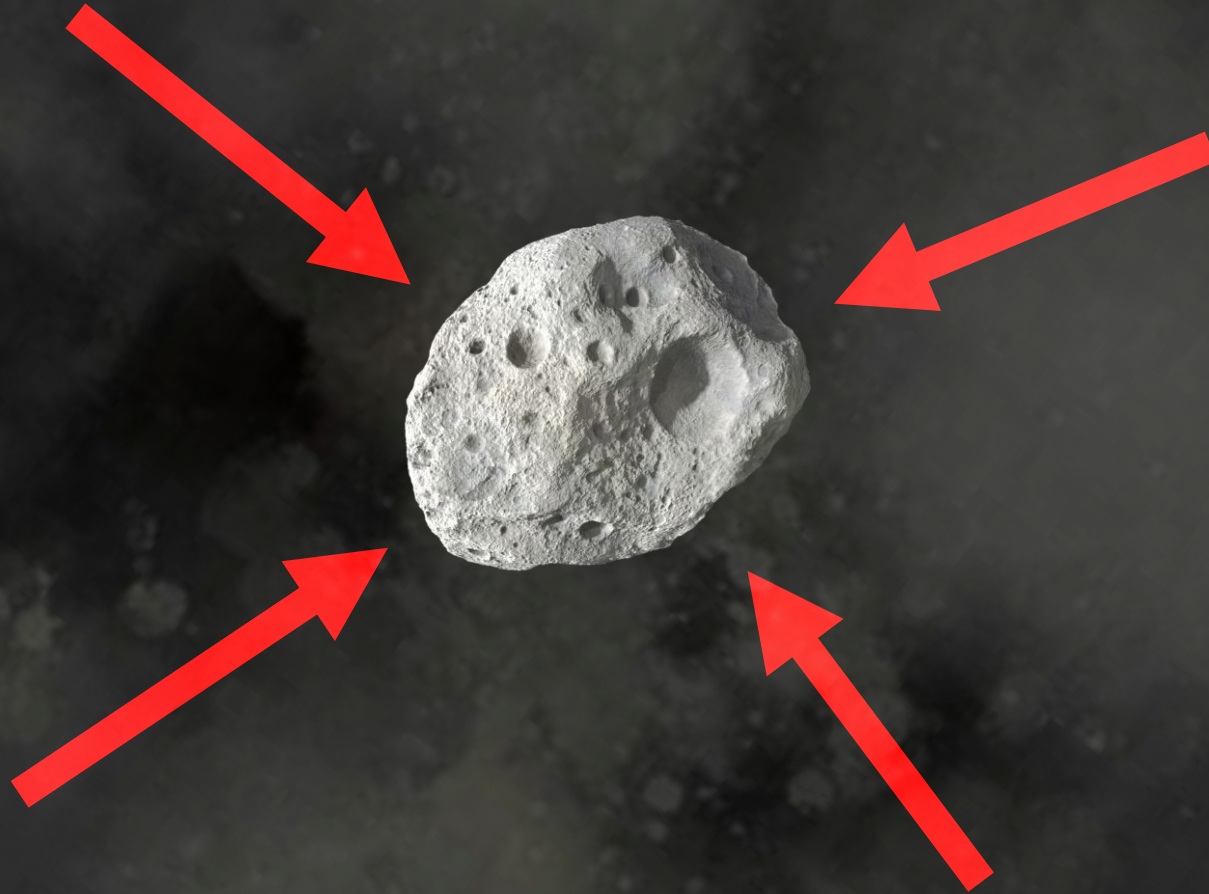
4 BYA



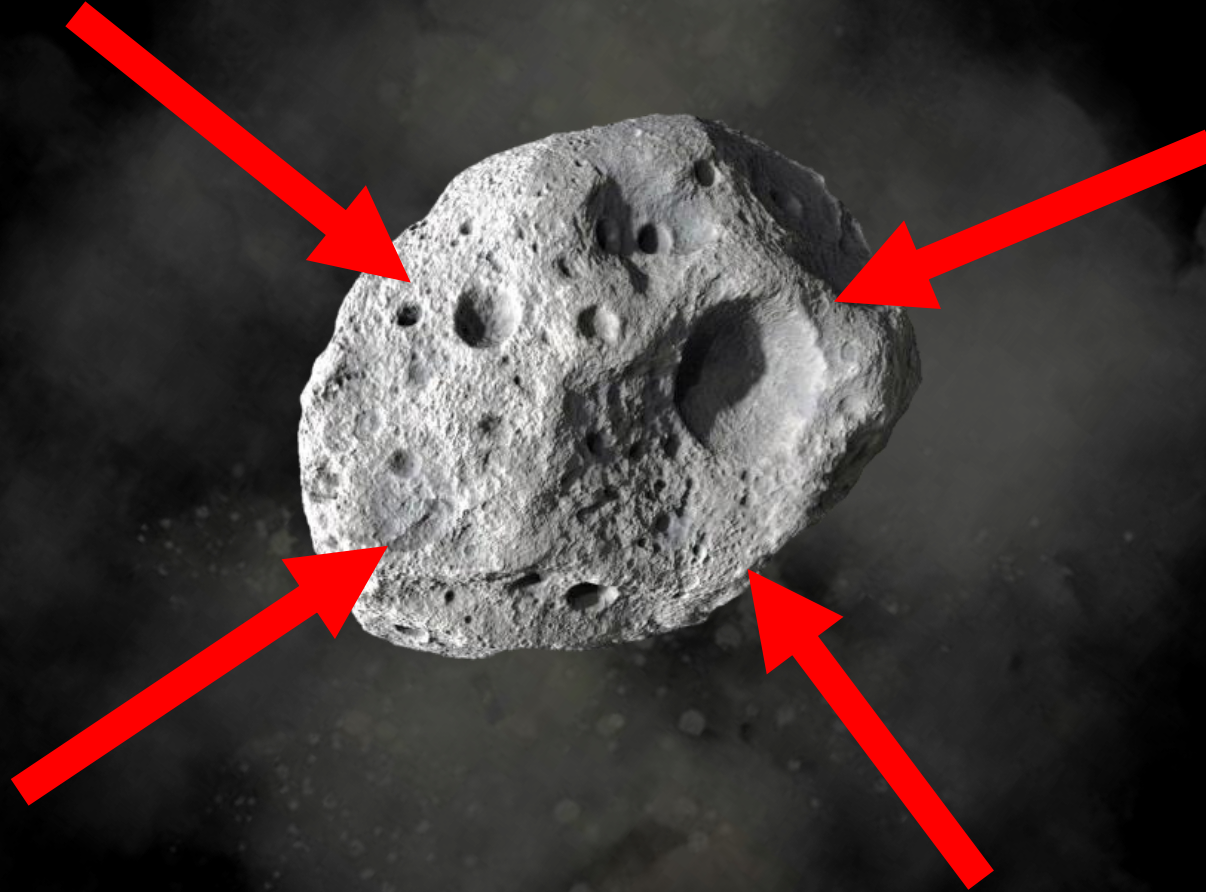


How was it formed?

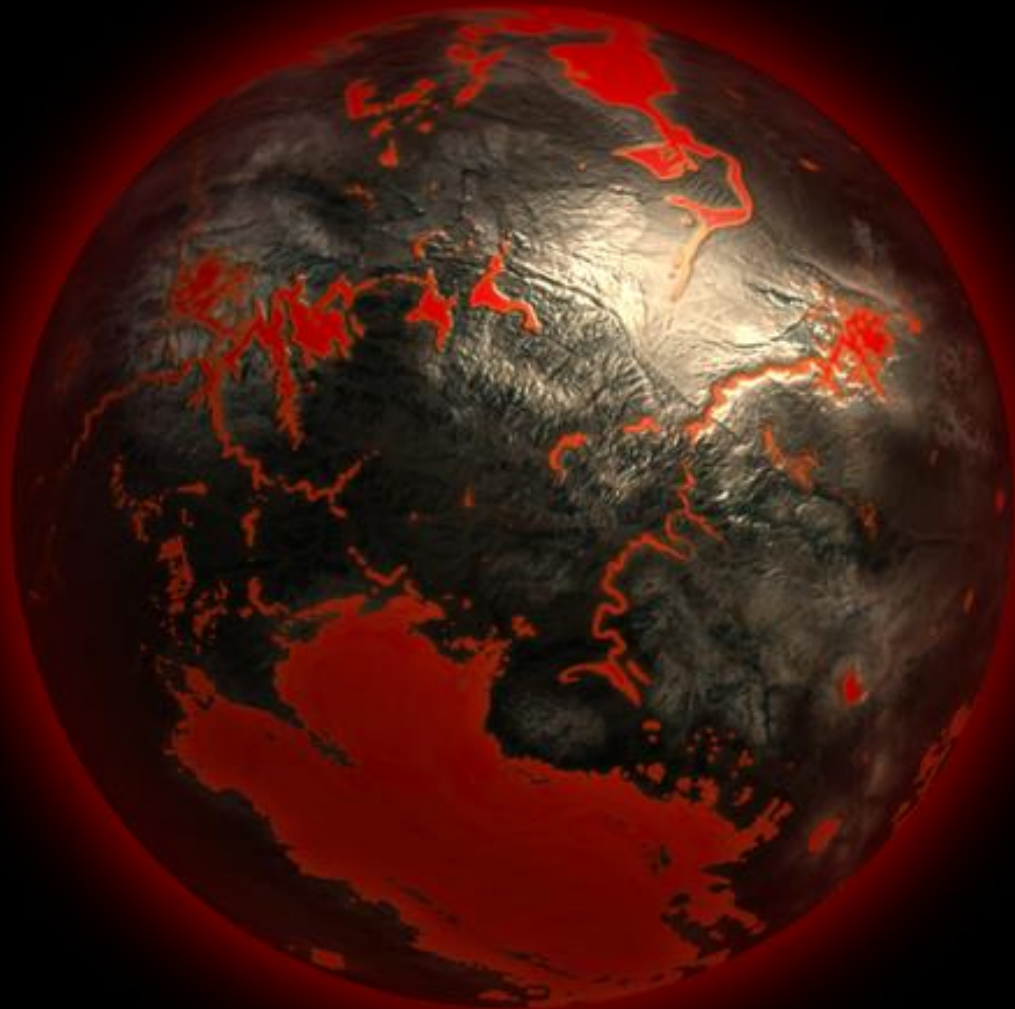
How was it formed?



How was it formed?

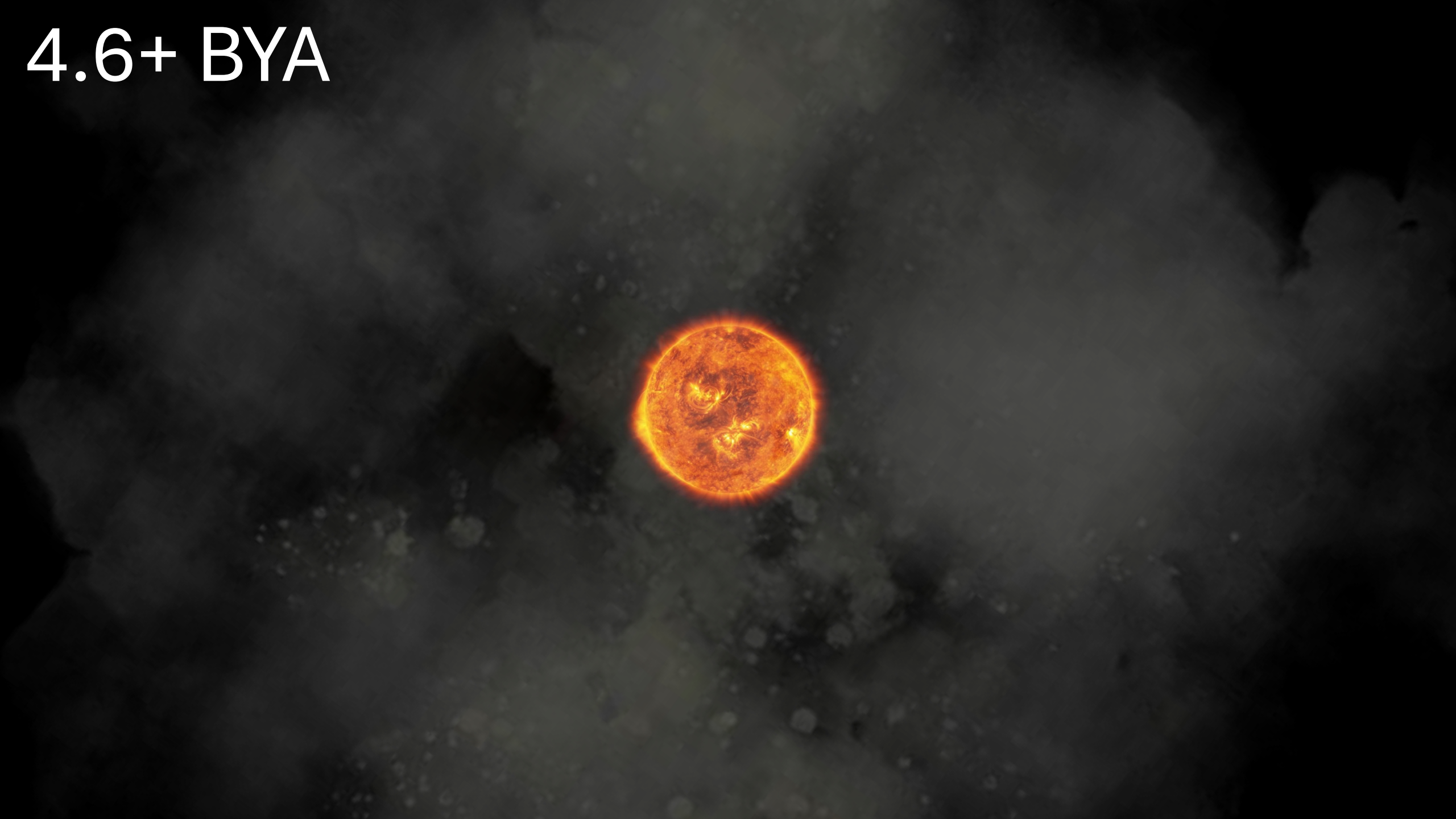


How was it formed?



4.6 BYA





4.6+ BYA

4.6+ BYA

What happens
when we go all the
way back to the very
beginning?



13.8 BYA

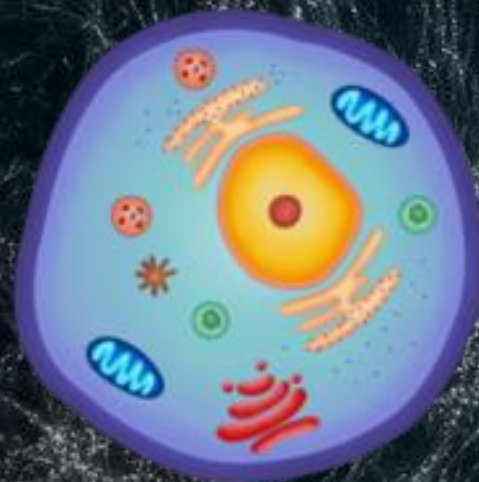
The Big Bang!

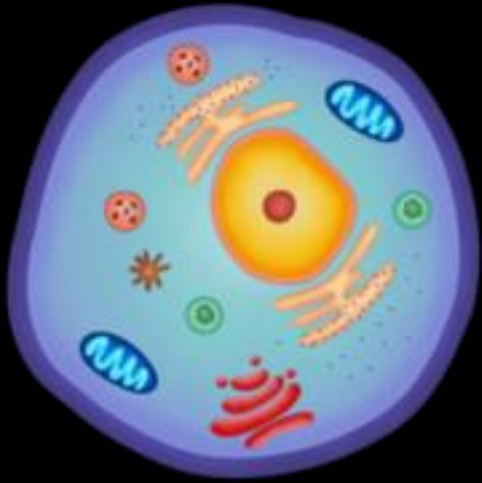
What is it?

What does it look like?

13.8 BYA

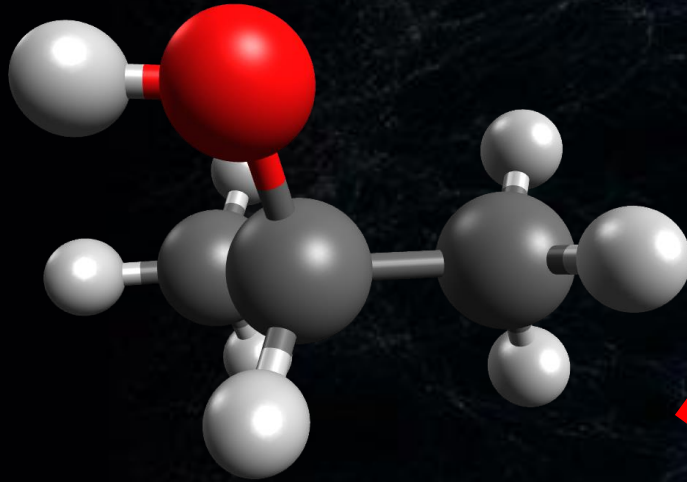
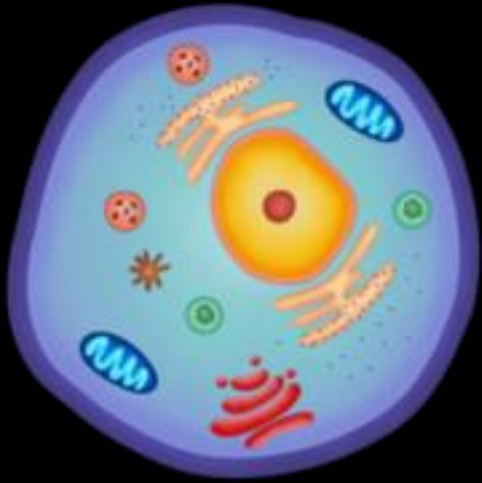
What is the smallest object
you can think of?



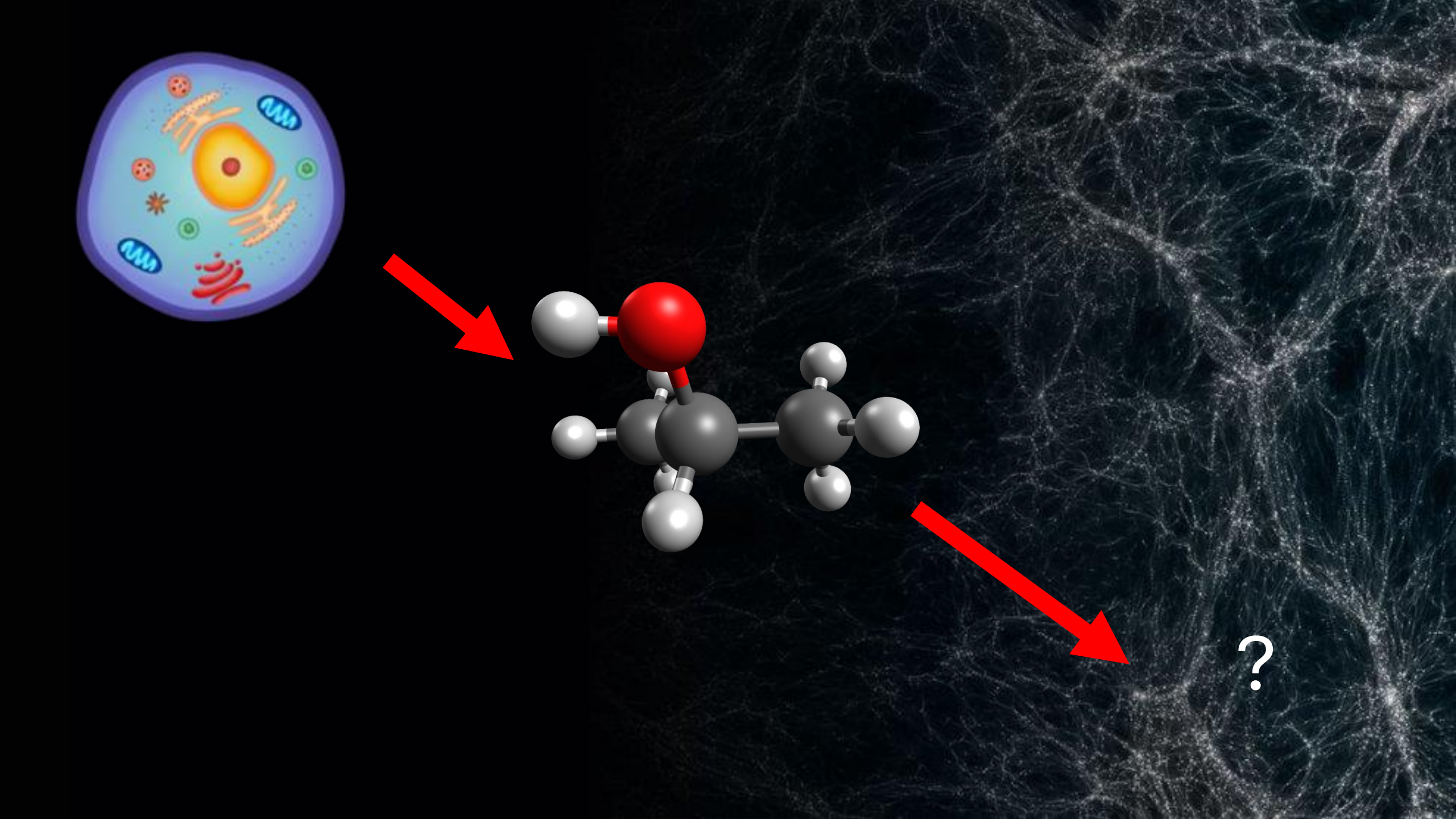


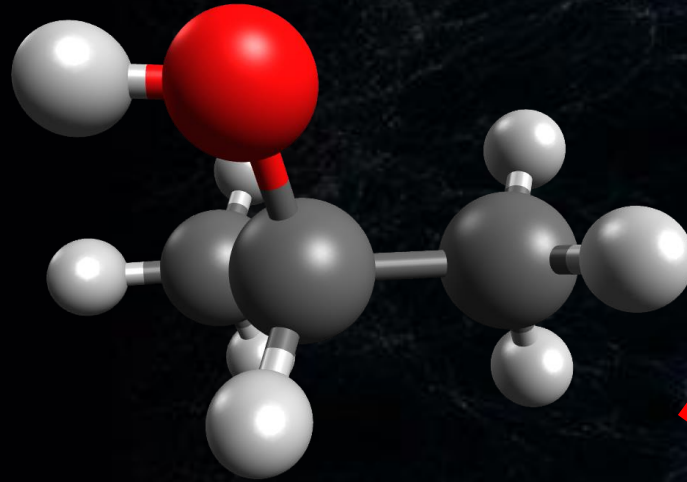
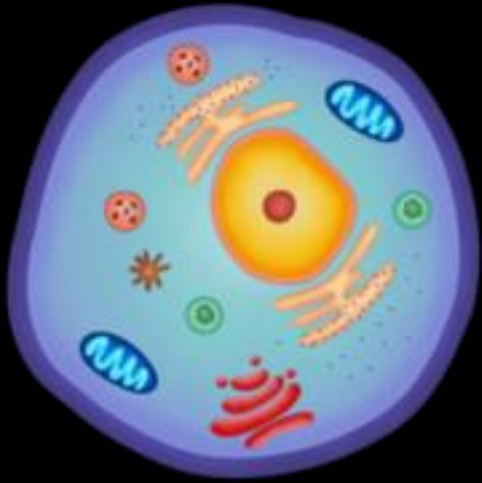
?





?



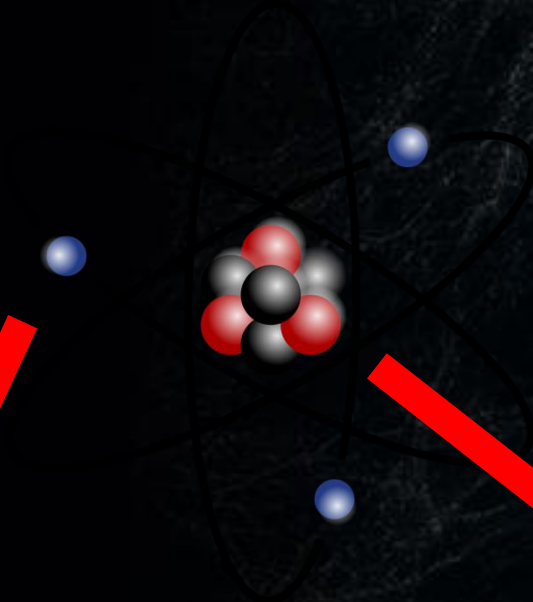


Is there something
even smaller?





=

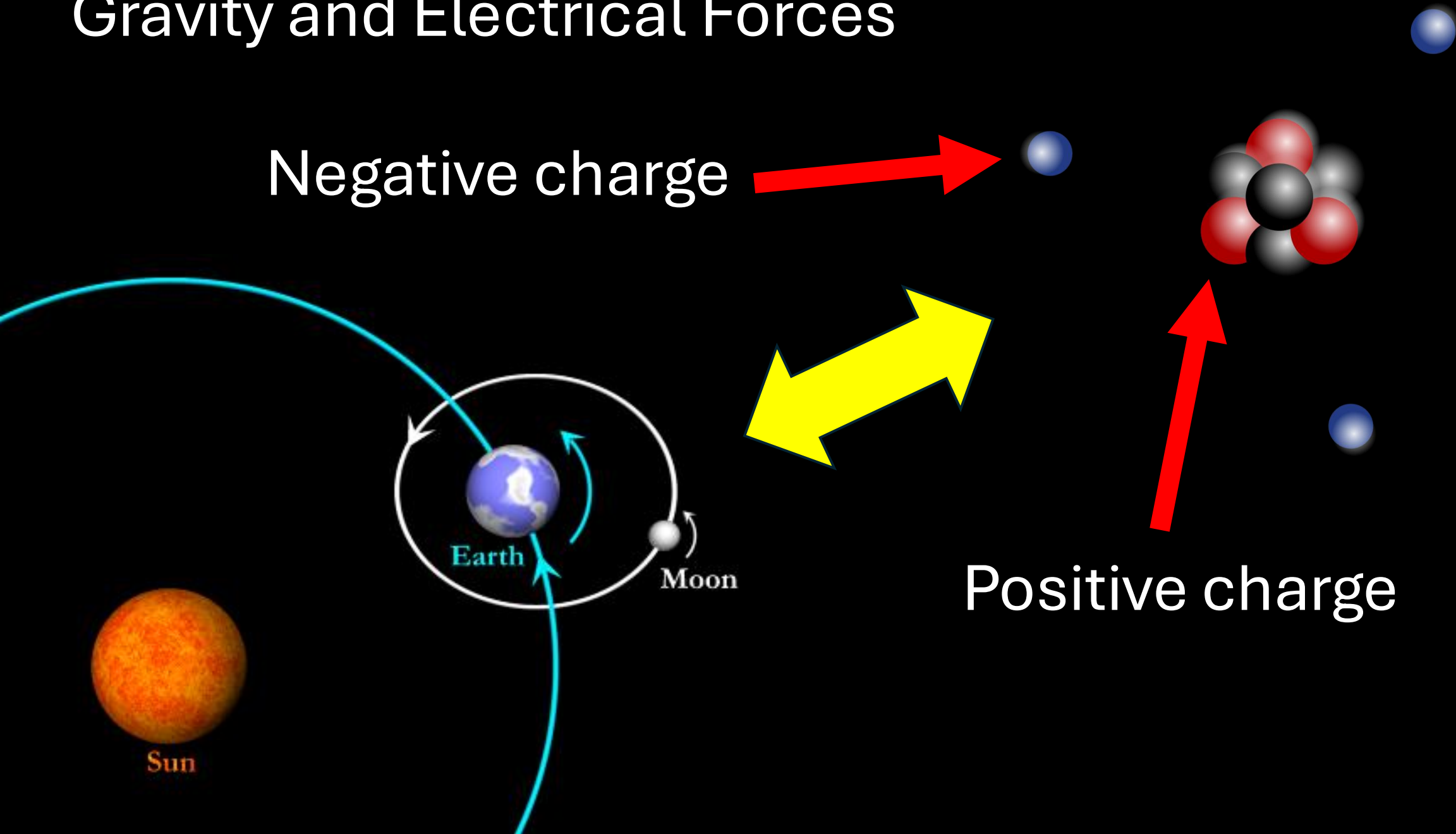


electron



proton

Gravity and Electrical Forces



Gravity and Electrical Forces

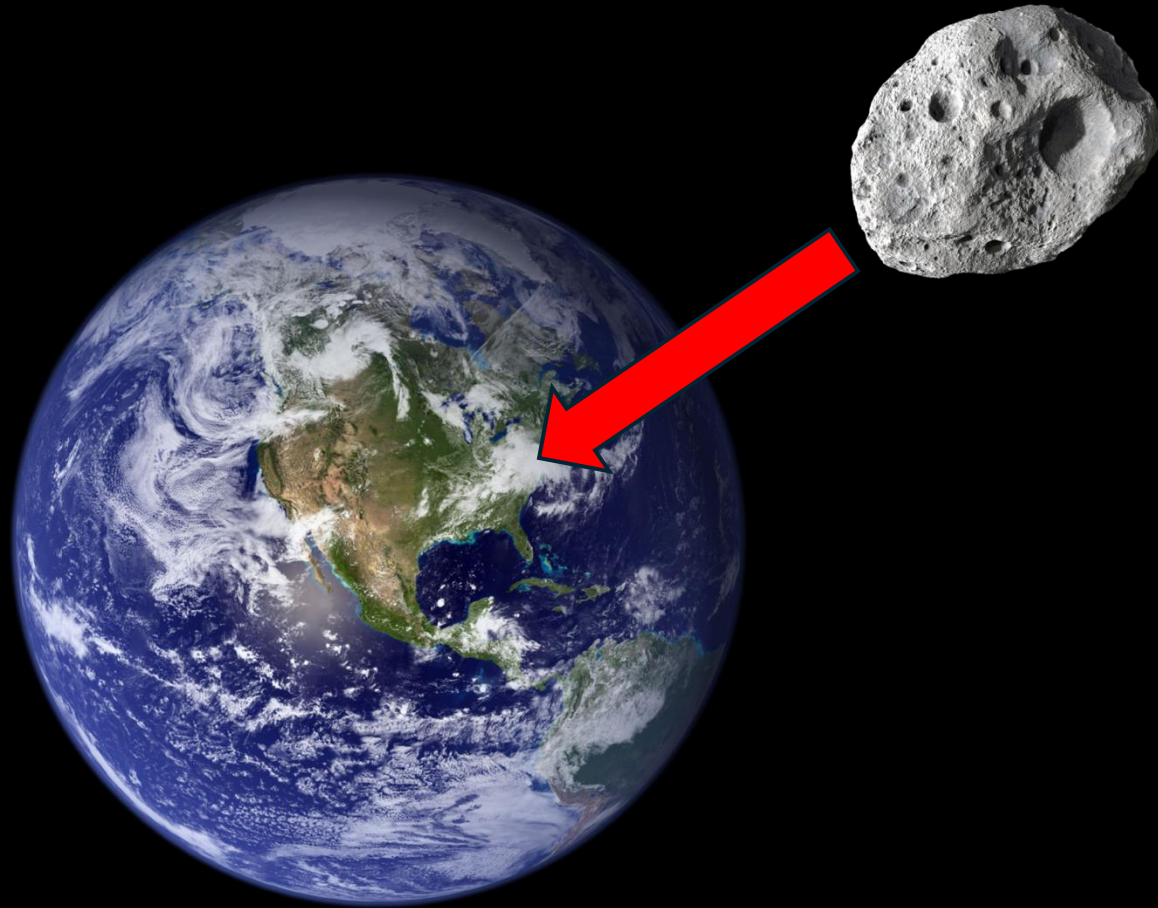
Asteroid

What trajectories can
the asteroid take?





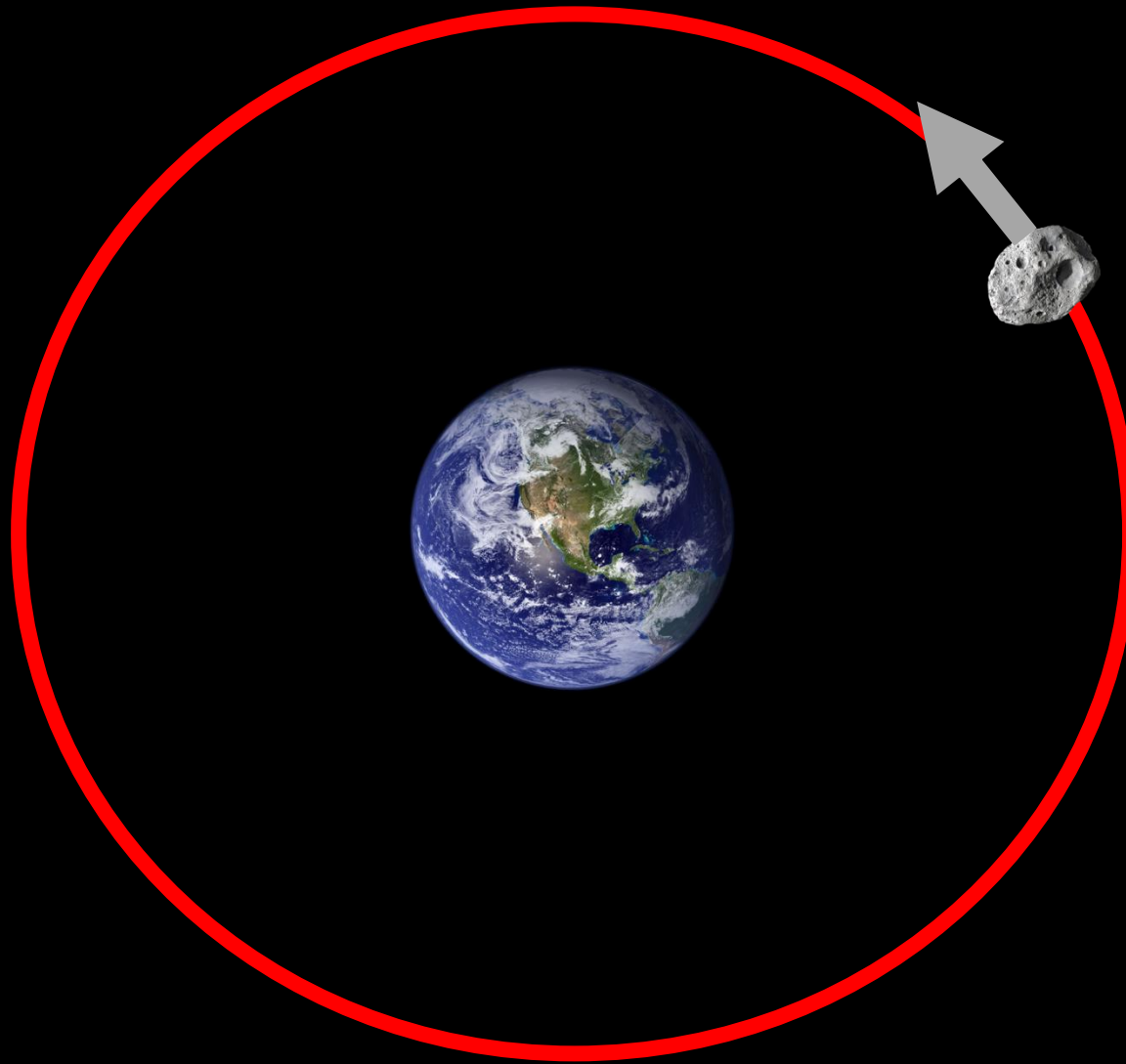
Gravity and Electrical Forces







Gravity and Electrical Forces



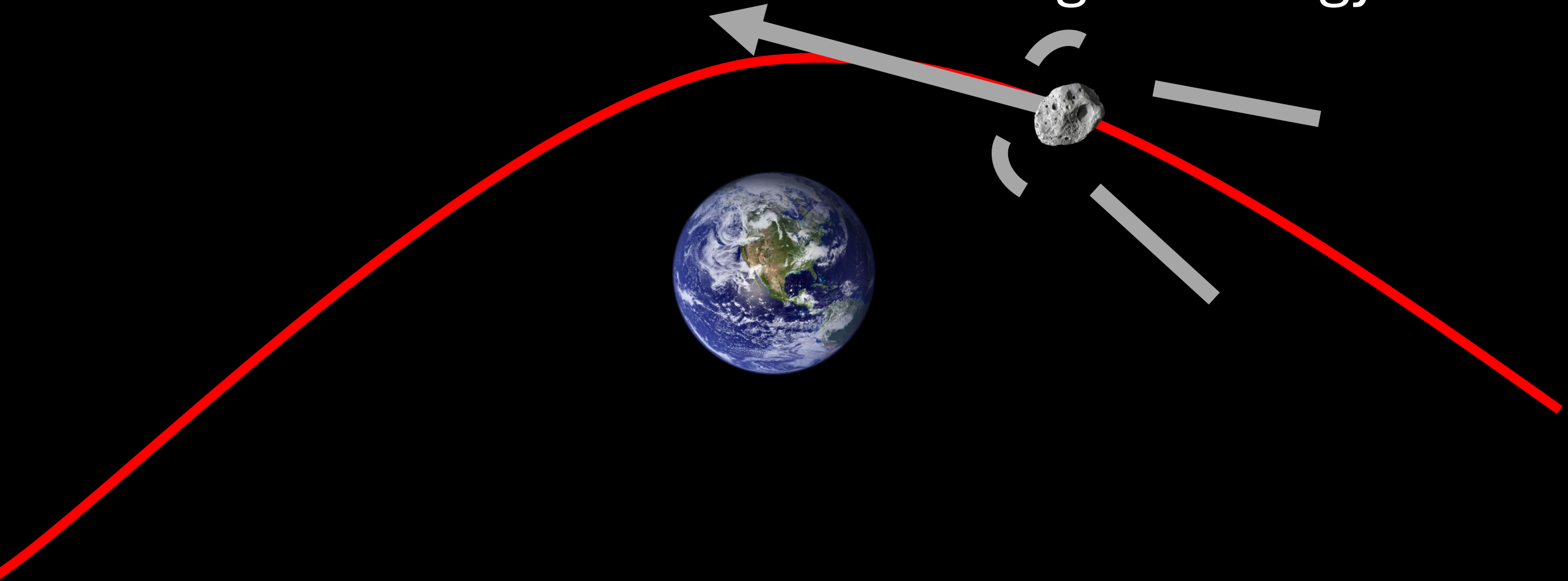
Low energy ->
Low speed

Comet C/2011 L4



Gravity and Electrical Forces

Much higher energy!



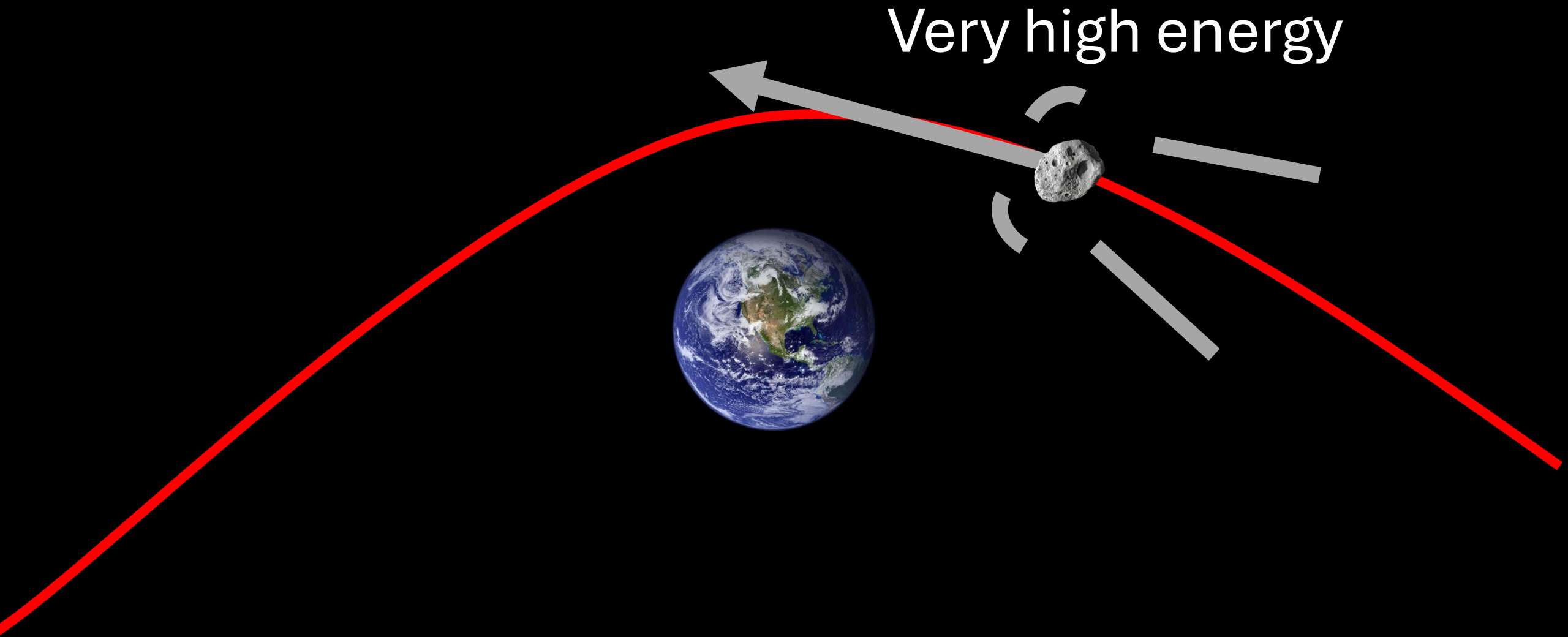
Ex

Extremely short
time after the big
bang

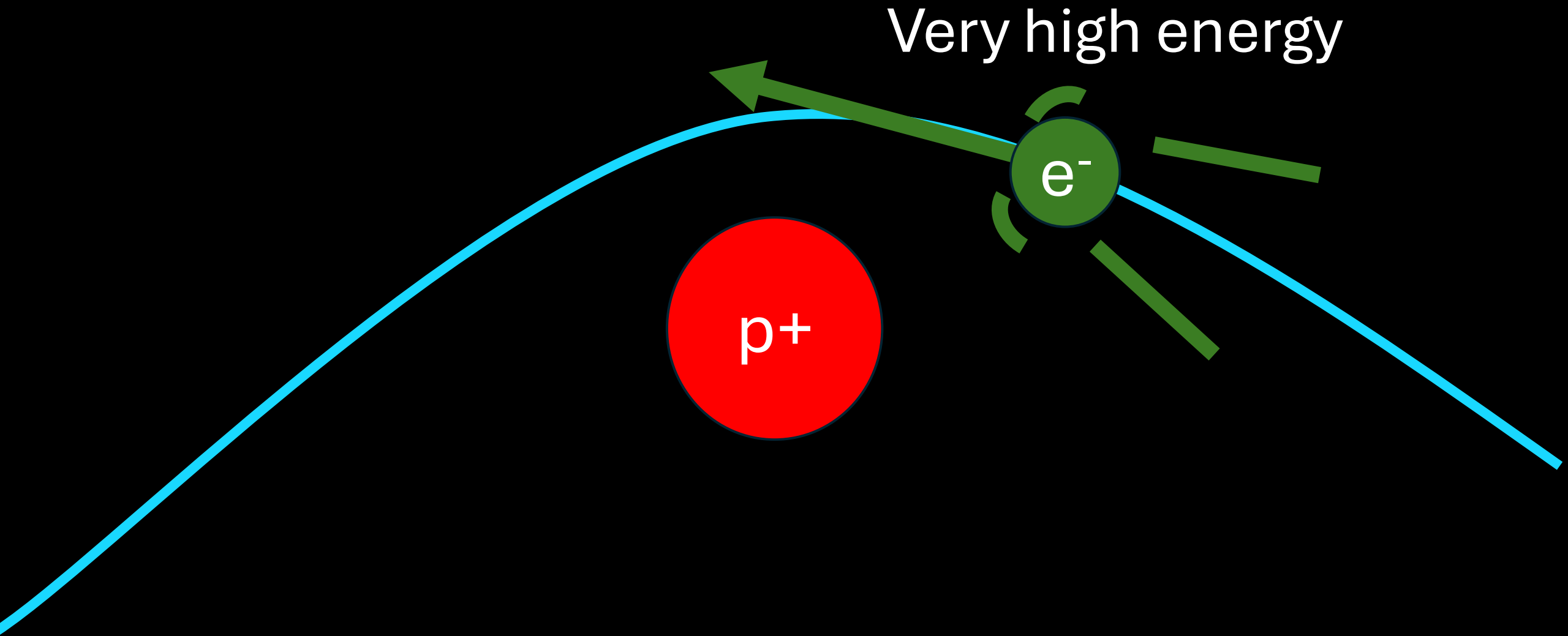
Very high energy



Gravity and Electrical Forces



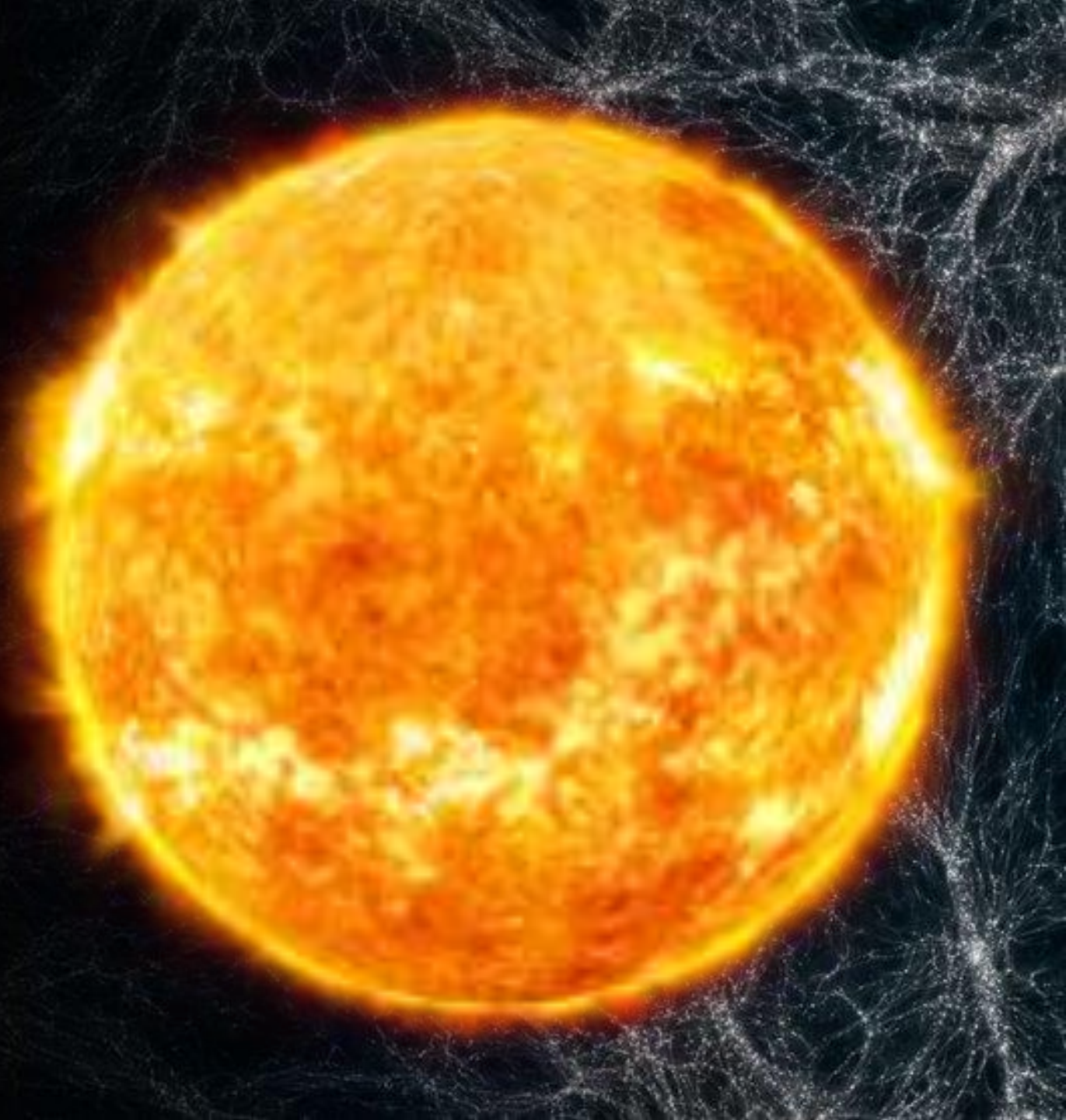
Gravity and Electrical Forces



Plasma

Similar conditions
as in the sun

What would it look
like if you were
there?

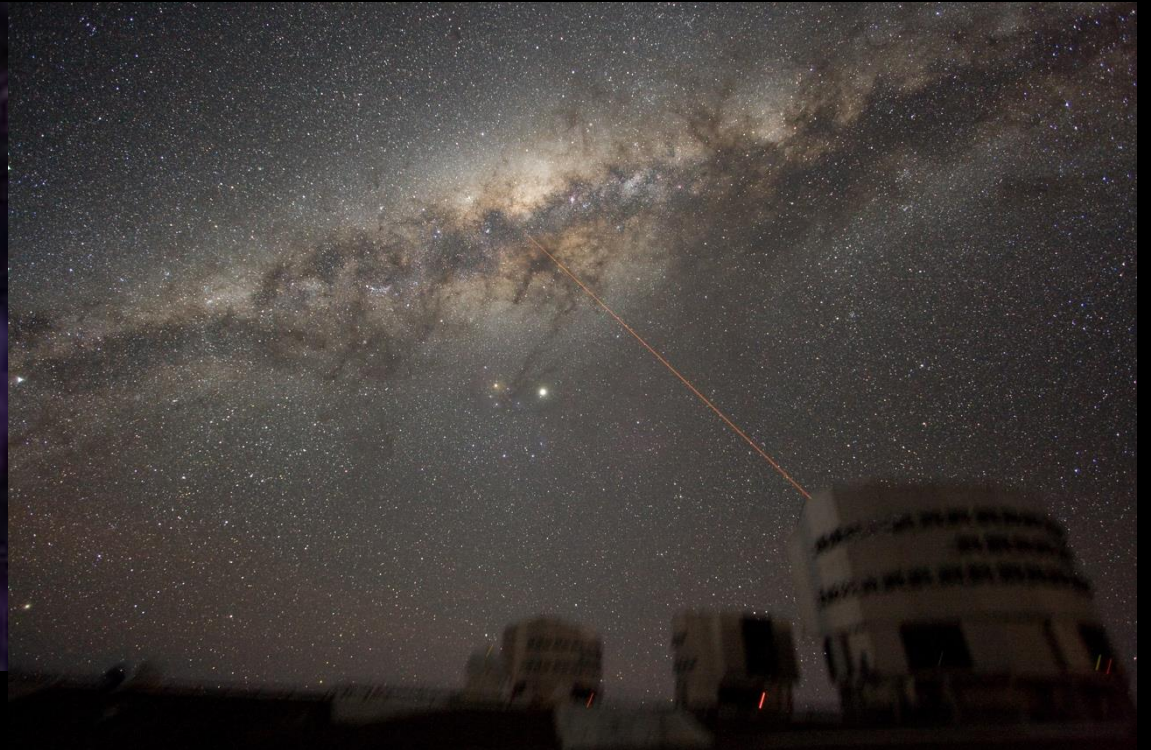


13.8 BYA

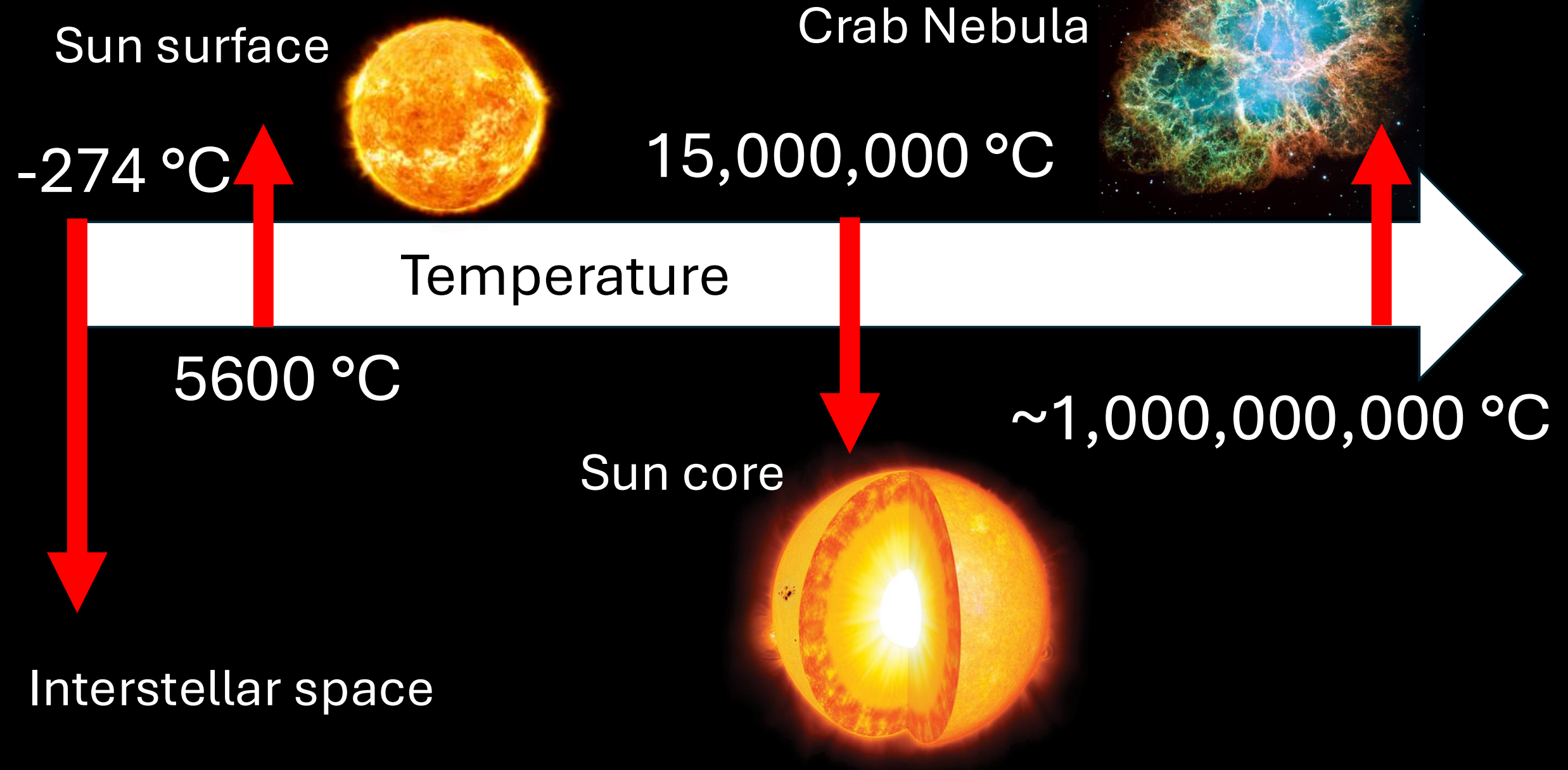
The Temperature of the Universe

First atoms form:
~3000 °C

Today:
? °C

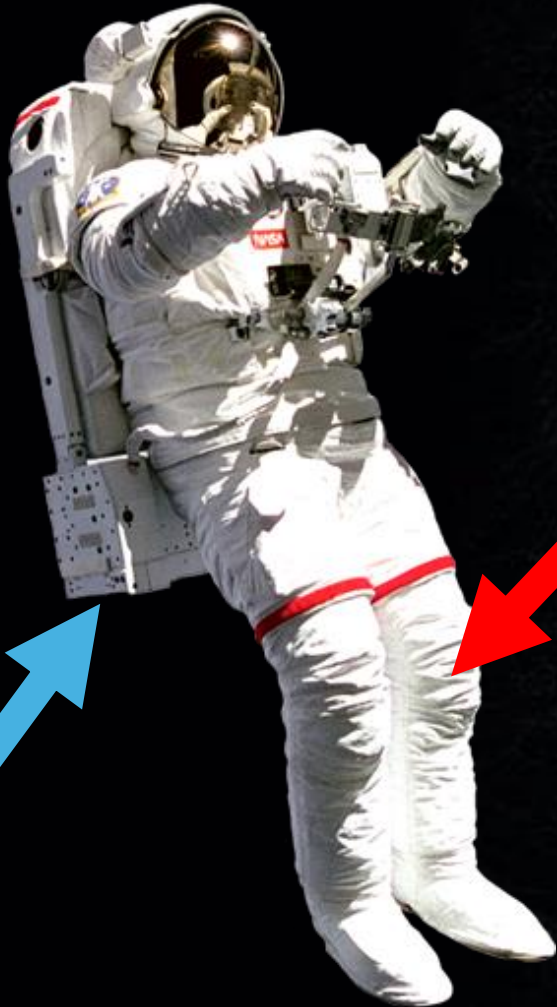


The Temperature of the Universe



The Temperature of the Universe

-250 °C



250 °C

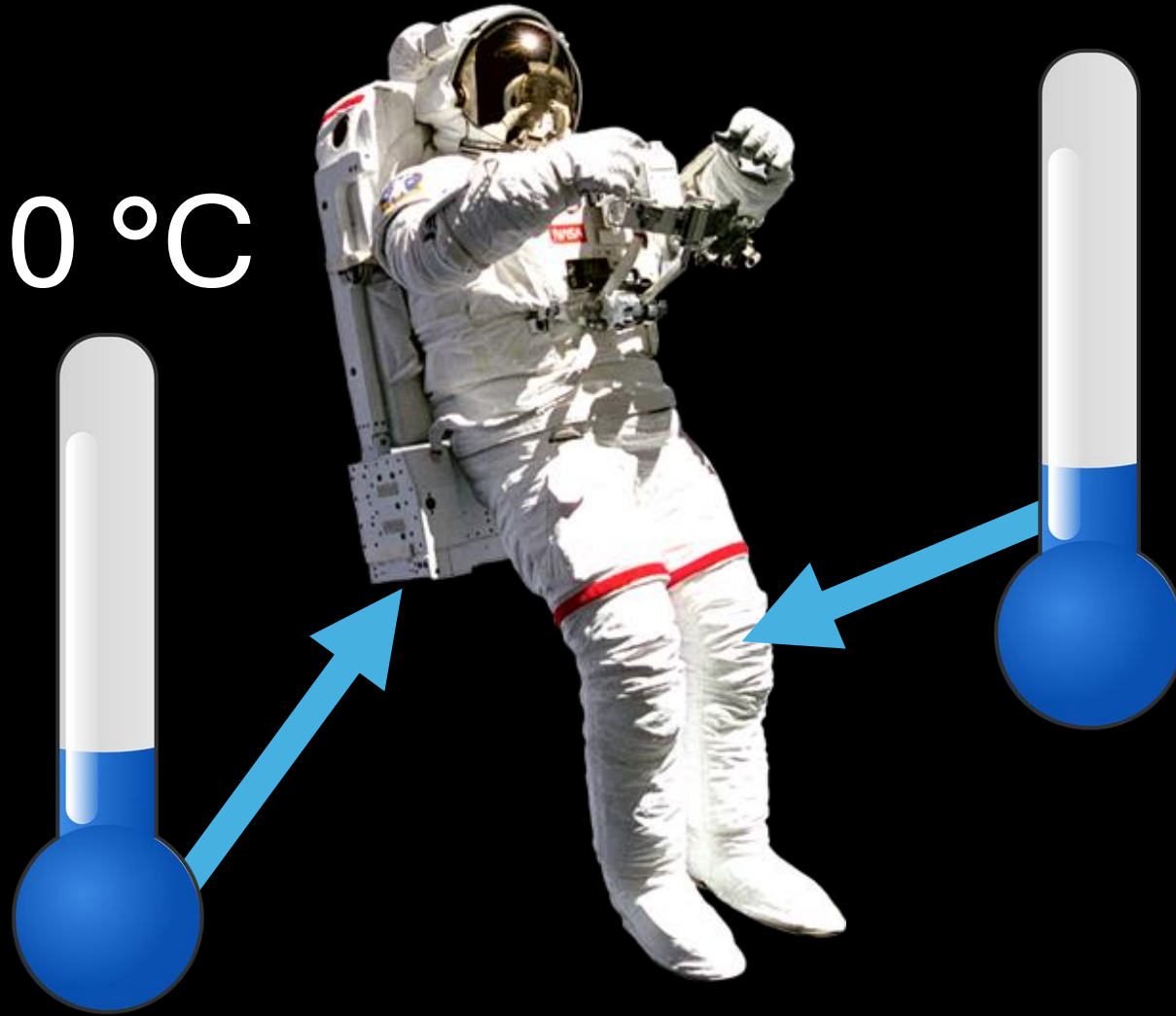


5600 °C

The Temperature of the Universe

-270 °C

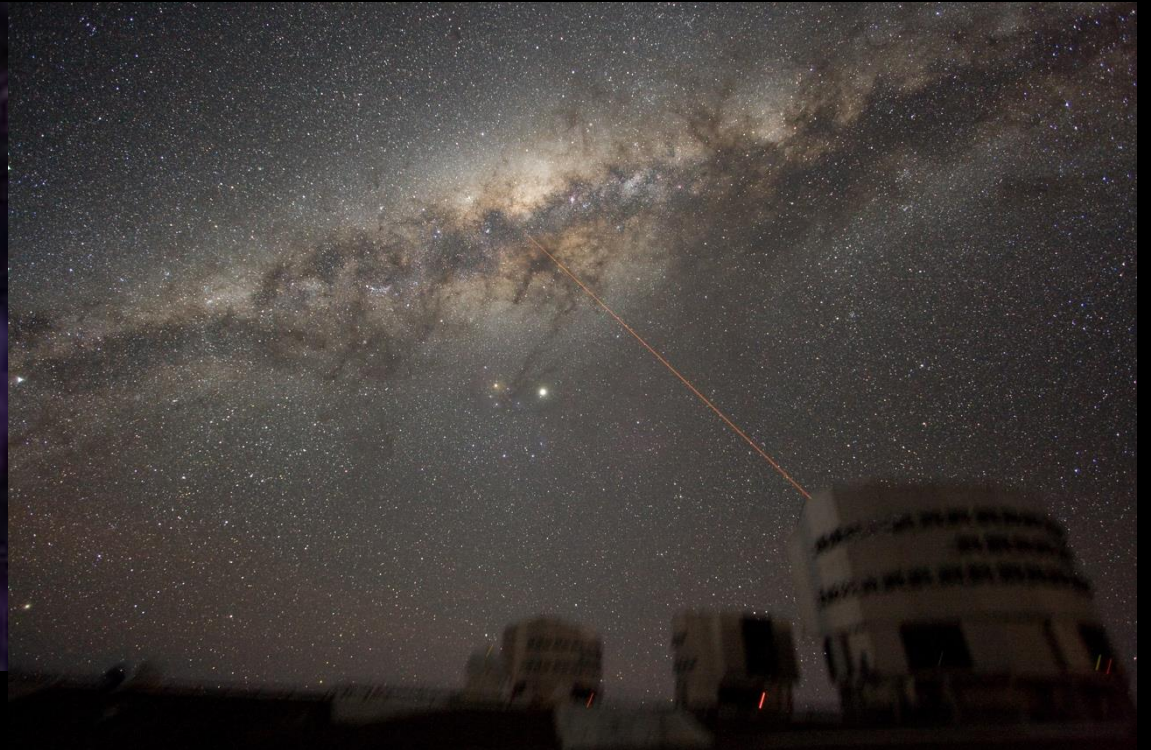
-270 °C



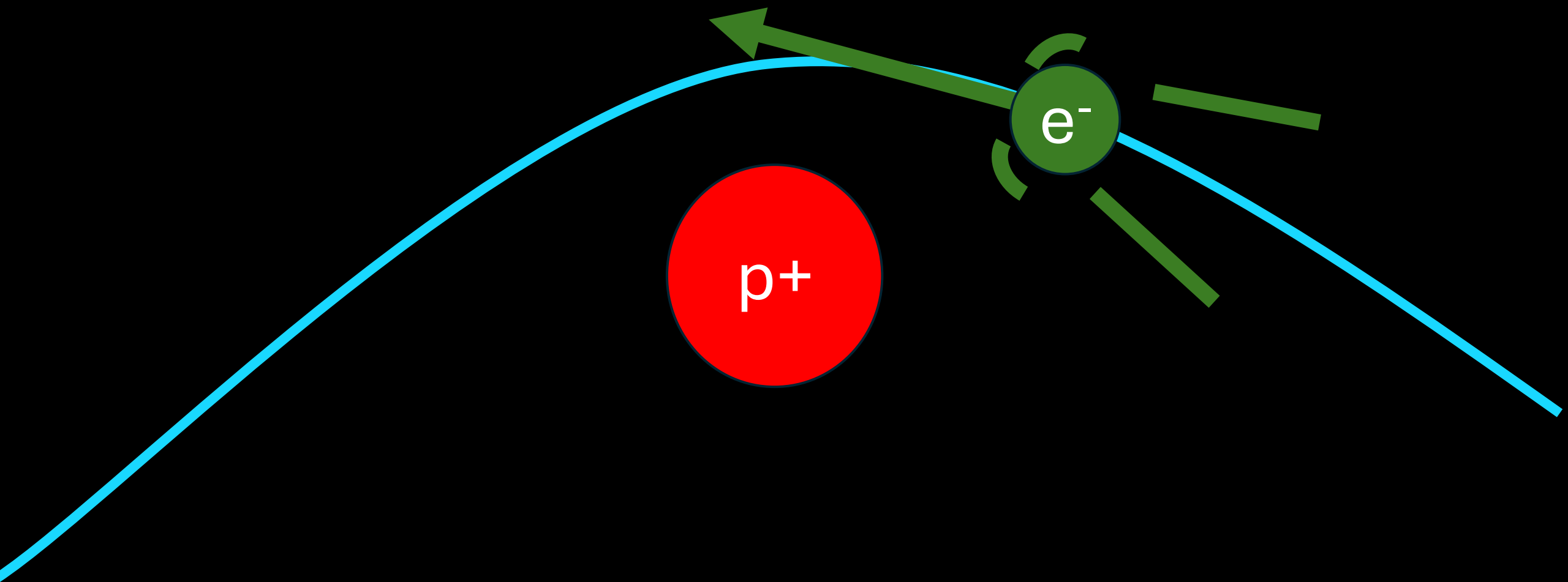
The Temperature of the Universe

First atoms form:
~3000 °C

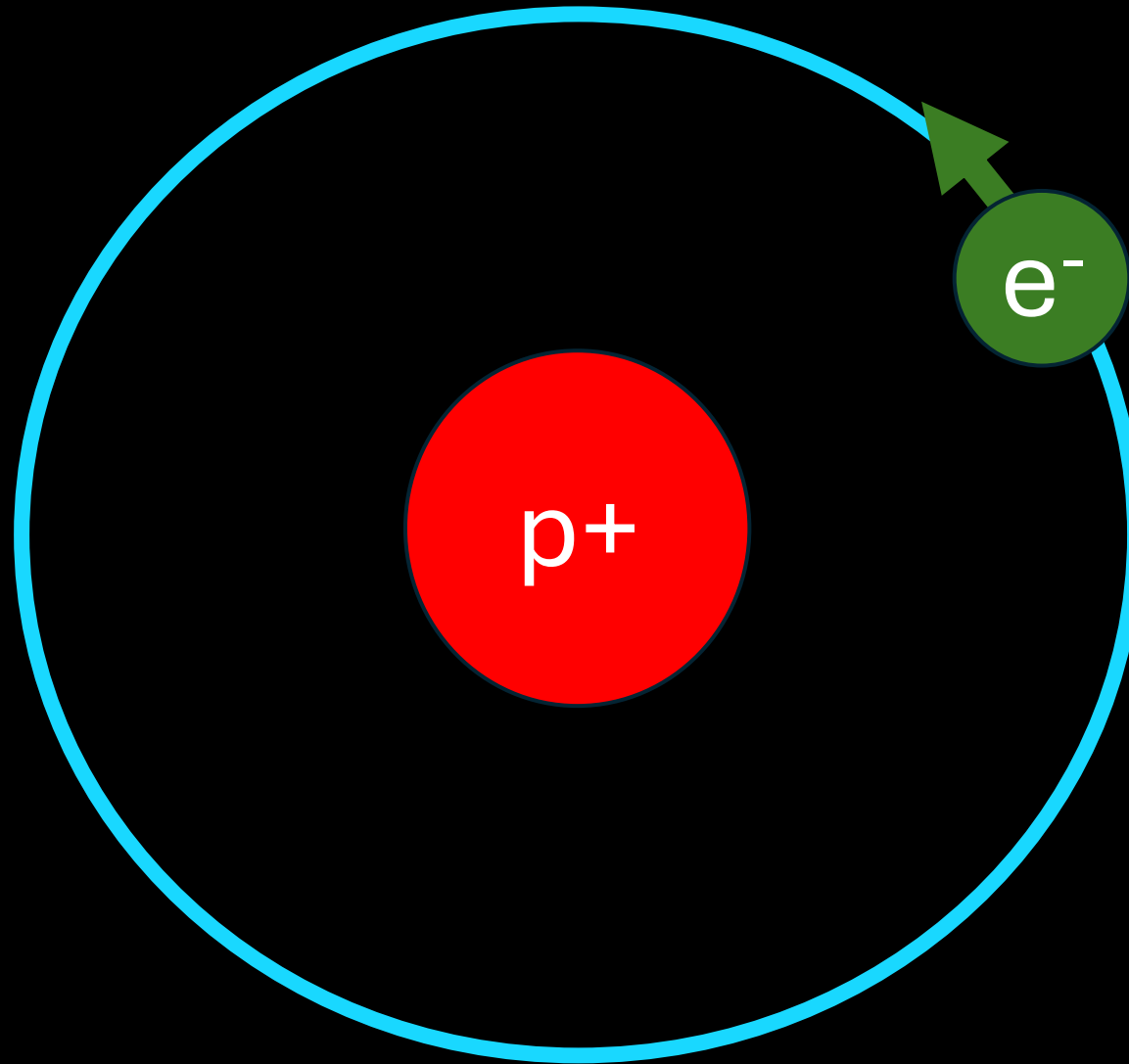
Today:
-270 °C



Loss of energy



Loss of energy



Low energy
state

Plasma

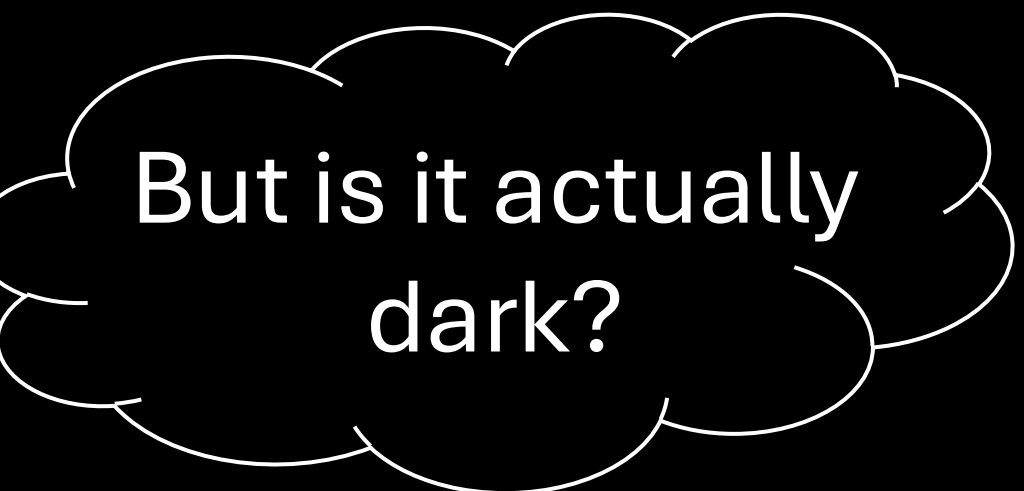


13.6 BYA

?????

13.6 BYA

The Dark Ages of the Universe

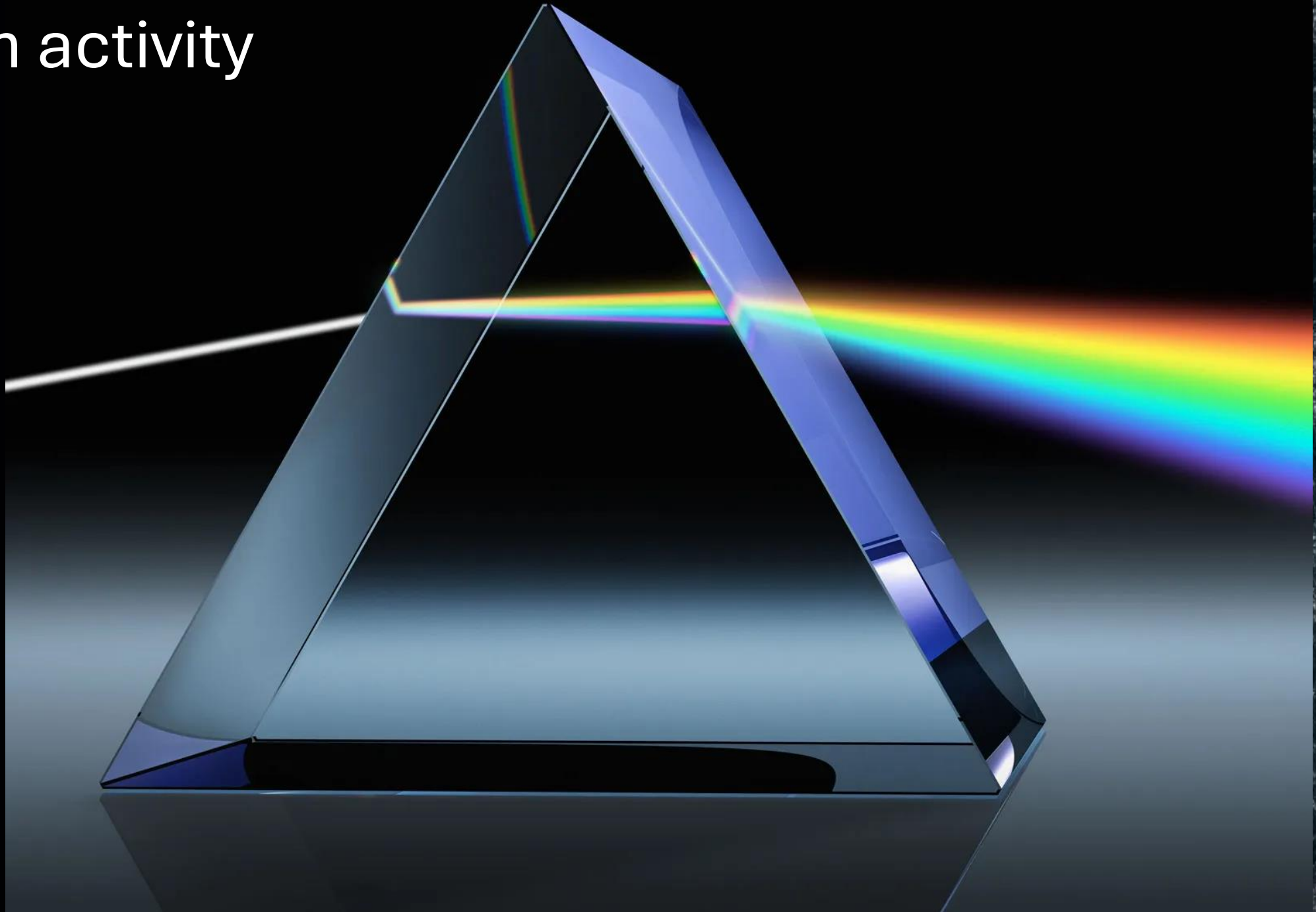


But is it actually
dark?

We observe the universe using light.
Do we really have the whole picture?



Prism activity



Prism activity

Group 1:

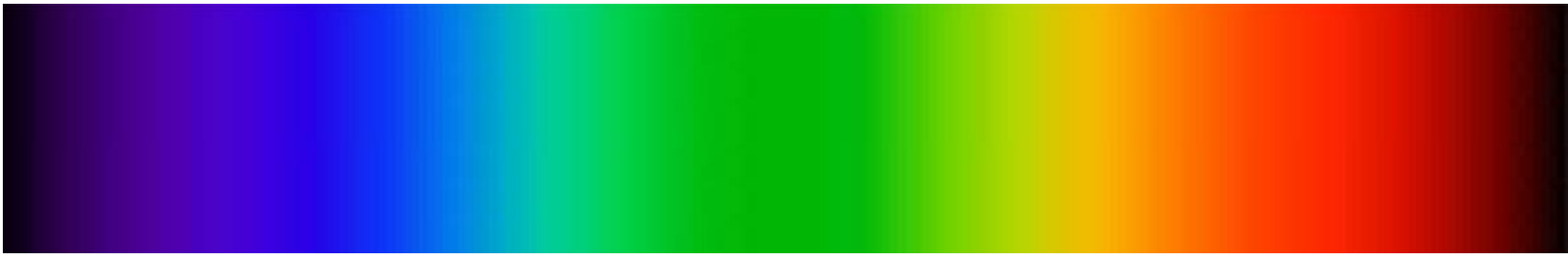
- 5 minutes
- 10 people
- Diffraction glasses

What did you
observe?
Why?

Group 2:

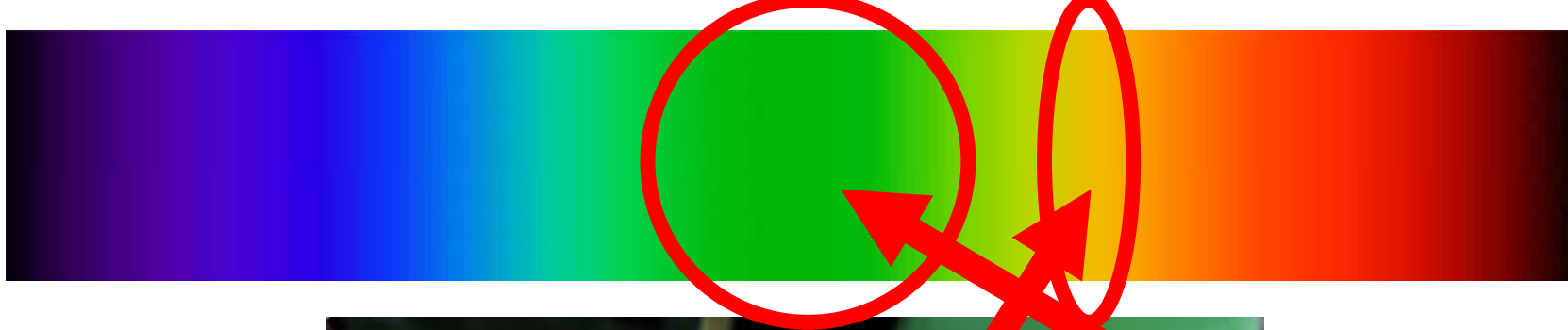
- 5 minutes
- 10 people
- Diffraction prisms





What do you
see? Identify the
colours on the
spectrum above

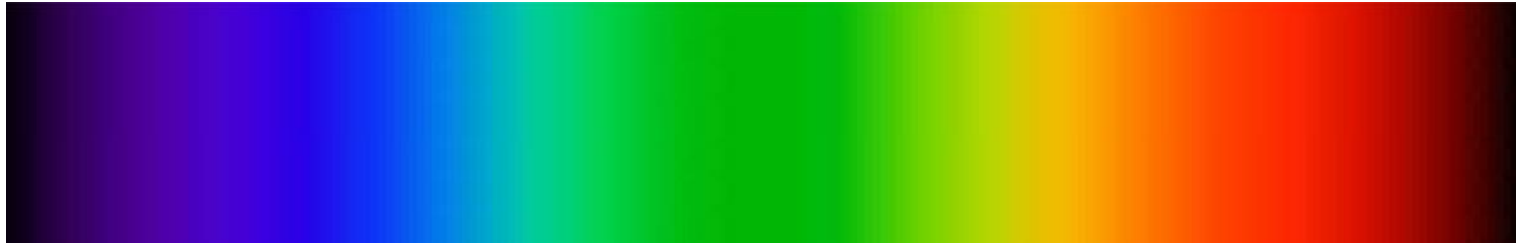




What do you
see? Identify the
colours on the
spectrum above

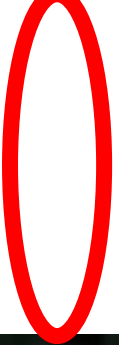


What do
you see?

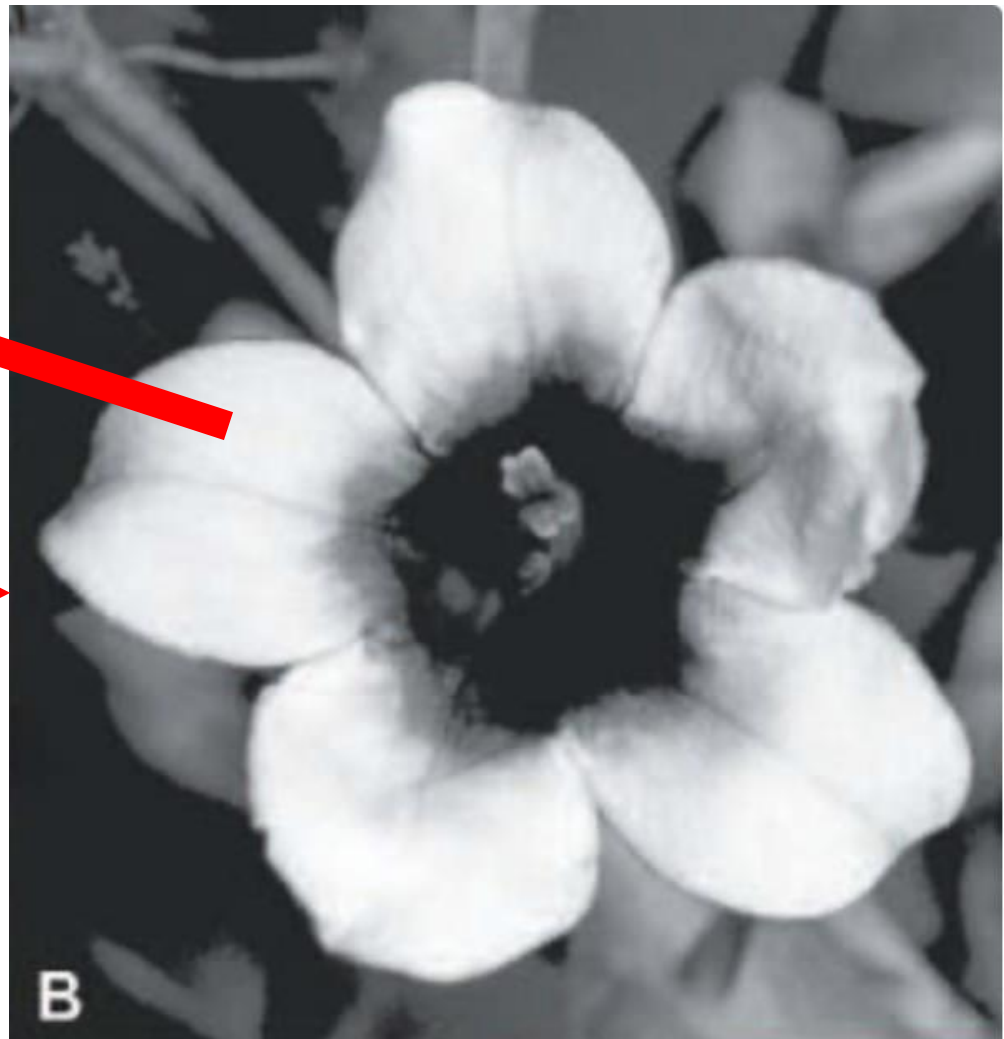


What a bee
sees





What a bee
sees



Radio

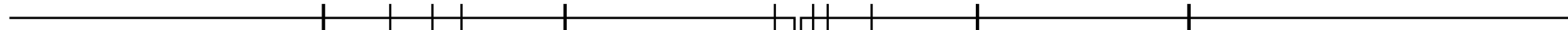
Microwave

Infrared

Ultraviolet

X-Rays

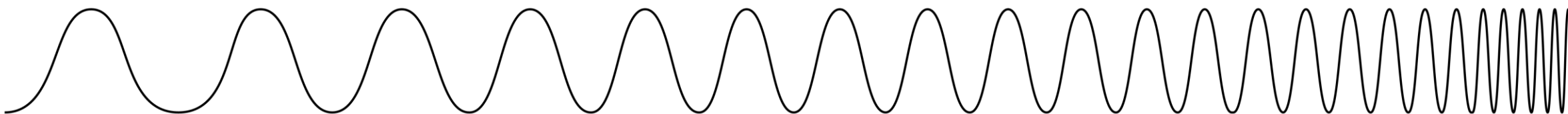
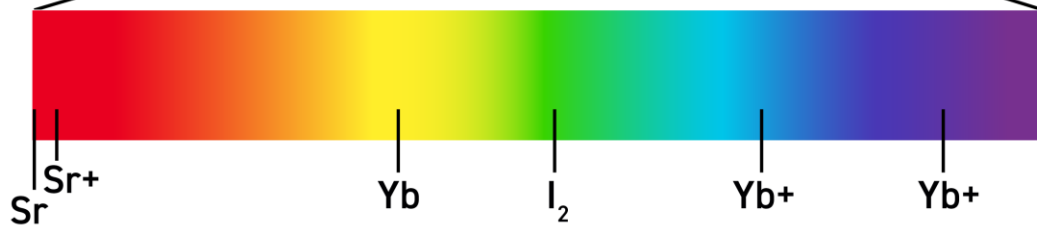
Gamma Rays



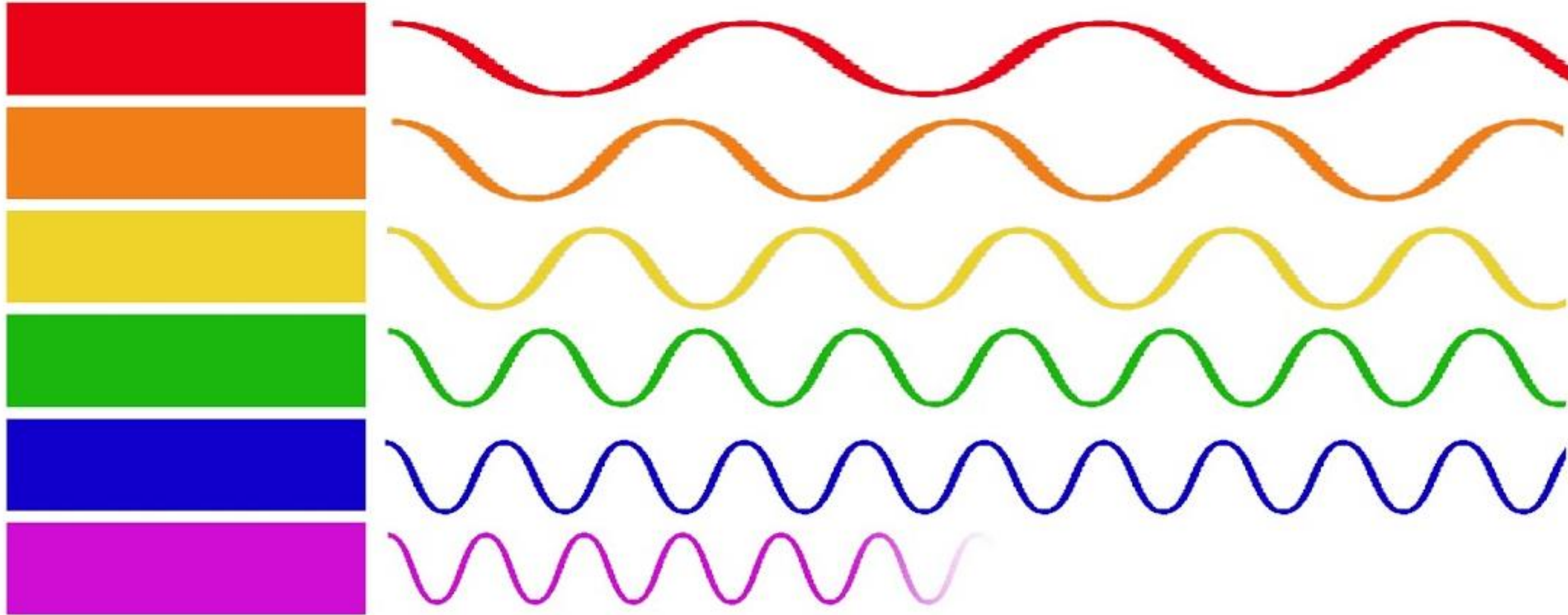
longer wavelength
lower frequency
lower energy

Visible Light

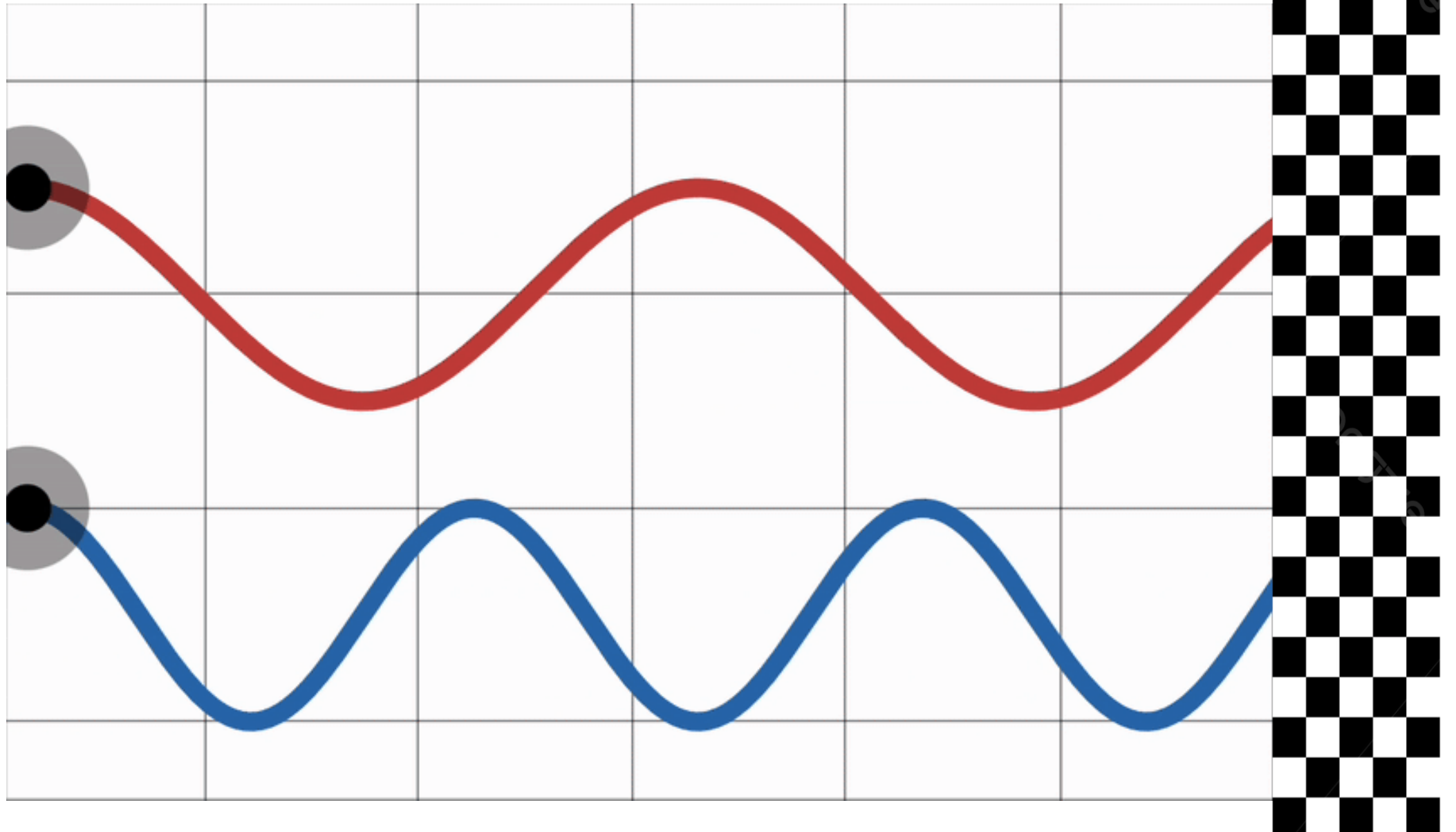
shorter wavelength
higher frequency
higher energy



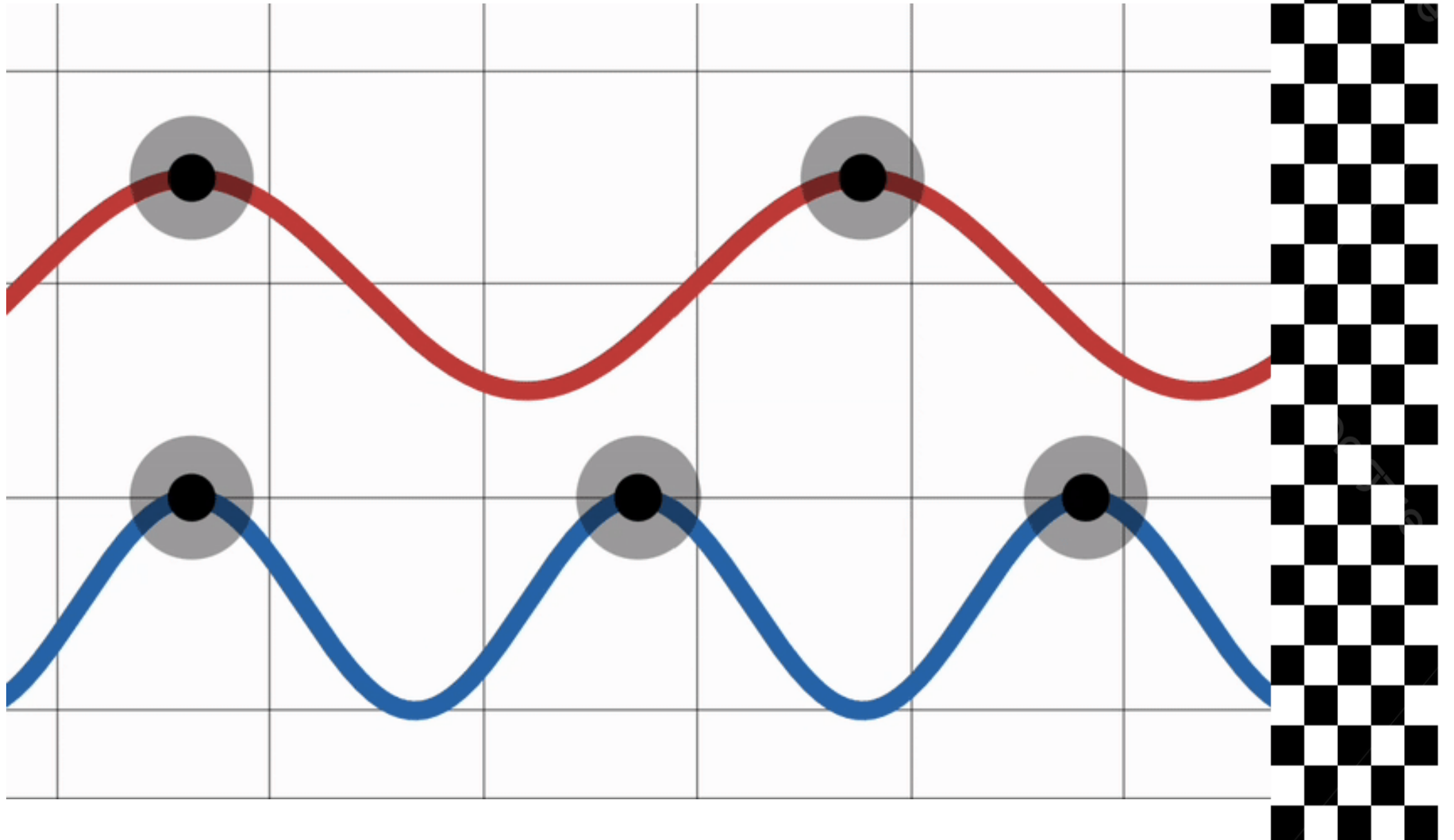
Light as a wave

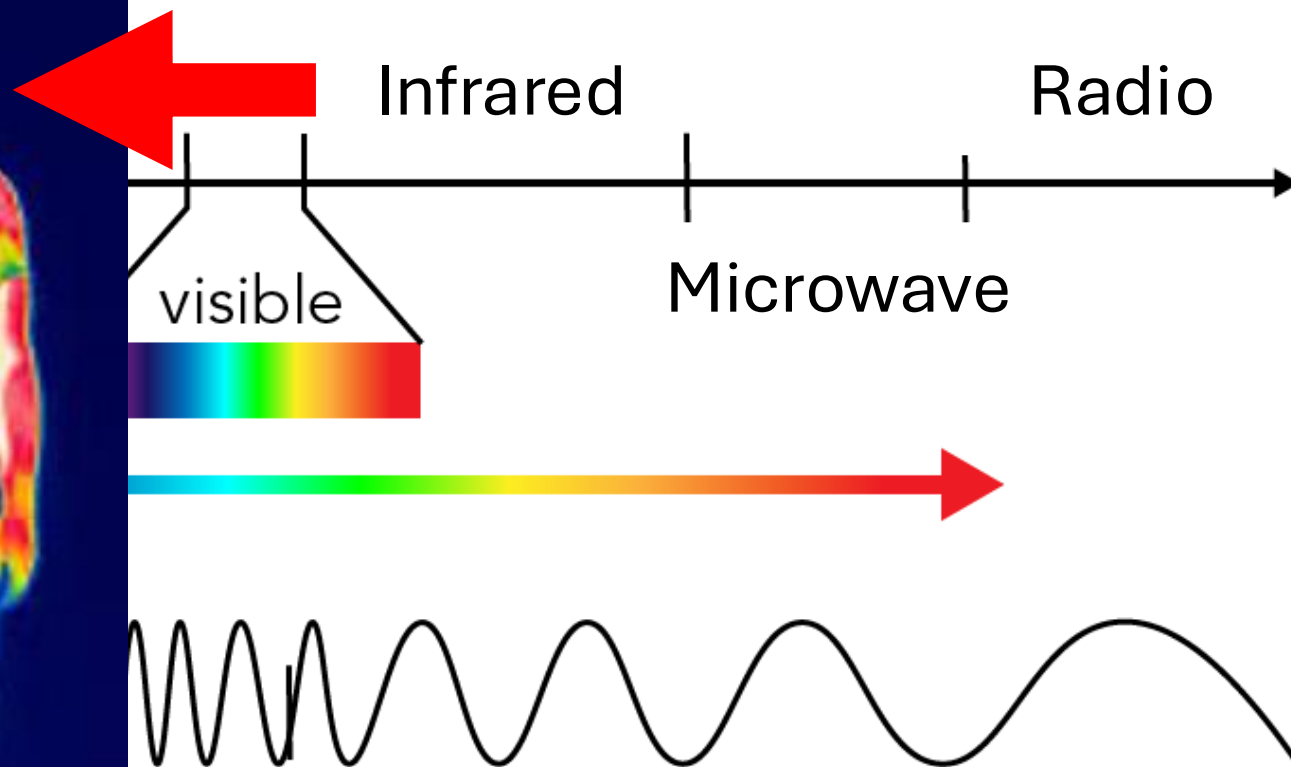


Speed of...



Frequency and Colour



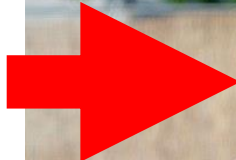


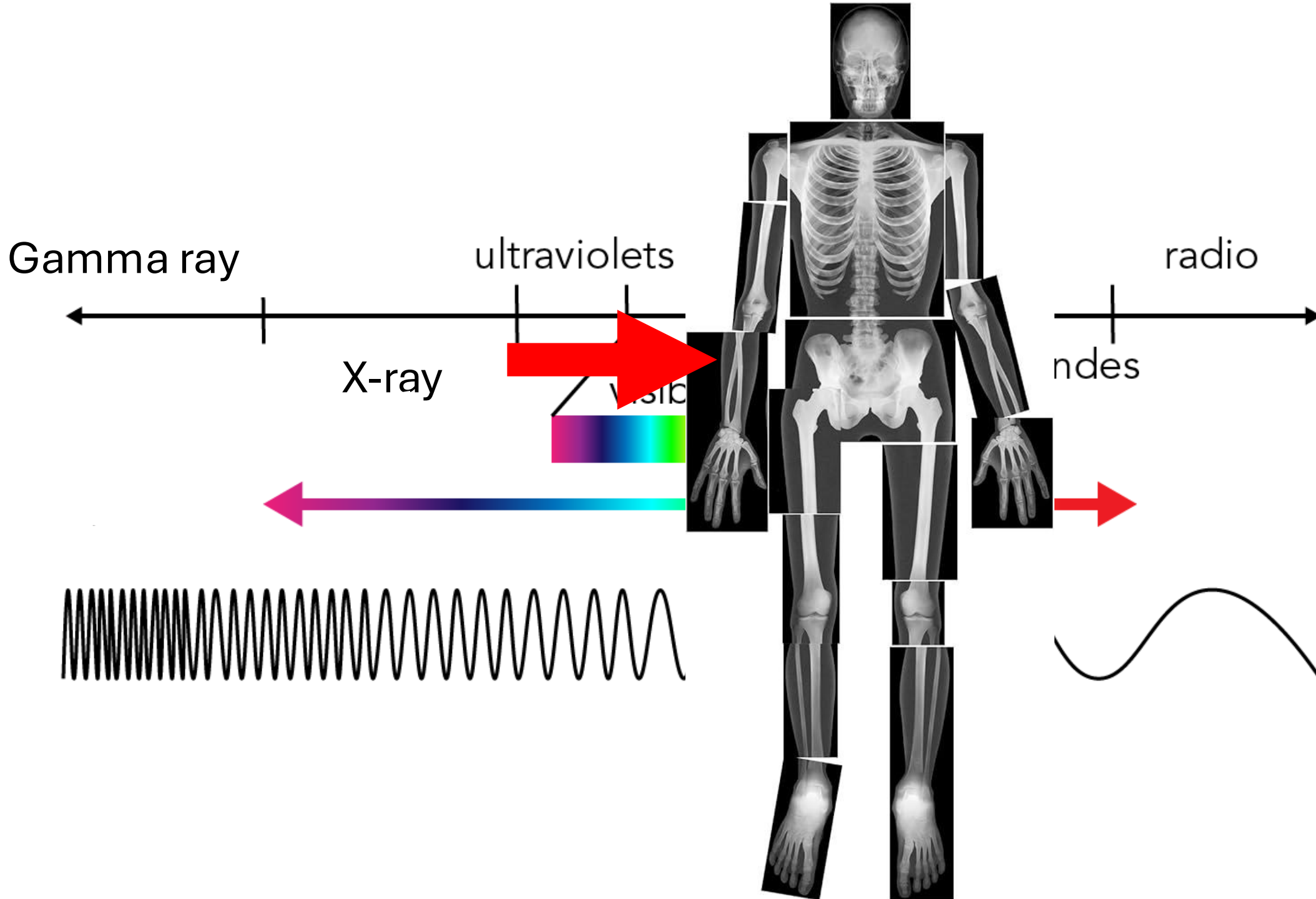
Gamma ray

ultraviolets

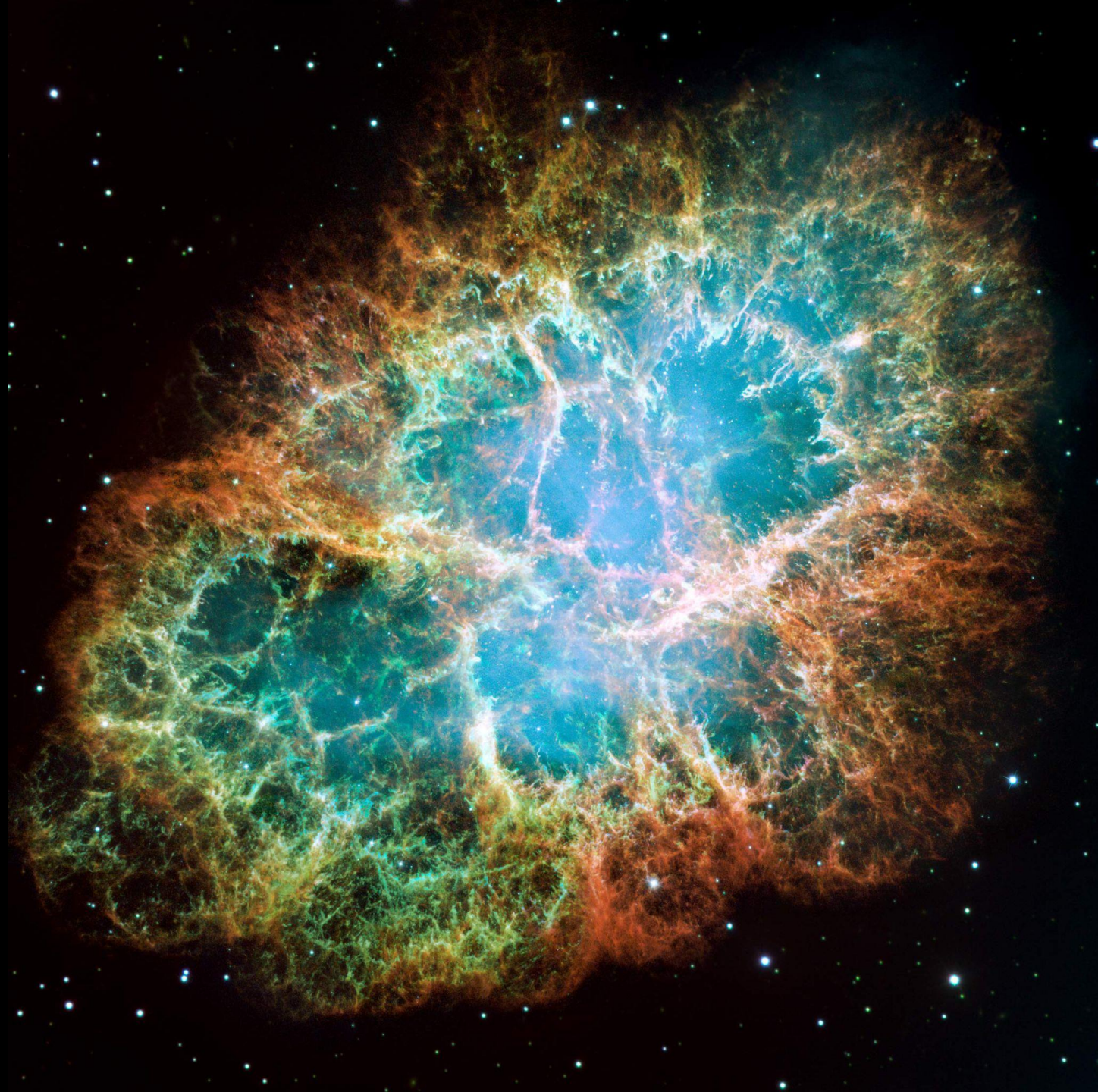
X-ray

visib

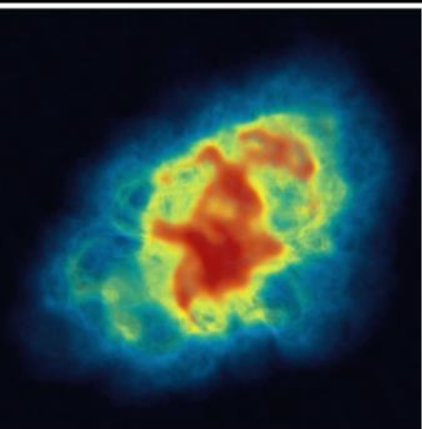




Crab nebula (visible)



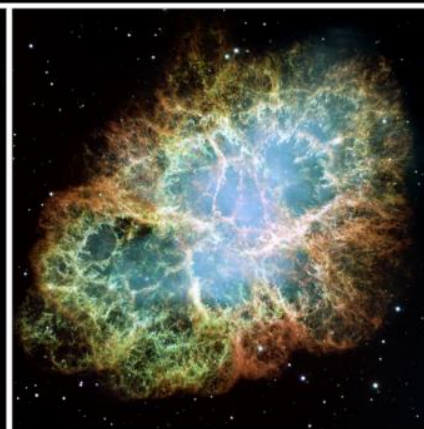
CRAB NEBULA



RADIO



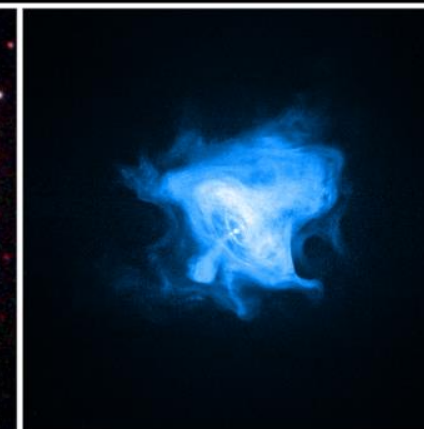
INFRARED



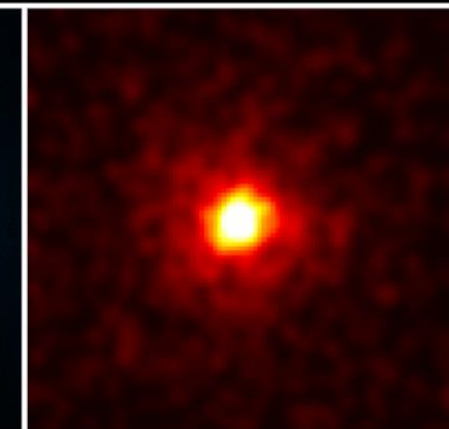
VISIBLE LIGHT



ULTRAVIOLET



X-RAYS



GAMMA RAYS

13.6 BYA

?????

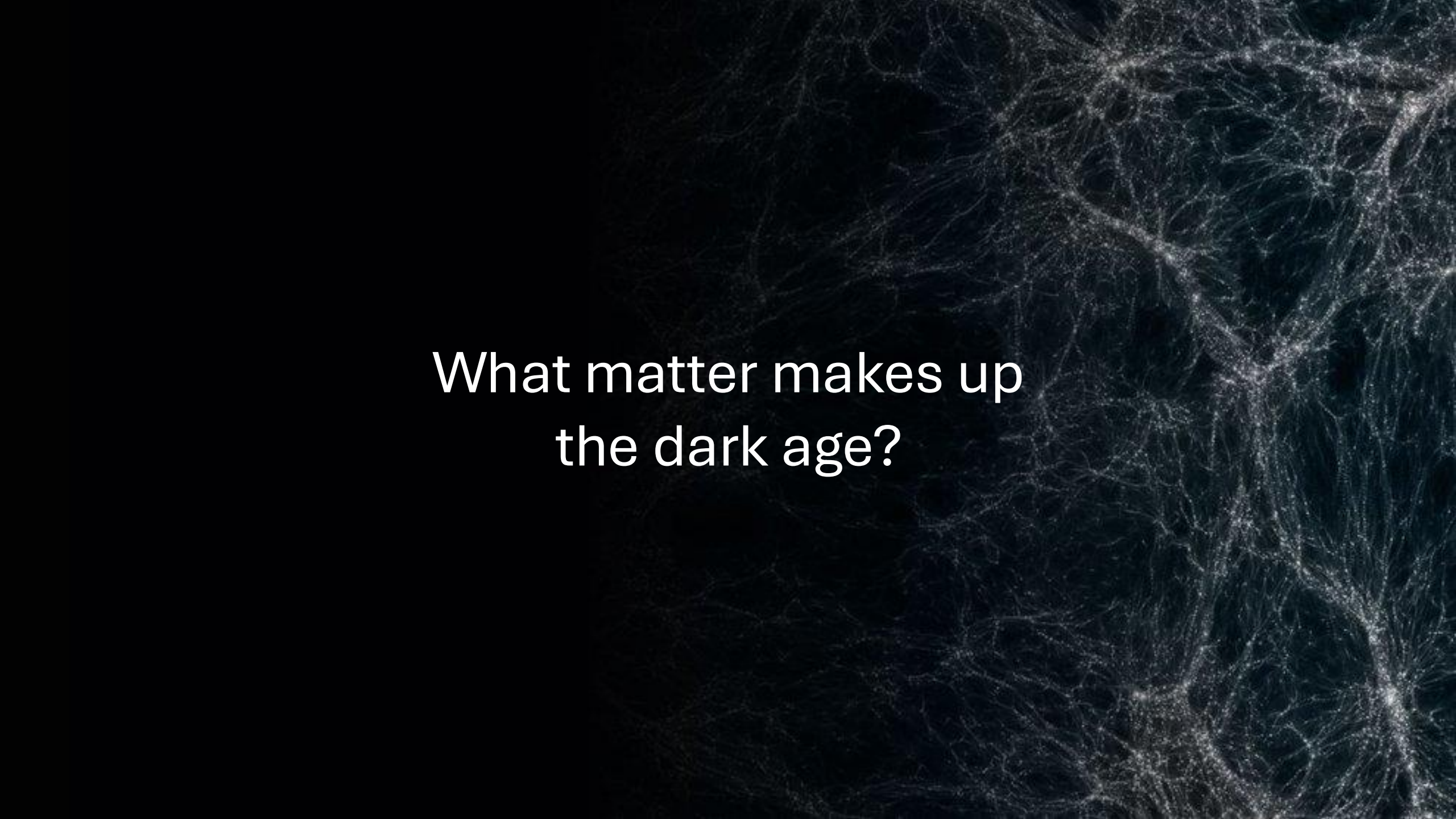
13.6 BYA

13.6

Radio

13.6 BYA

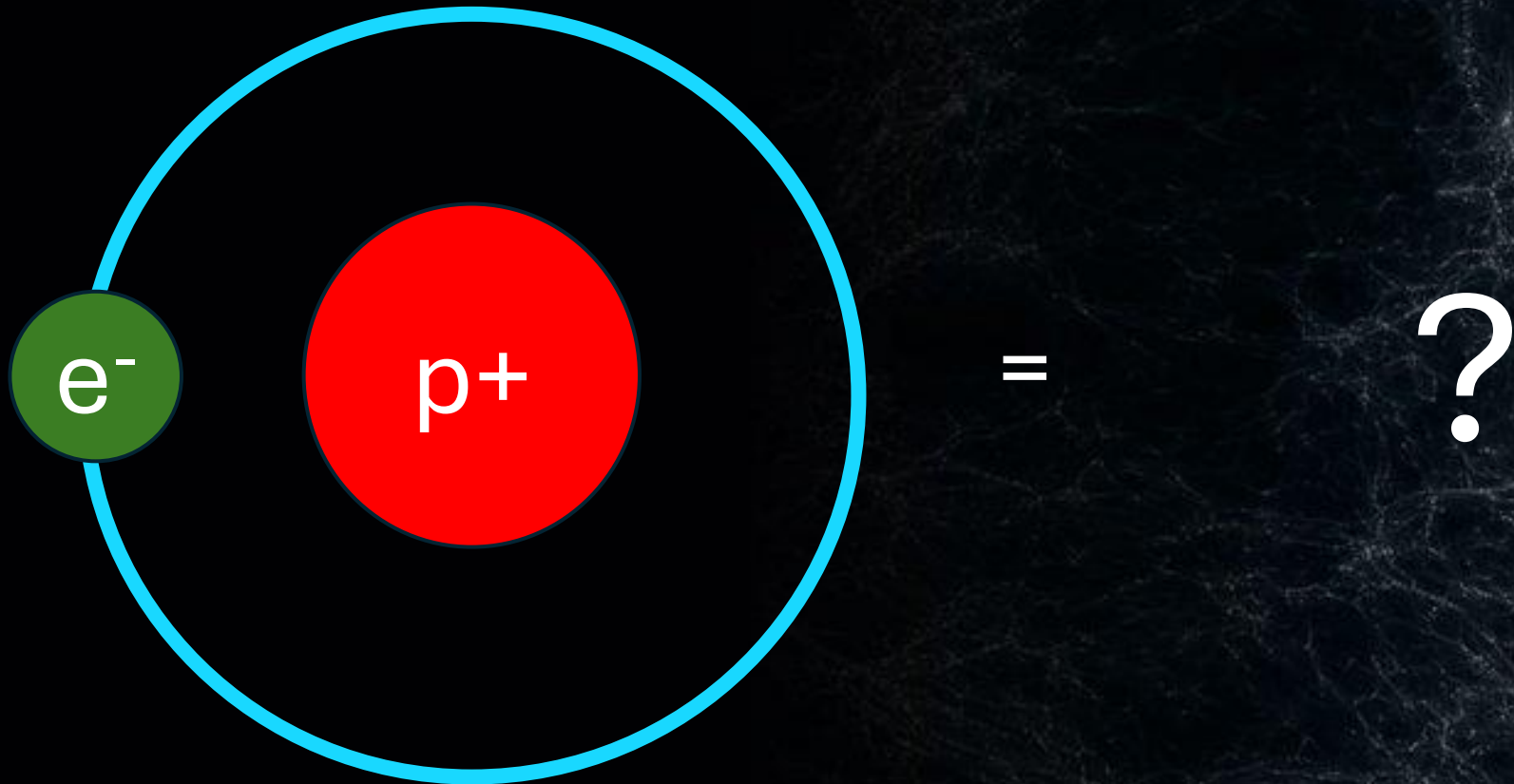
What do we
see at this
moment?

A visualization of the cosmic web, showing a complex network of dark matter filaments and galaxy clusters. The filaments are represented by thin, glowing lines of light, while the clusters are denser regions of light. The background is a deep black, emphasizing the structure of the universe.

What matter makes up
the dark age?

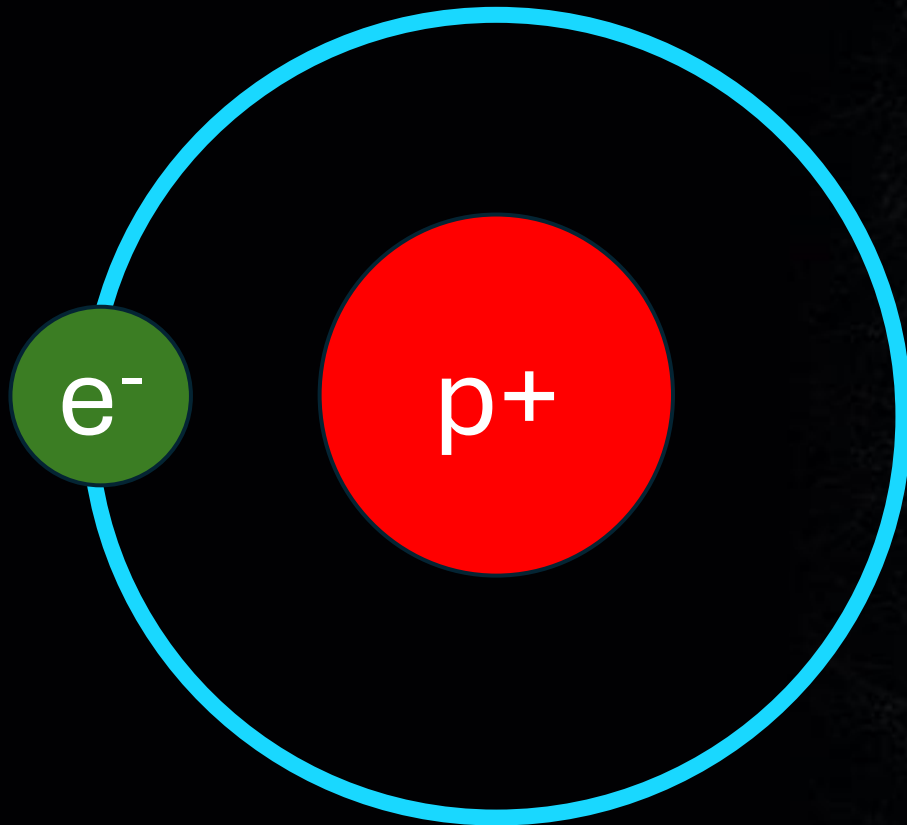
What matter makes up the dark age?

Cooler Temperatures $\rightarrow$ Recombination

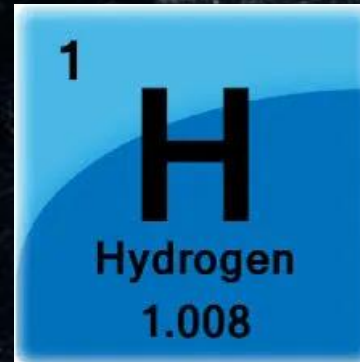


What matter makes up the dark age?

Cooler Temperatures → Recombination



=



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Quantum states of Hydrogen



$|\uparrow \text{ ou } \downarrow\rangle$



$|\uparrow \text{ ou } \downarrow\rangle$



||



||



Hydrogen Spin Activity

Teams of 1 or 2

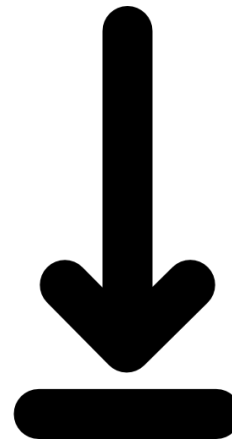
5 minutes

Come get your hydrogen atom!

Which configuration is more stable?



= spin up



= spin down

Quantum states of Hydrogen

Which configuration is
more stable?

A solid green circle with a thin dark blue outline, containing the quantum state notation $|\uparrow \text{ou} \downarrow\rangle$ in white text.

$|\uparrow \text{ou} \downarrow\rangle$

A solid red circle with a thin dark blue outline, containing the quantum state notation $|\uparrow \text{ou} \downarrow\rangle$ in white text.

$|\uparrow \text{ou} \downarrow\rangle$

Most stable configuration



$|\uparrow\rangle$



$|\downarrow\rangle$

or...

Most stable configuration:
Antiparallel spin



Configuration stability

Which is more stable?

A)



B)

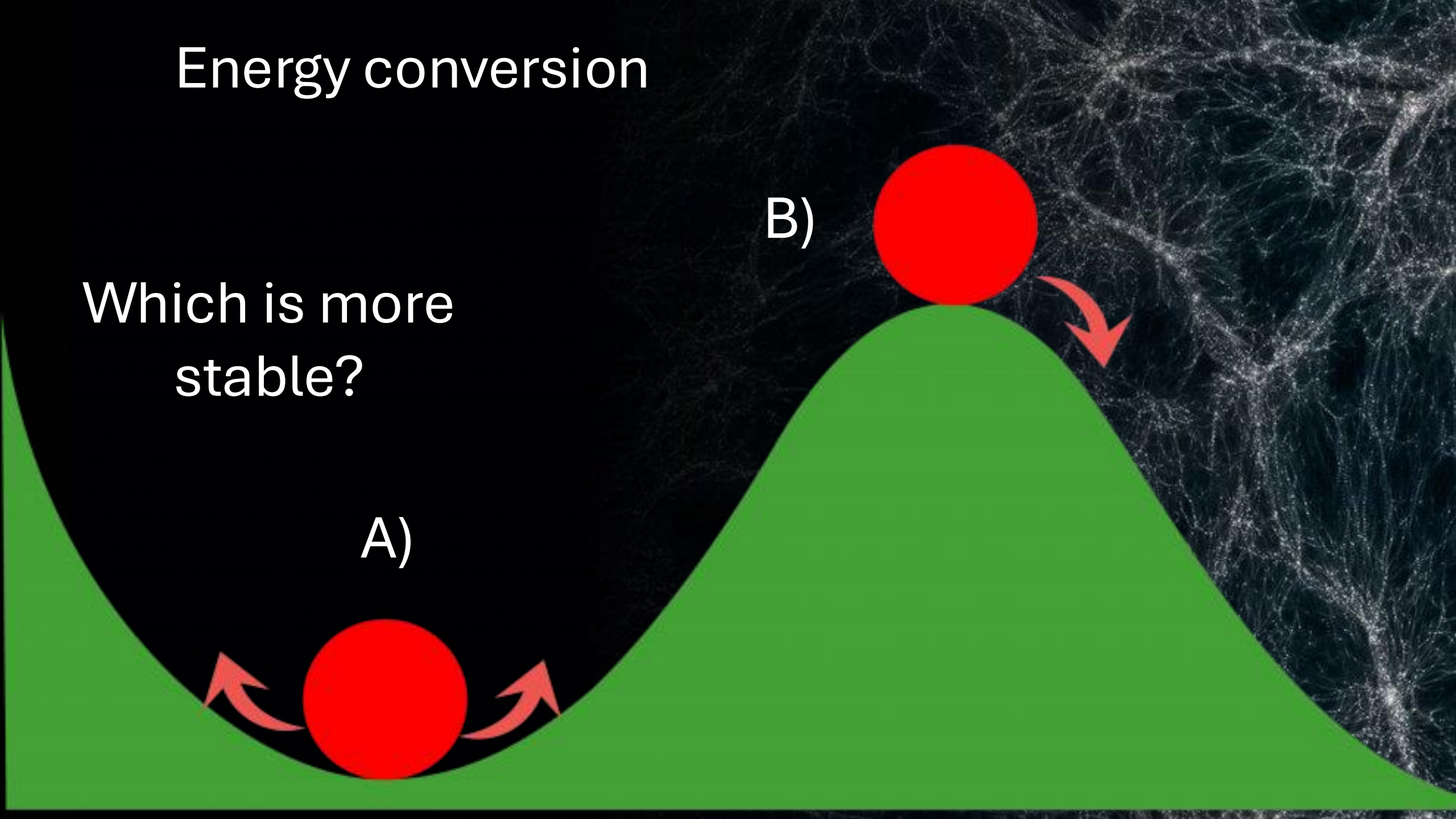


Energy conversion

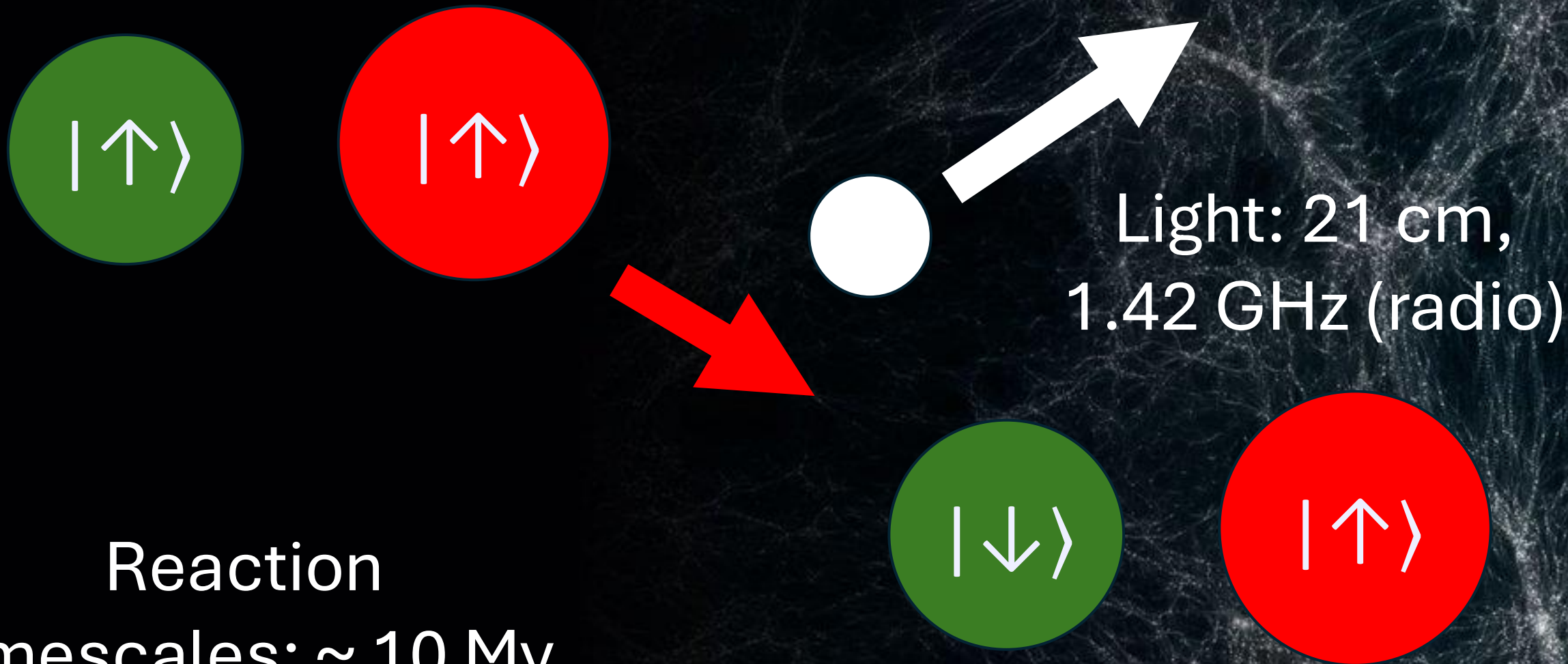
Which is more stable?

A)

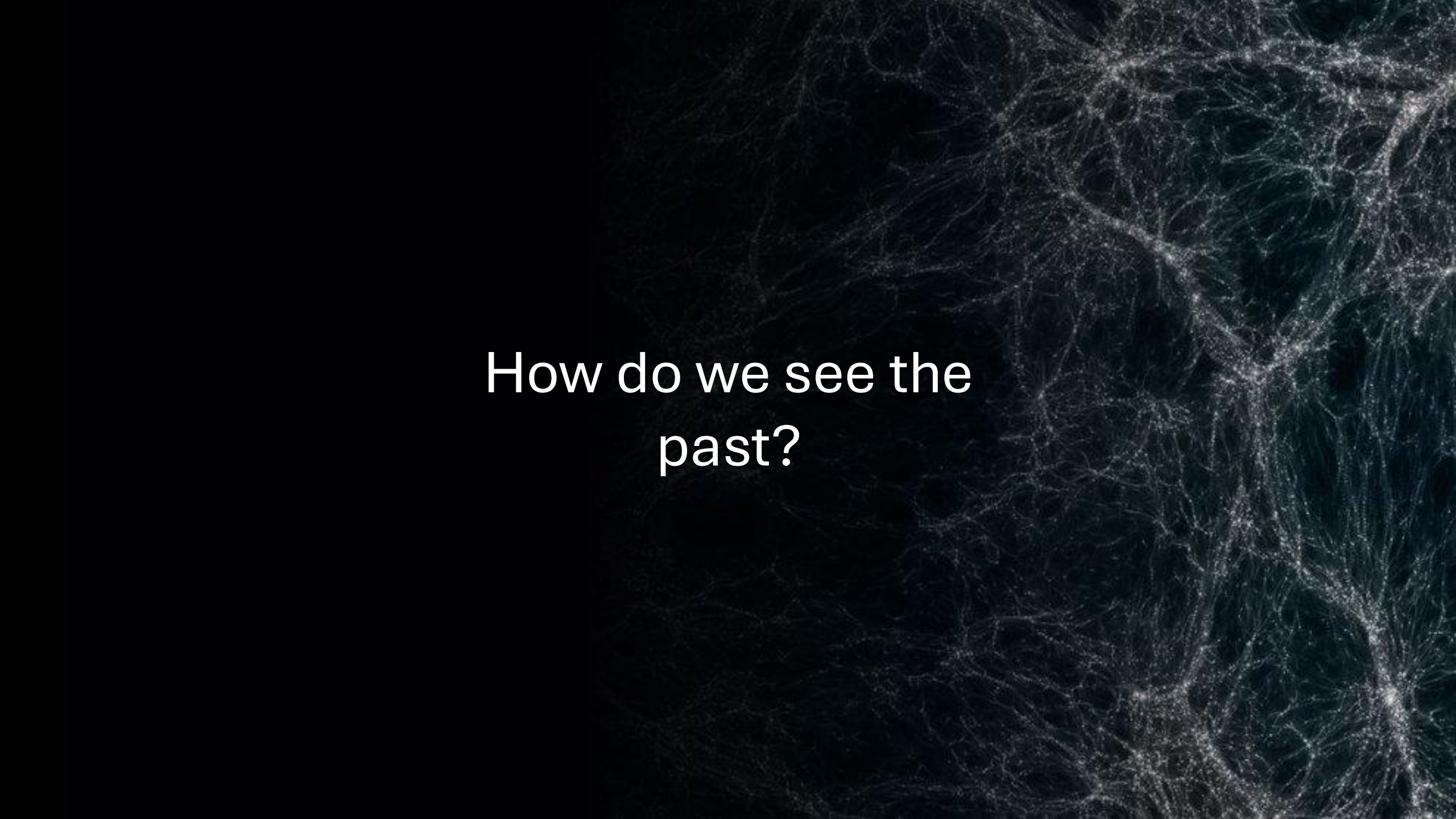
B)



Quantum states of Hydrogen

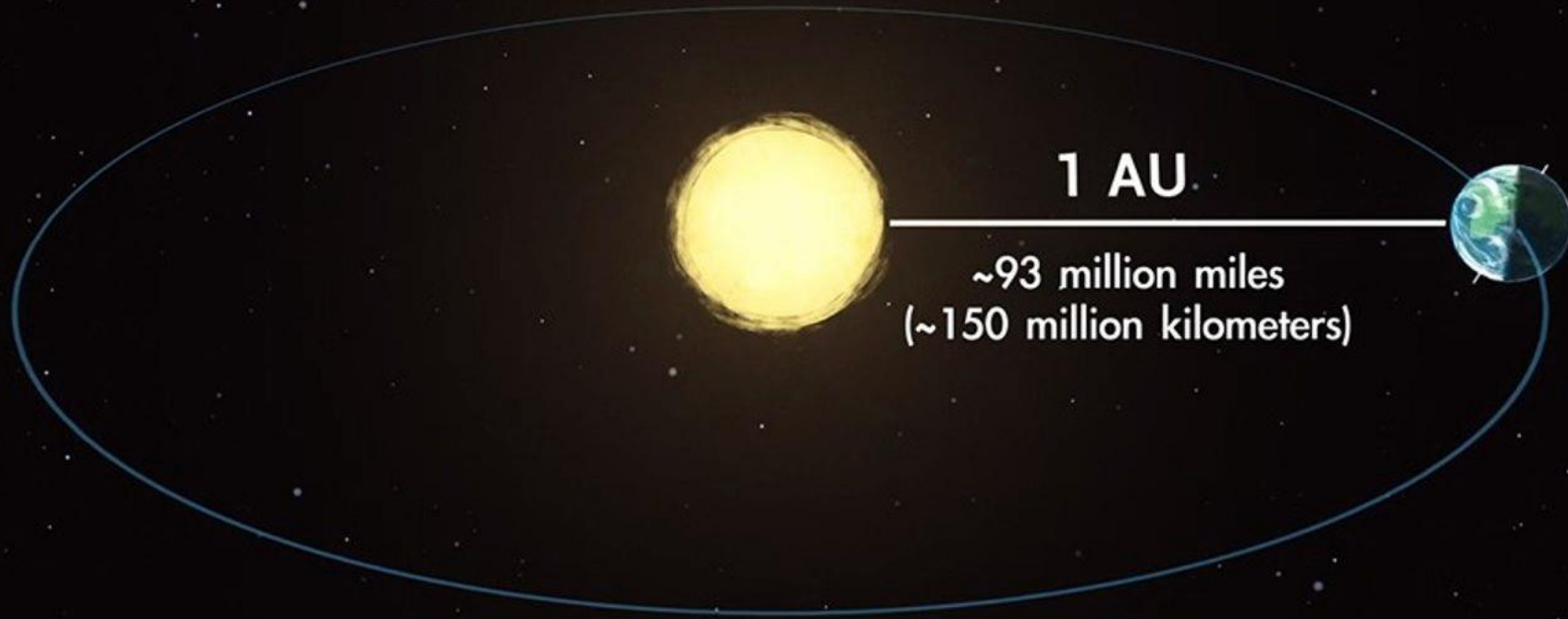


Reaction
timescales: ~ 10 My

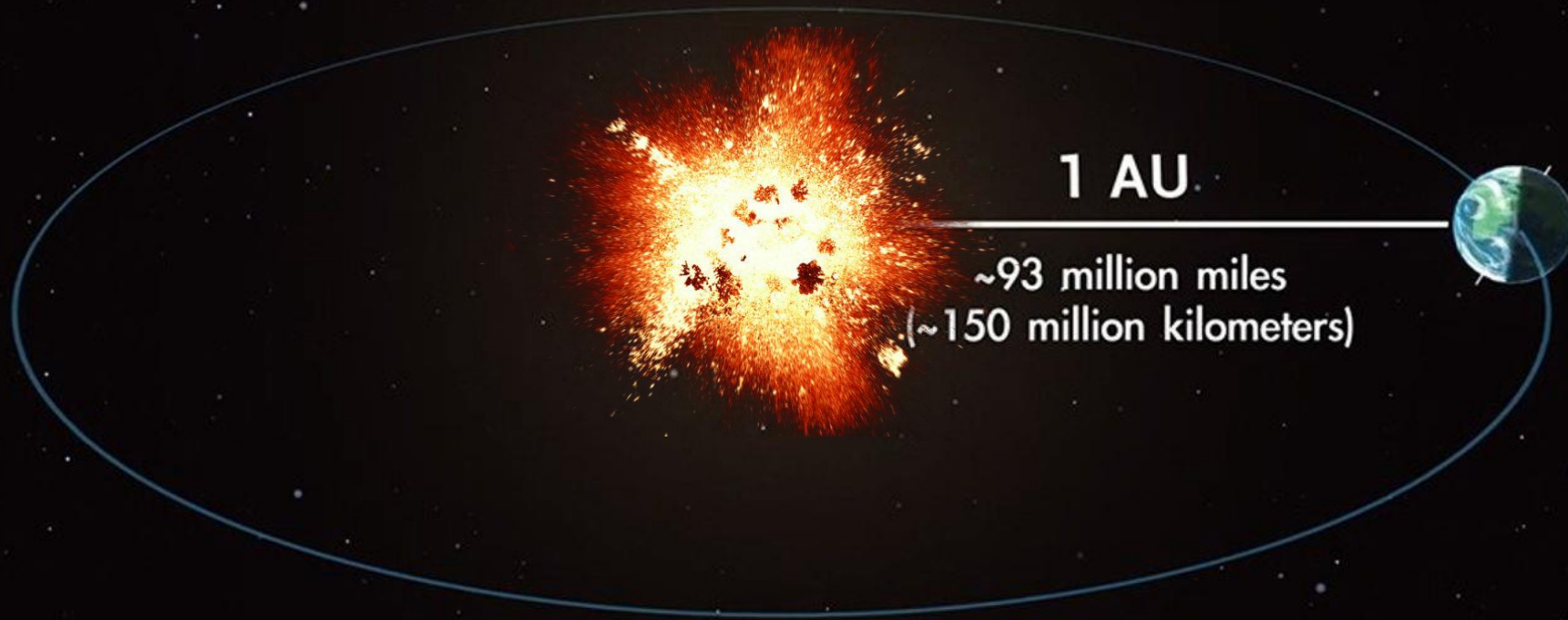
The background of the slide is a deep black space filled with a complex, glowing network of white and light blue filaments. These filaments represent the cosmic web, showing the large-scale structure of the universe with its dense clusters and elongated filaments of matter.

How do we see the
past?

Seeing into the past

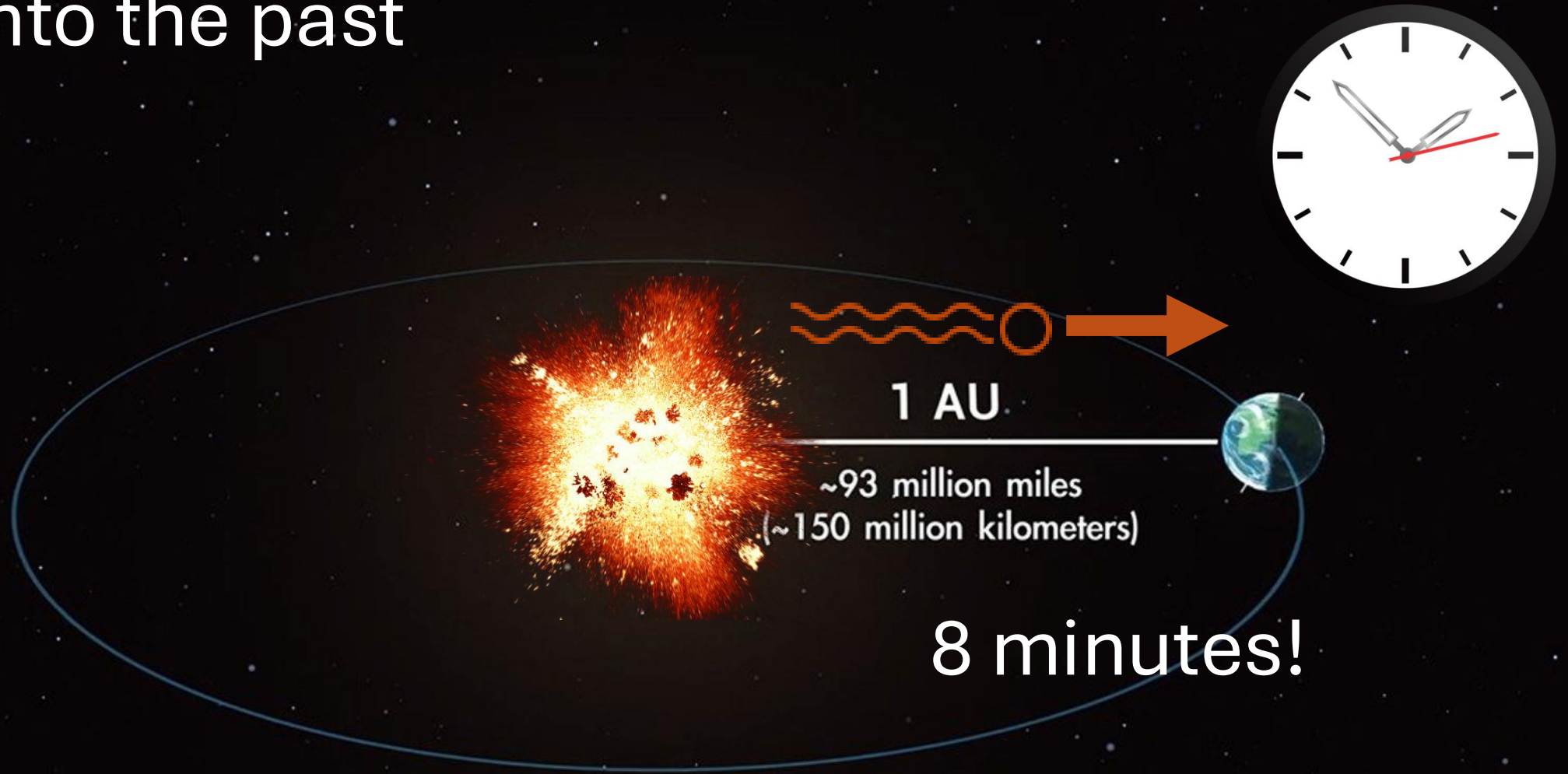


Seeing into the past



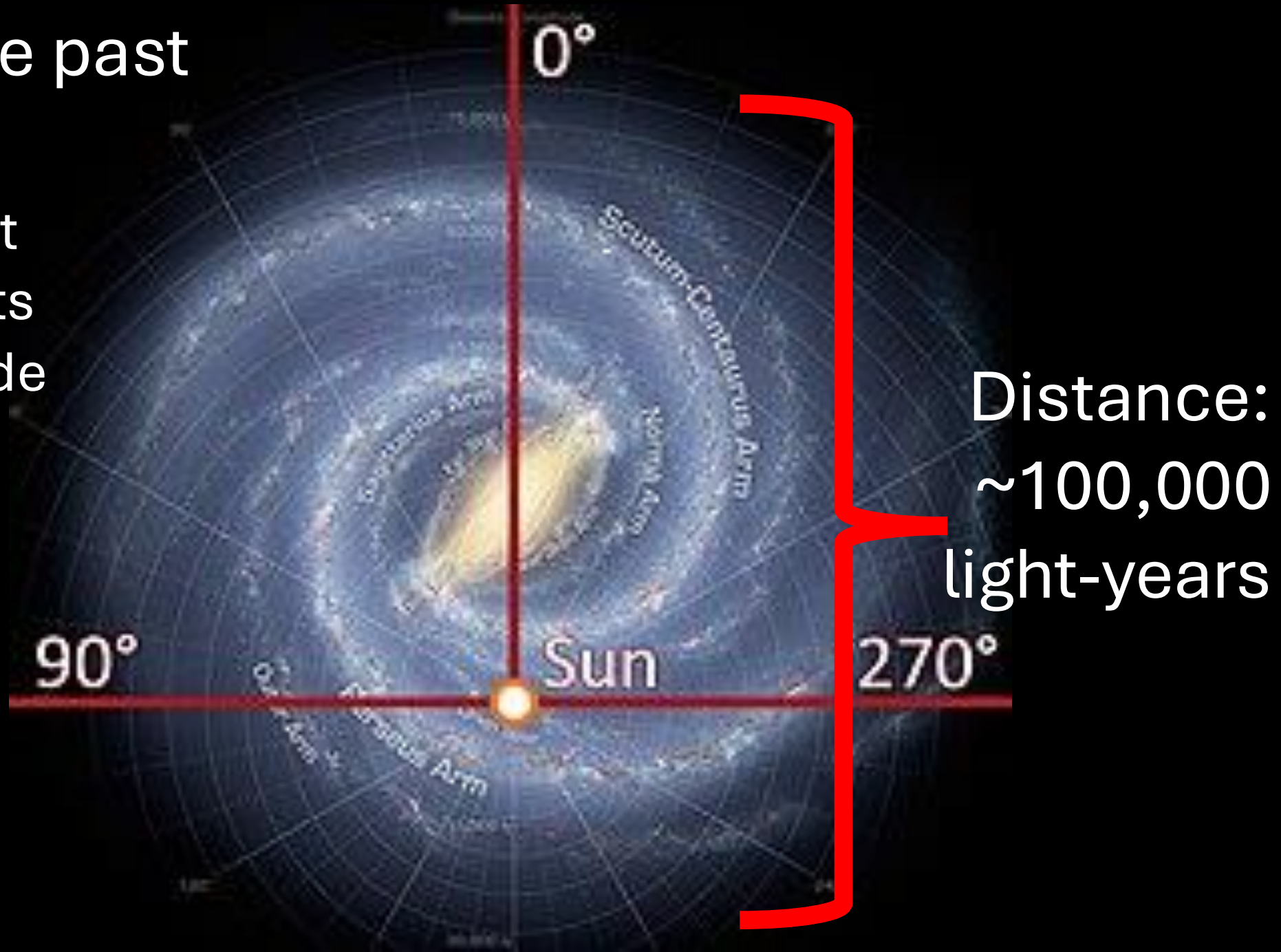
The sun just exploded!
What immediate effects do we feel?

Seeing into the past



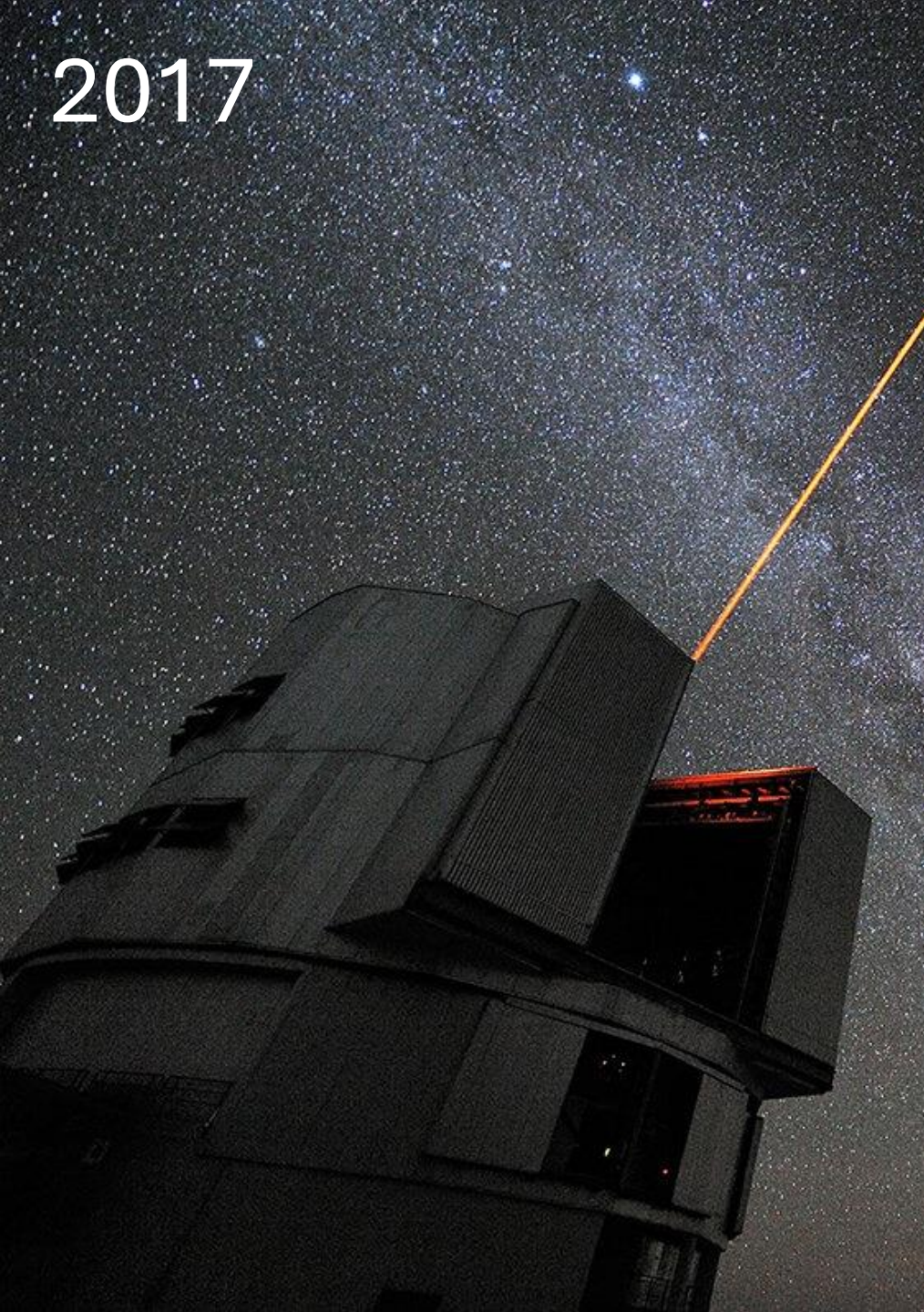
Seeing into the past

How long does it take to see events from the other side of the galaxy



2017

144 MYA (144M light-years away)

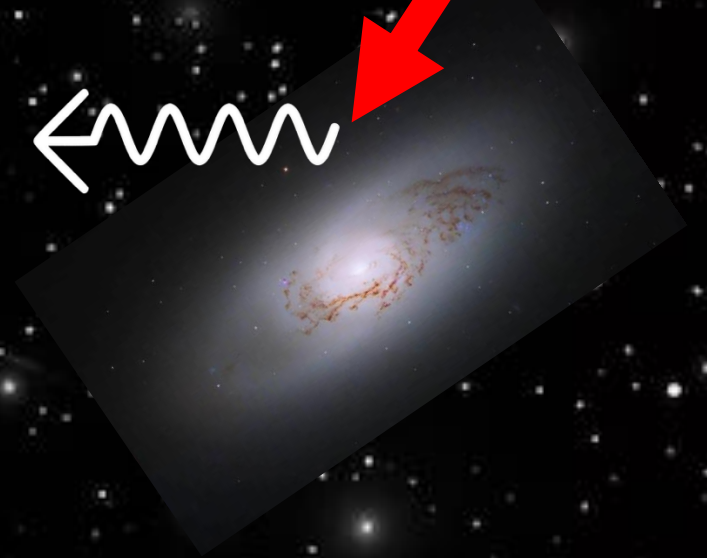
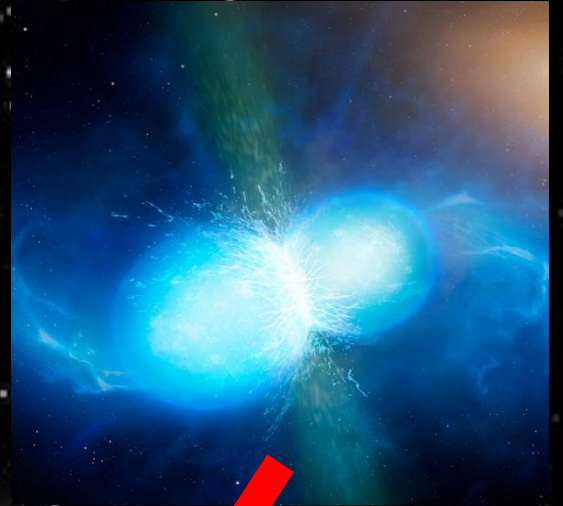


Stellar Merger GW170817

144 Million Years



Milky Way



NGC 4993 Galaxy

66 Million Years



56 Million Years



6 Million Years



100,000 Years



2017



Seeing into the past

What do we see?

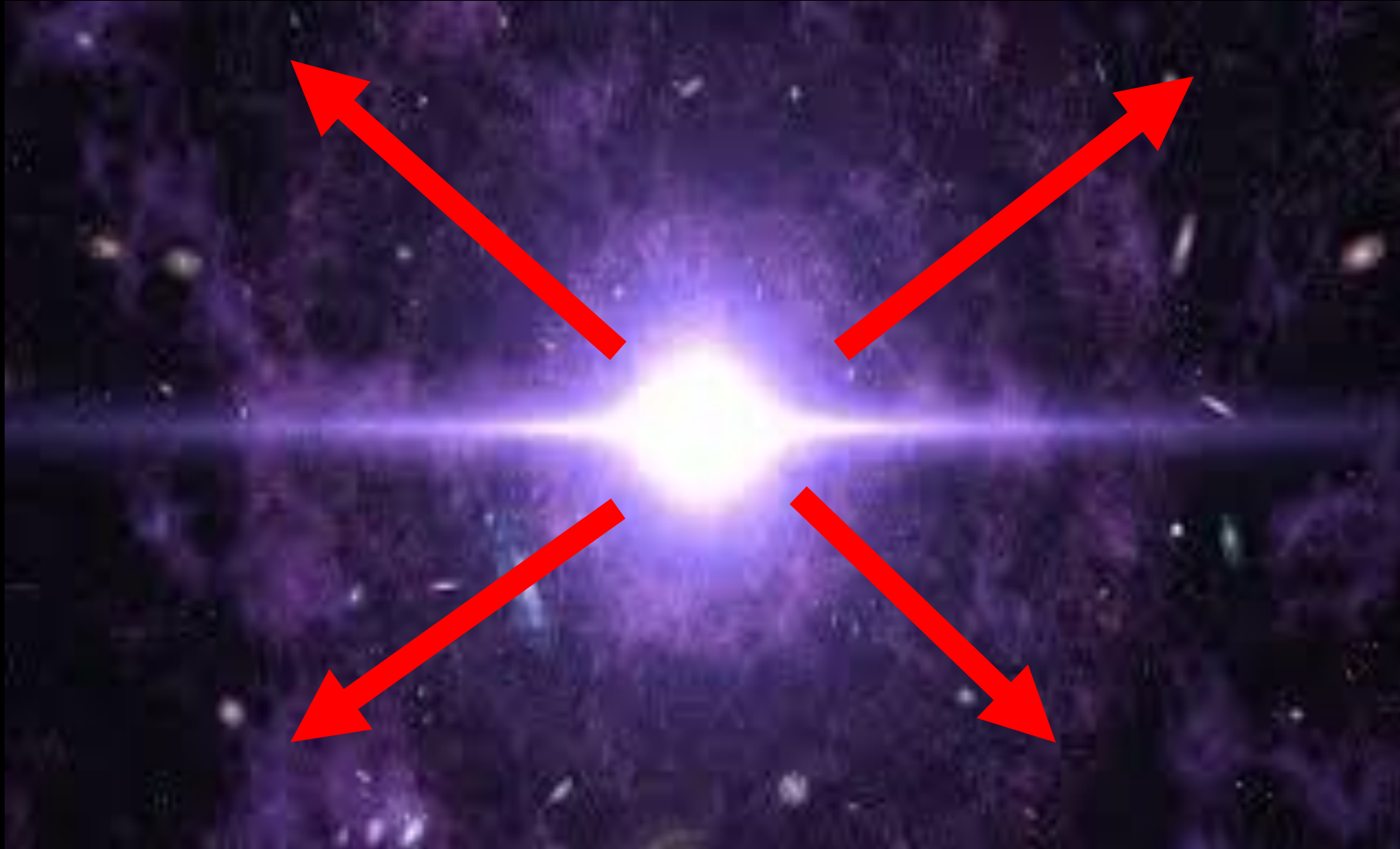


13.6 Billion light-years

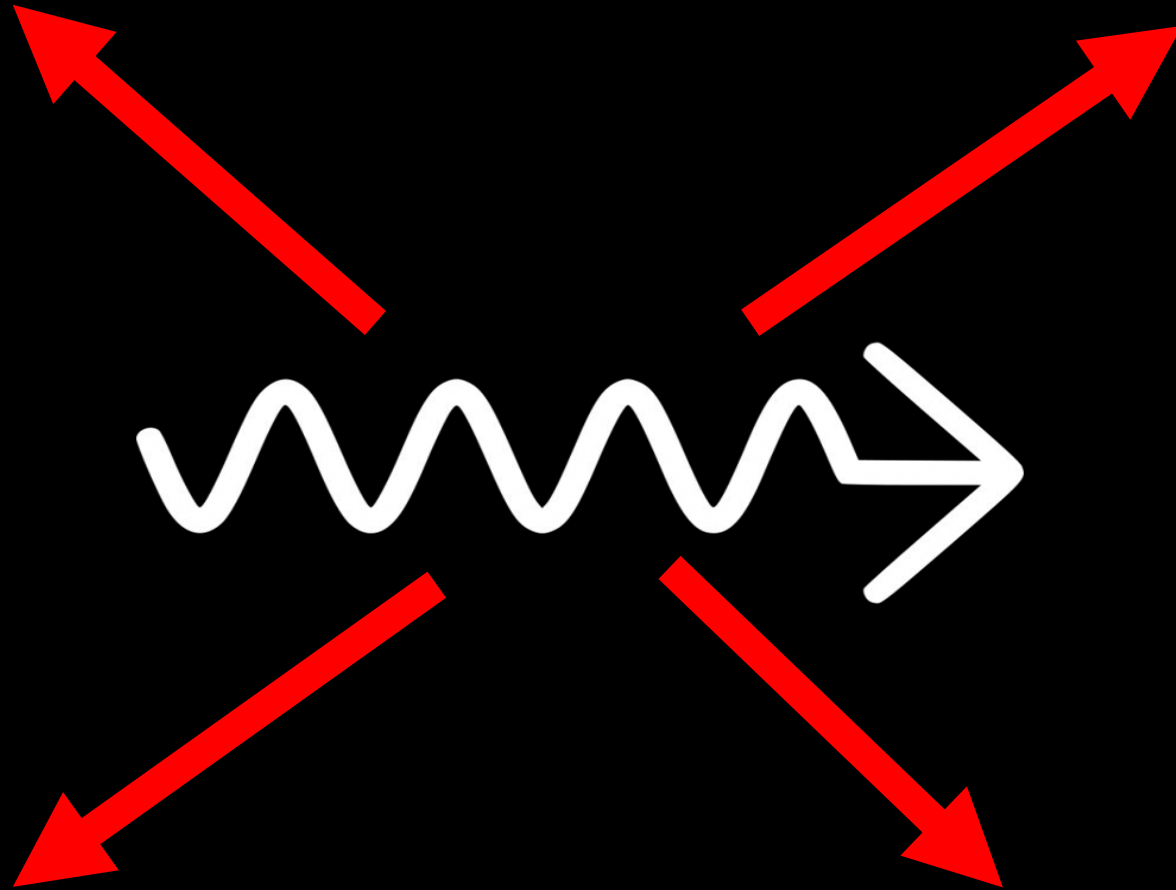
13.6 Billion light-years

Why red?

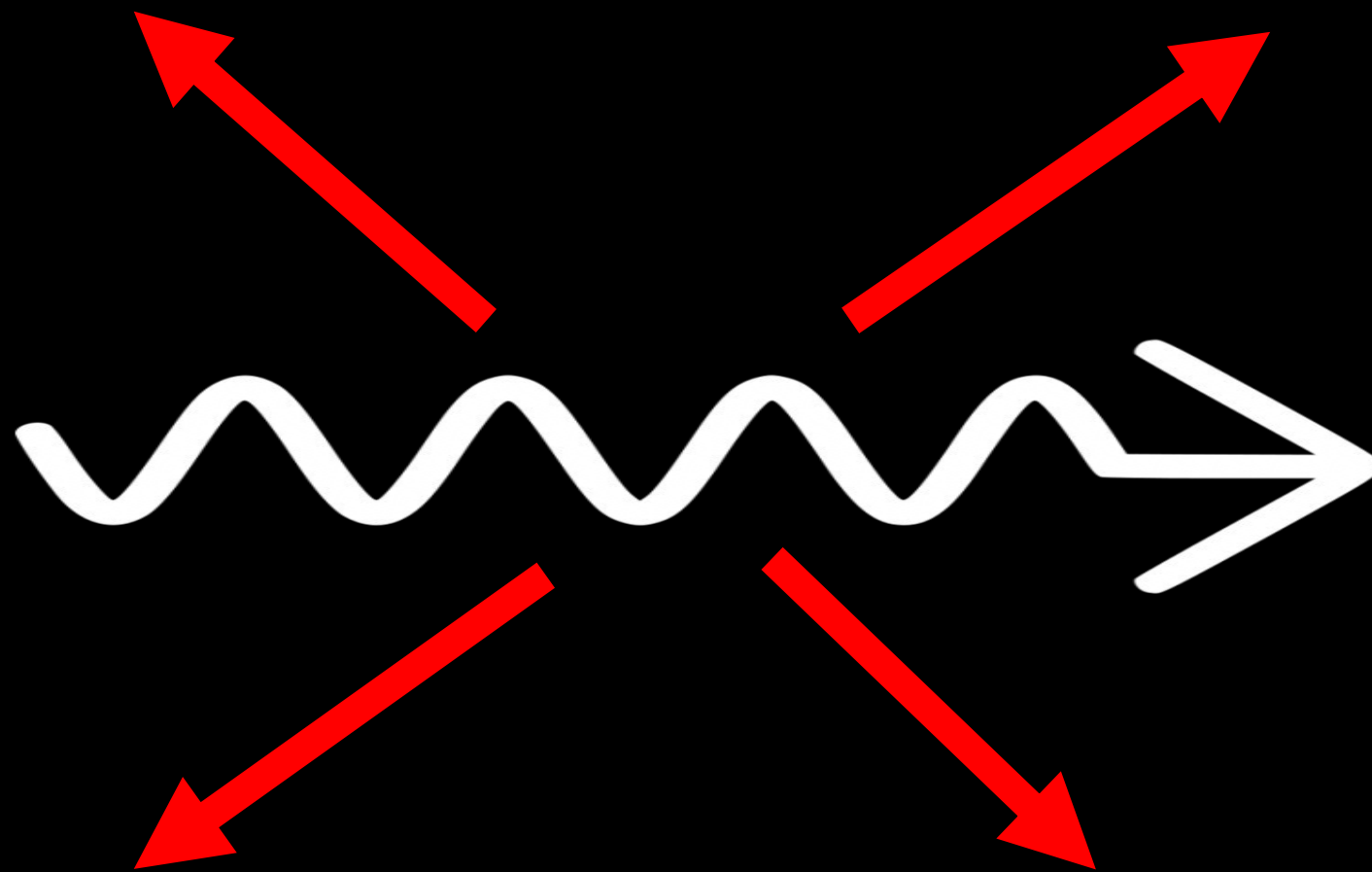
Expansion of the Universe



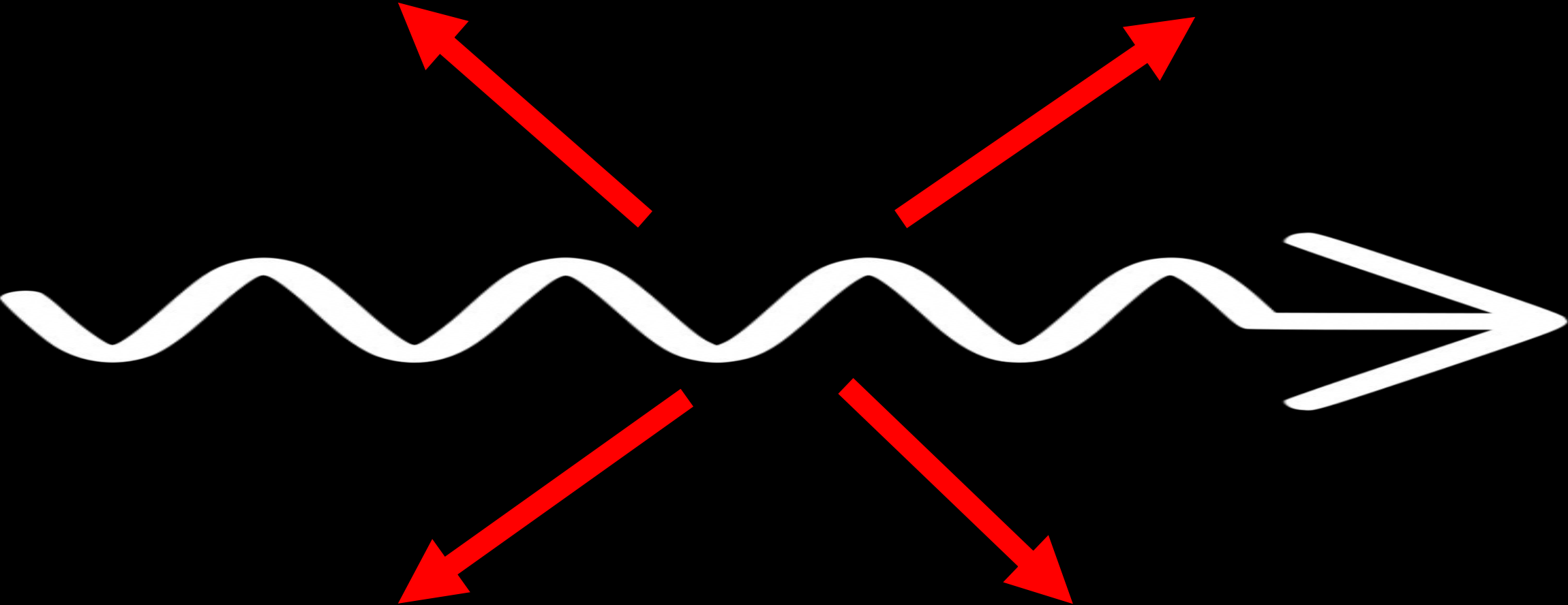
Expansion of the Universe



Expansion of the Universe



Expansion of the Universe

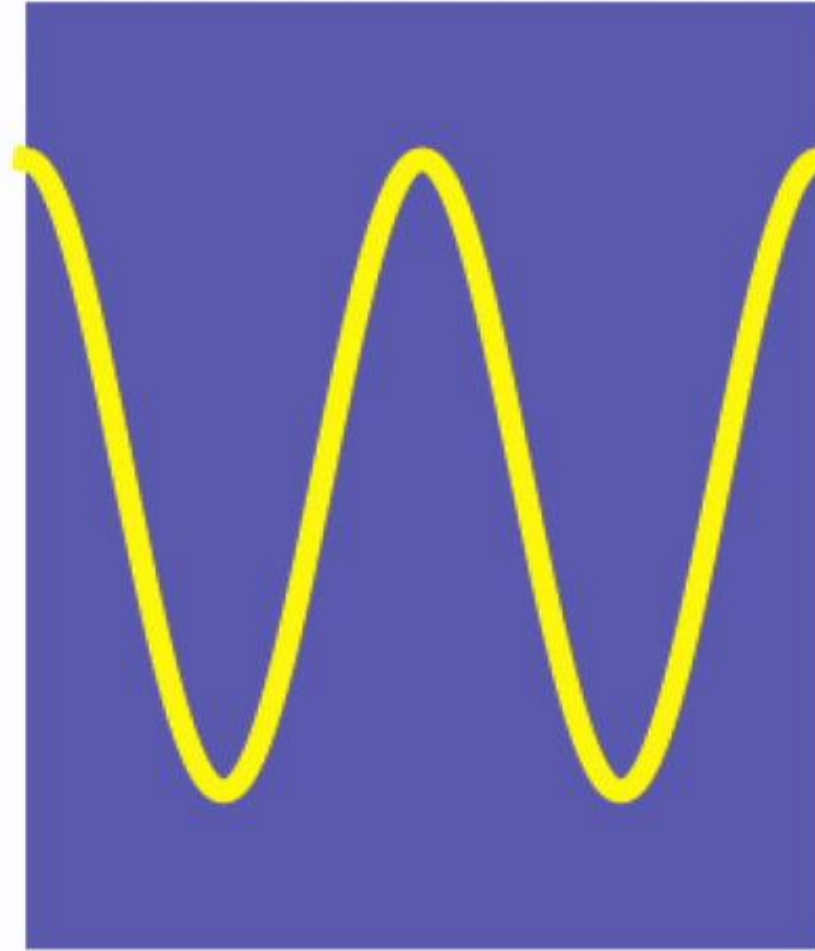


Expansion of the Universe

← More Energy ←
→ Less Energy →



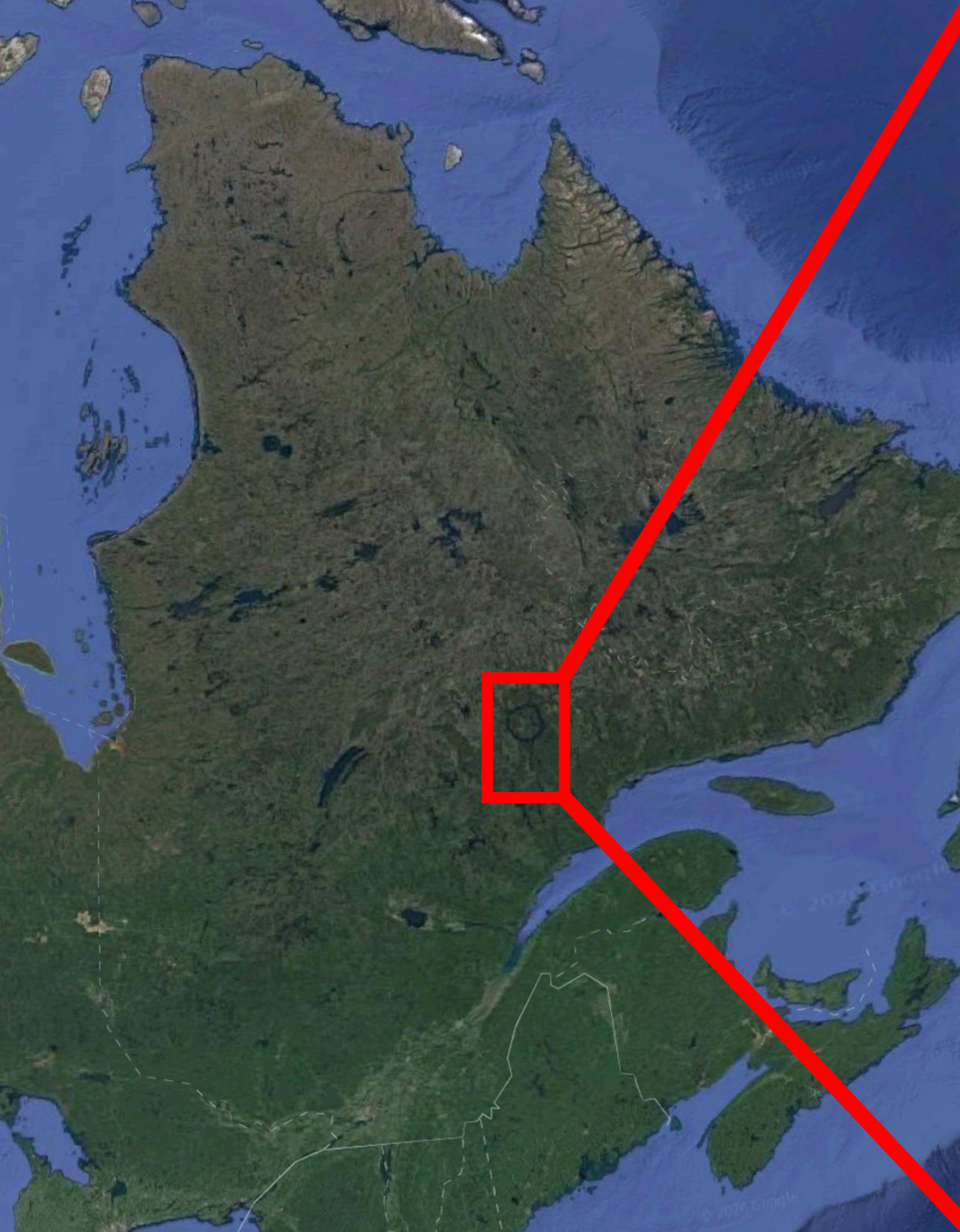
Photons
lose
energy!



McGill Radio Lab

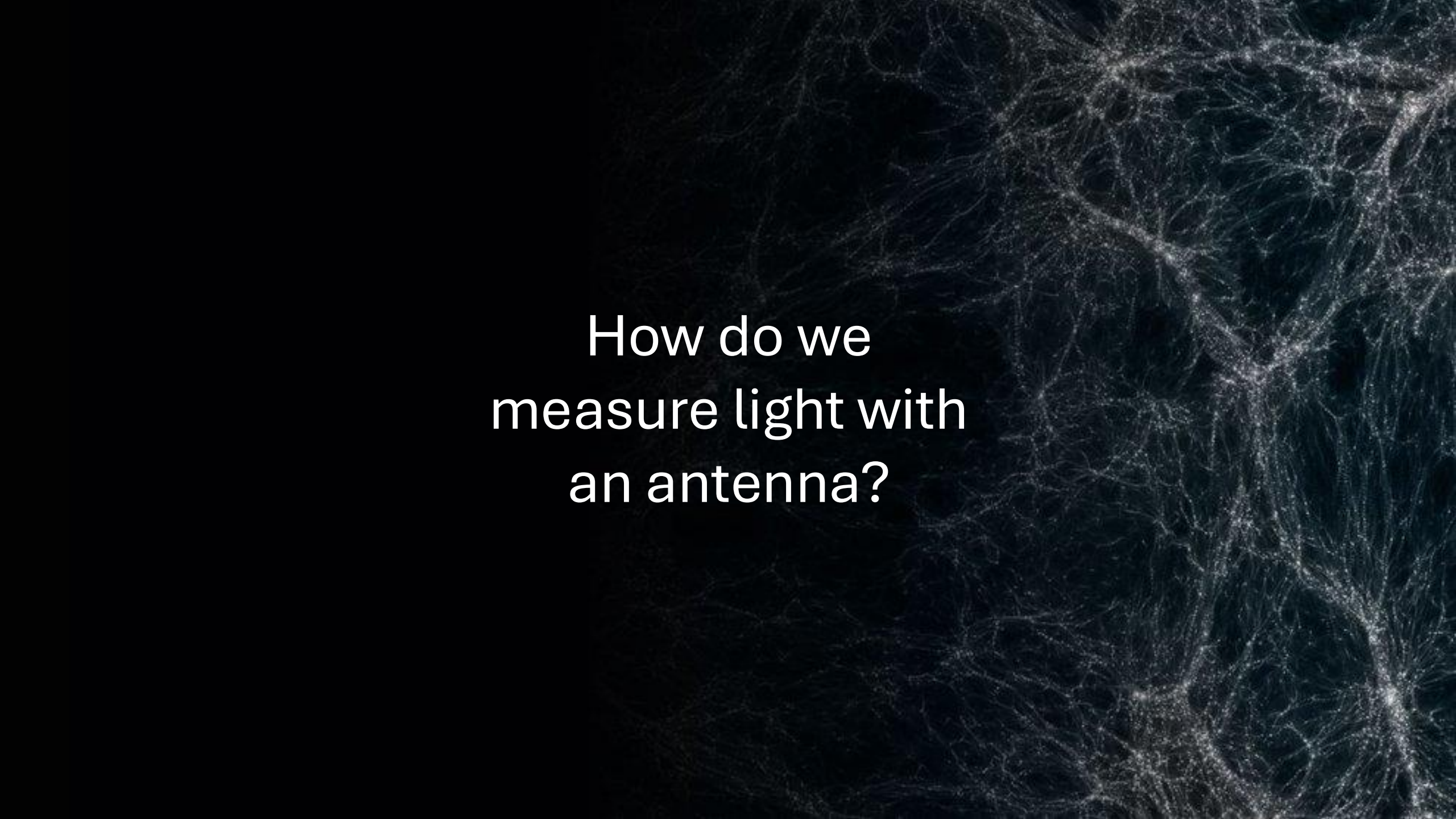
ALBATROS





Uapishka,
ALBATROS



The background of the slide is a deep black space filled with a complex, glowing network of white and light blue filaments. These filaments represent the cosmic web, showing the large-scale structure of the universe with its dense clusters and elongated filaments of galaxies and dark matter.

How do we
measure light with
an antenna?

Measuring Light

What is happening in
this image?

What might happen if they
touched someone else?



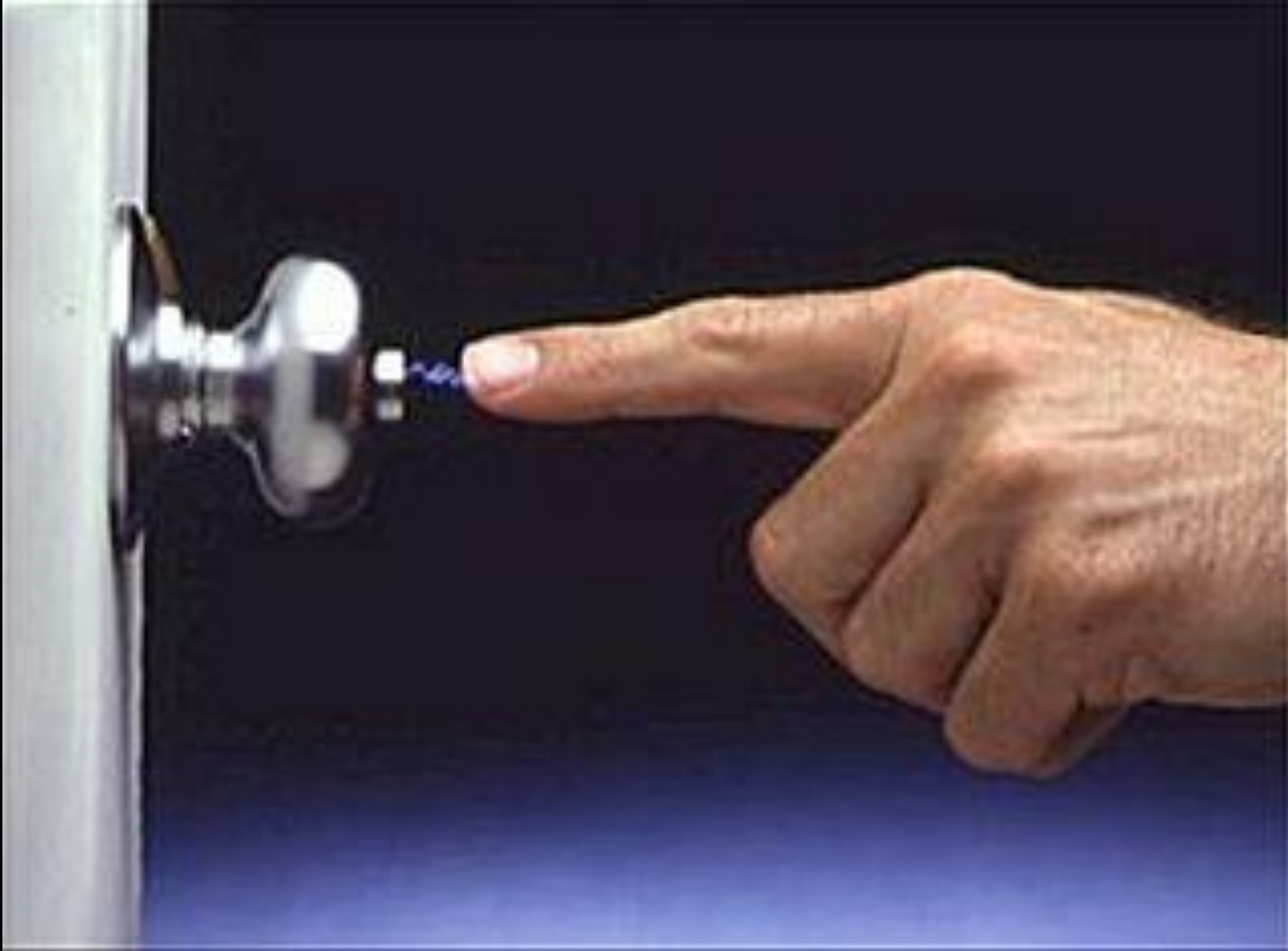
Measuring Light



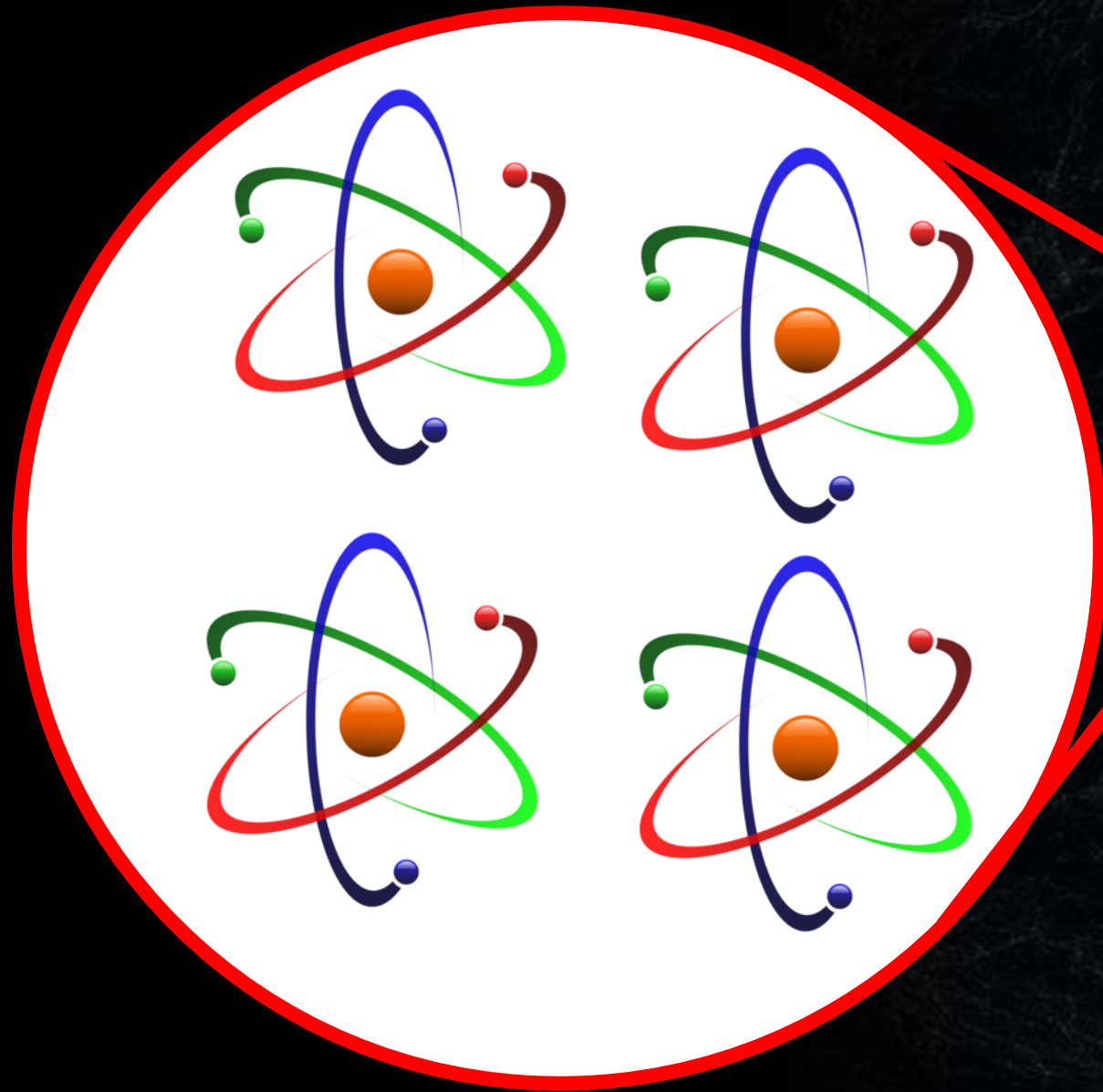
Measuring Light



Measuring Light

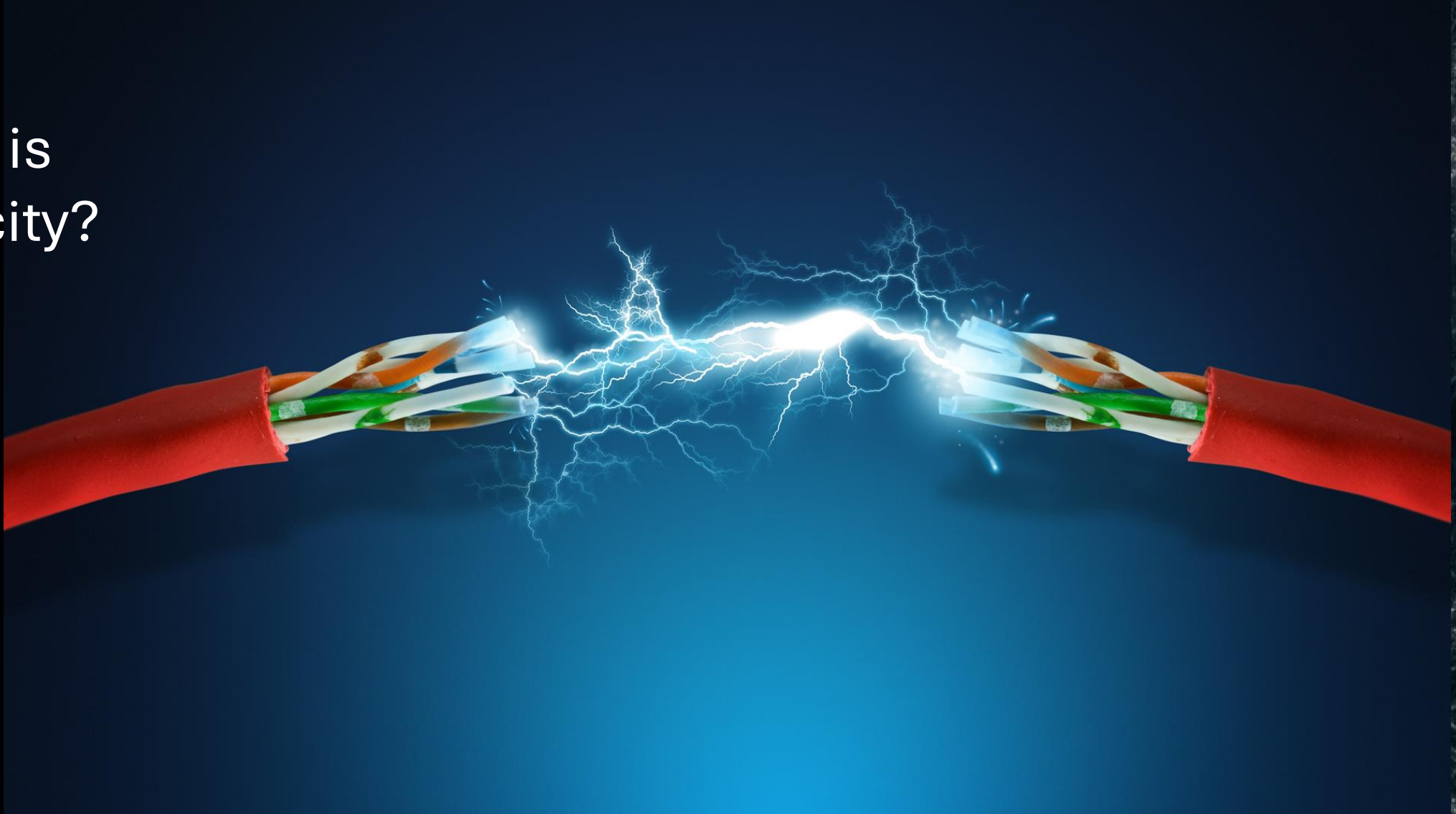


Measuring Light



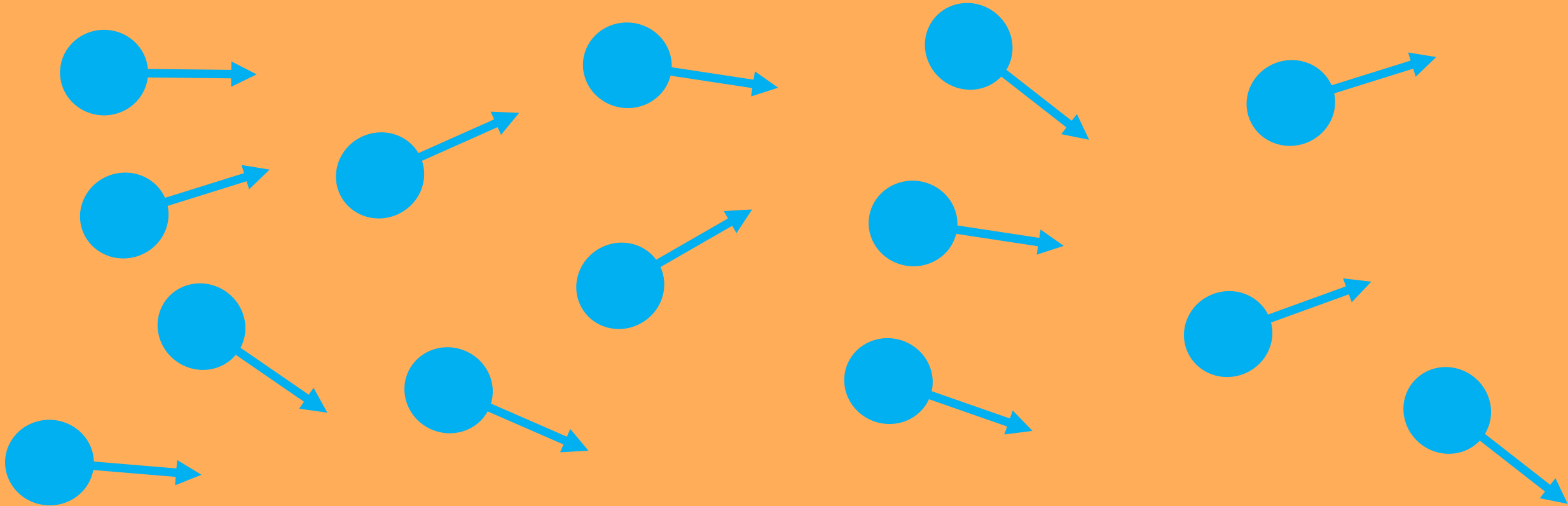
Measuring Light

What is
electricity?

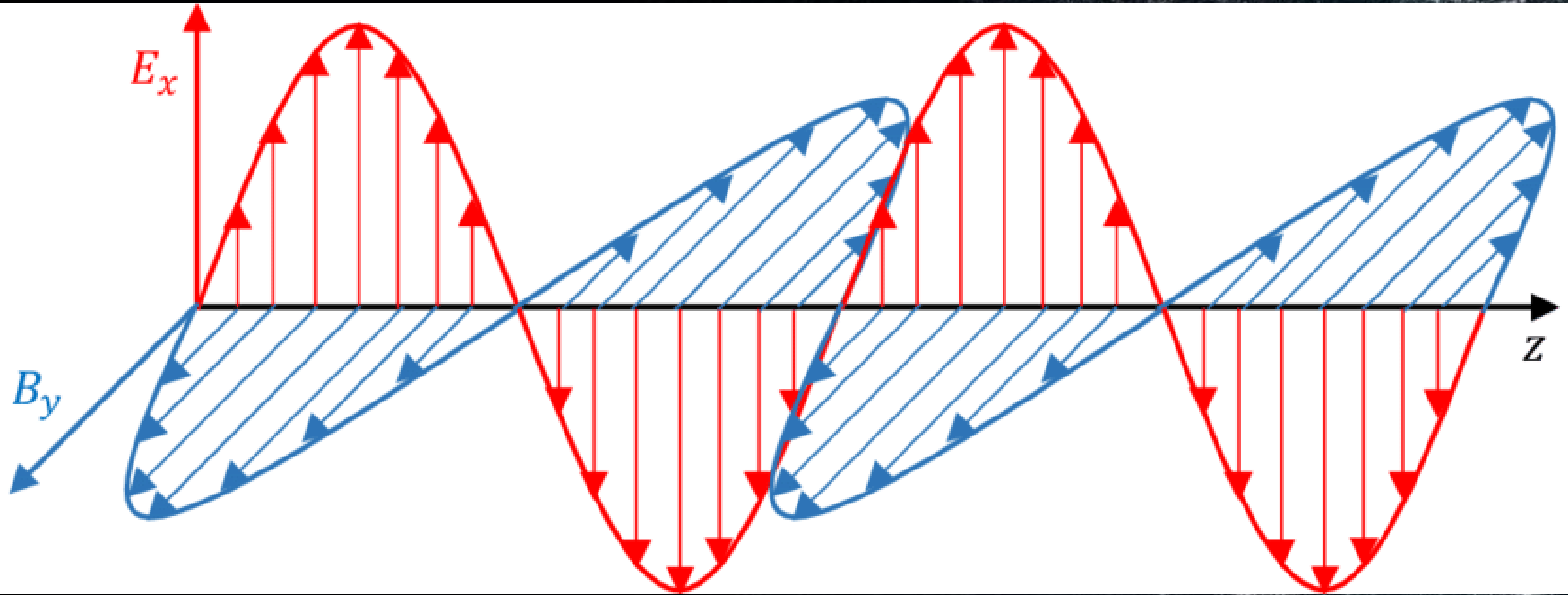


Measuring Light

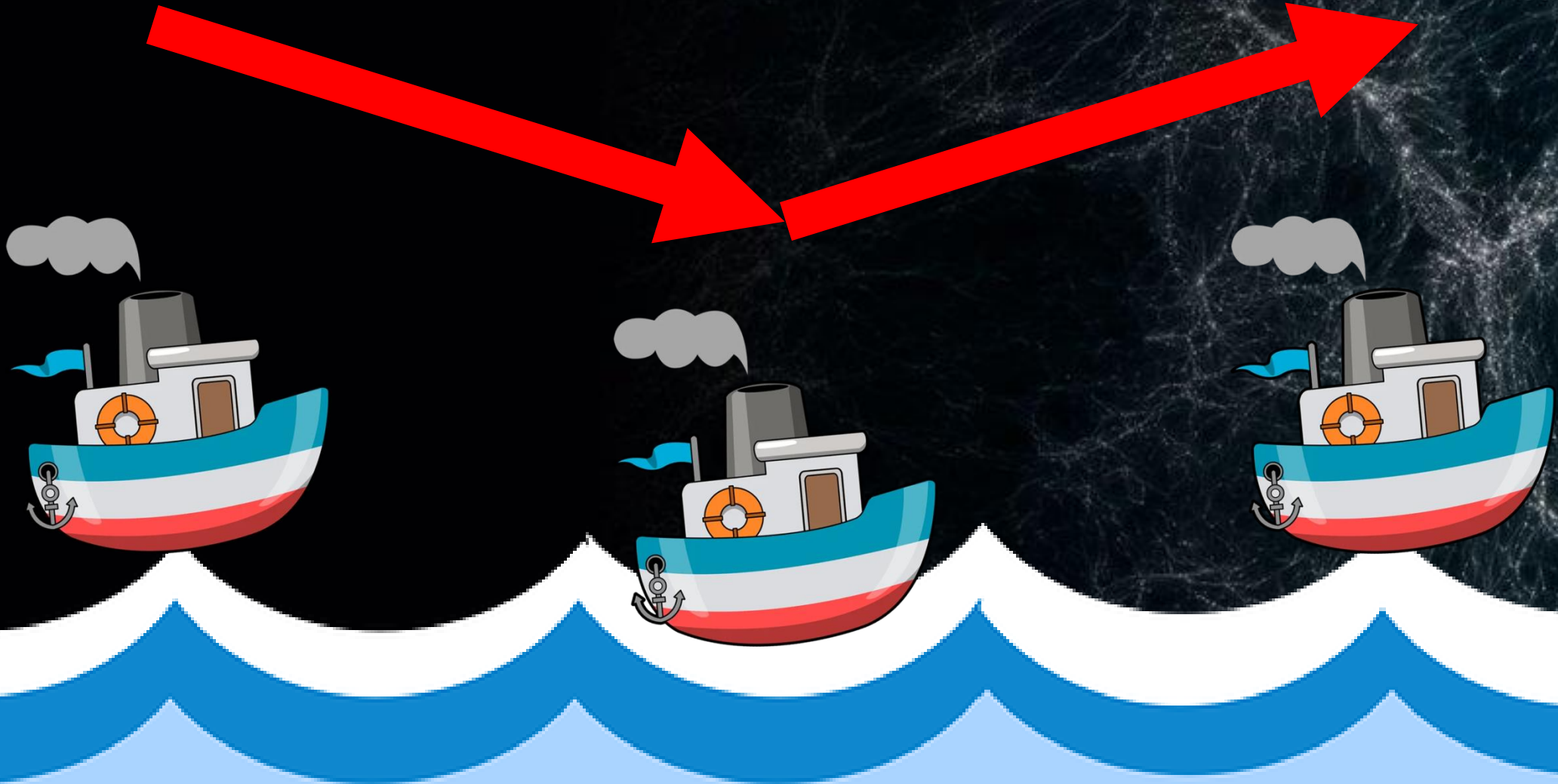
Charged Particles



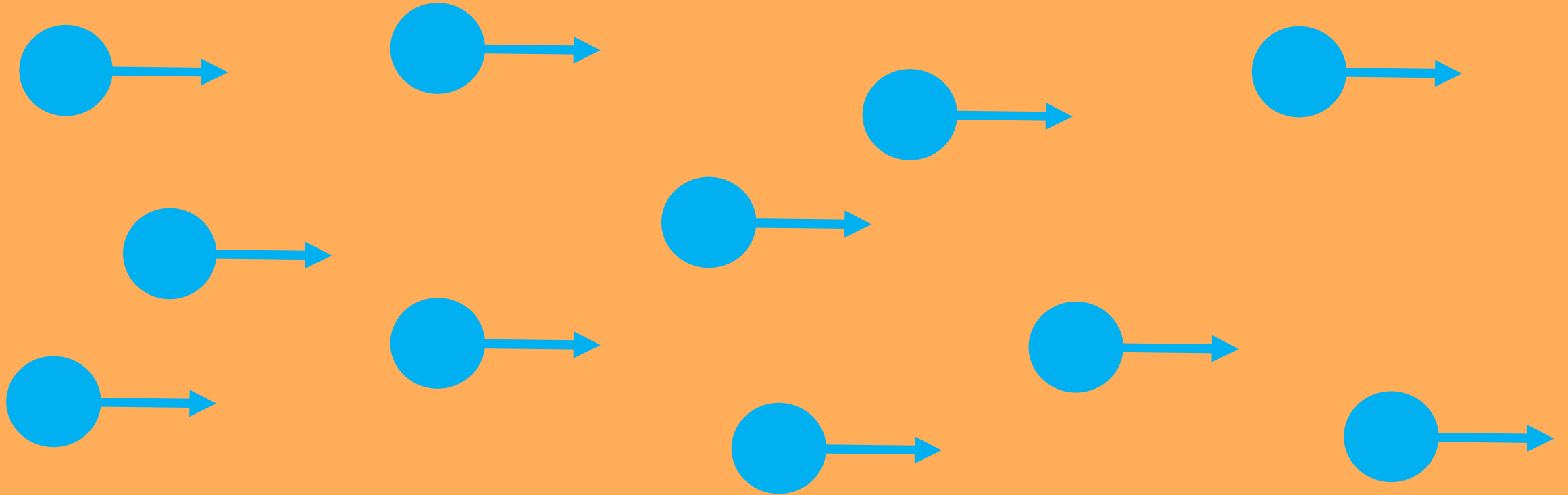
Measuring Light



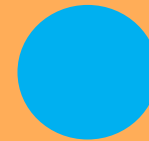
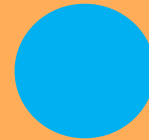
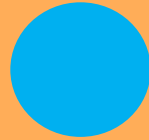
Measuring Light



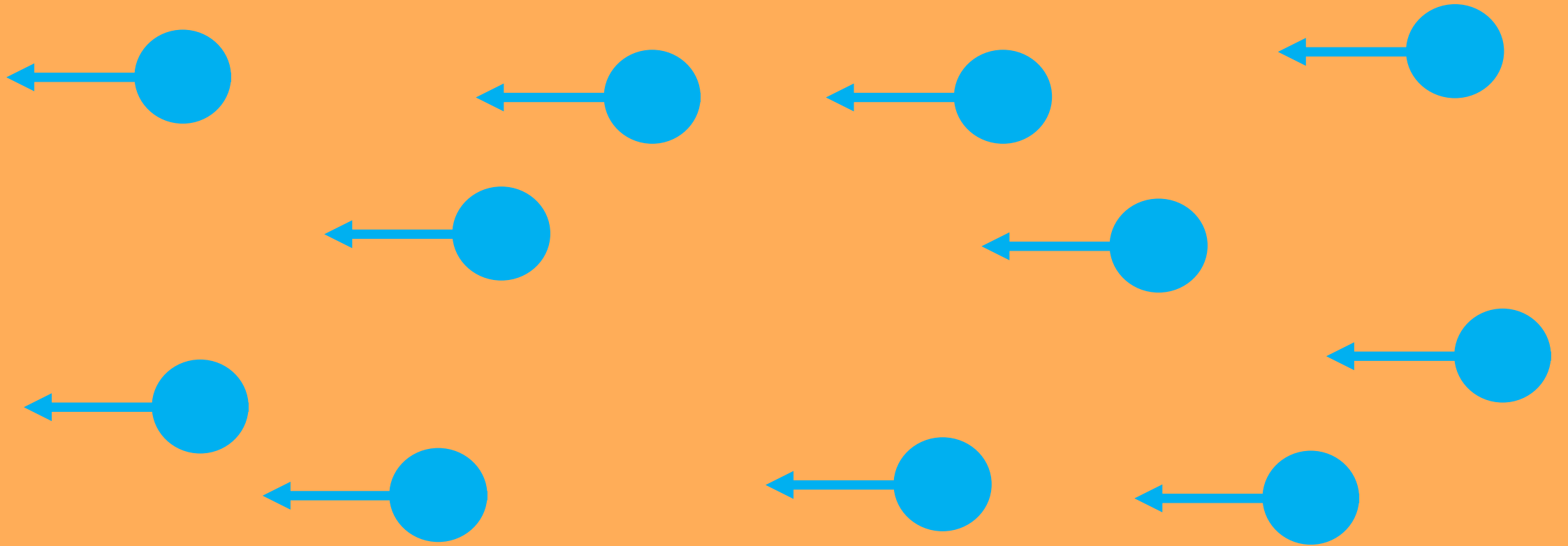
Measuring Light



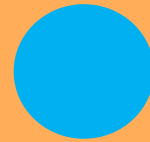
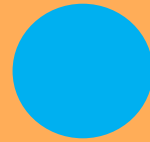
Measuring Light



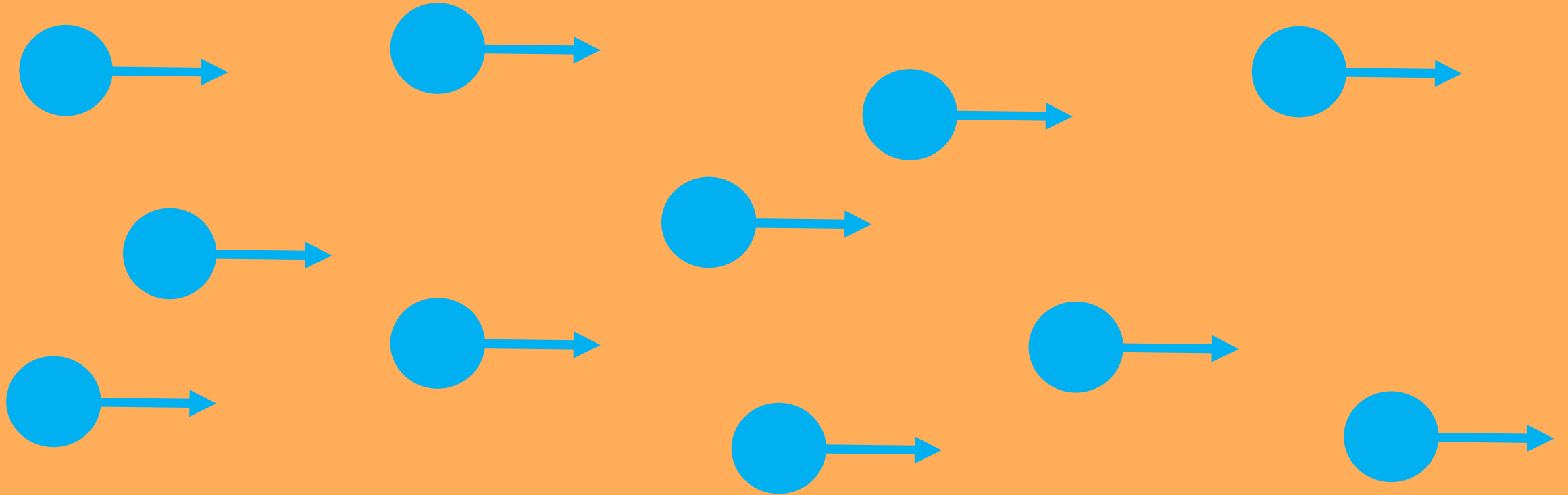
Measuring Light



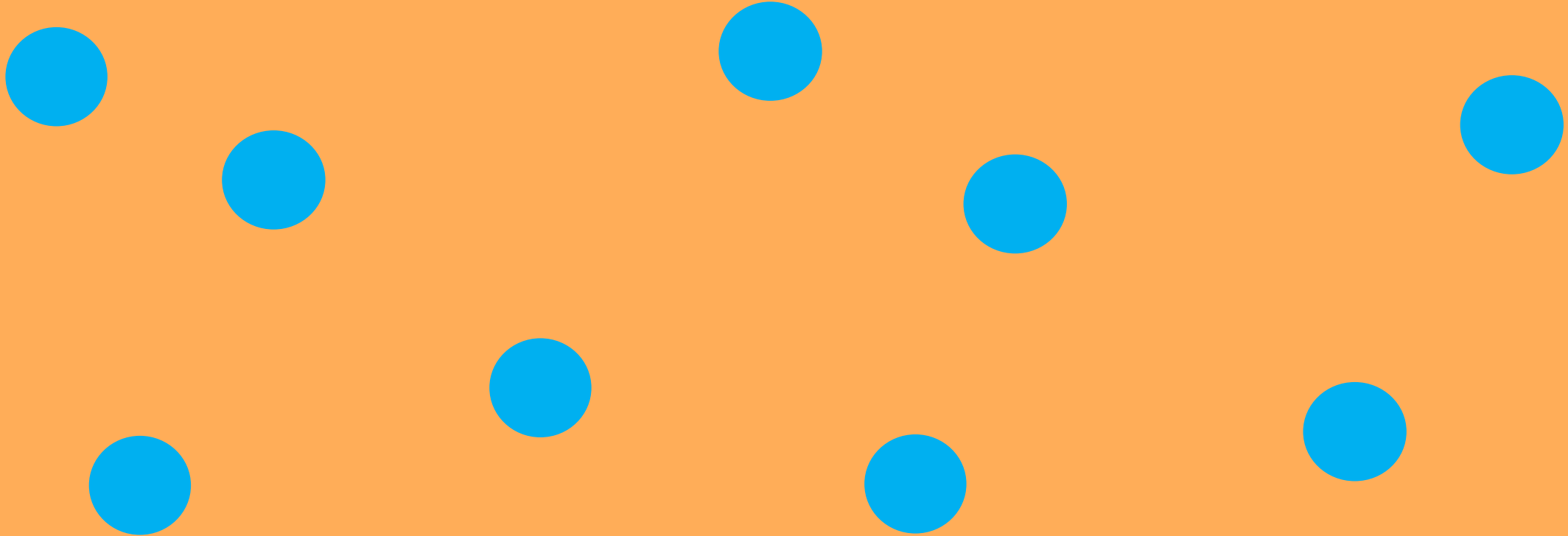
Measuring Light



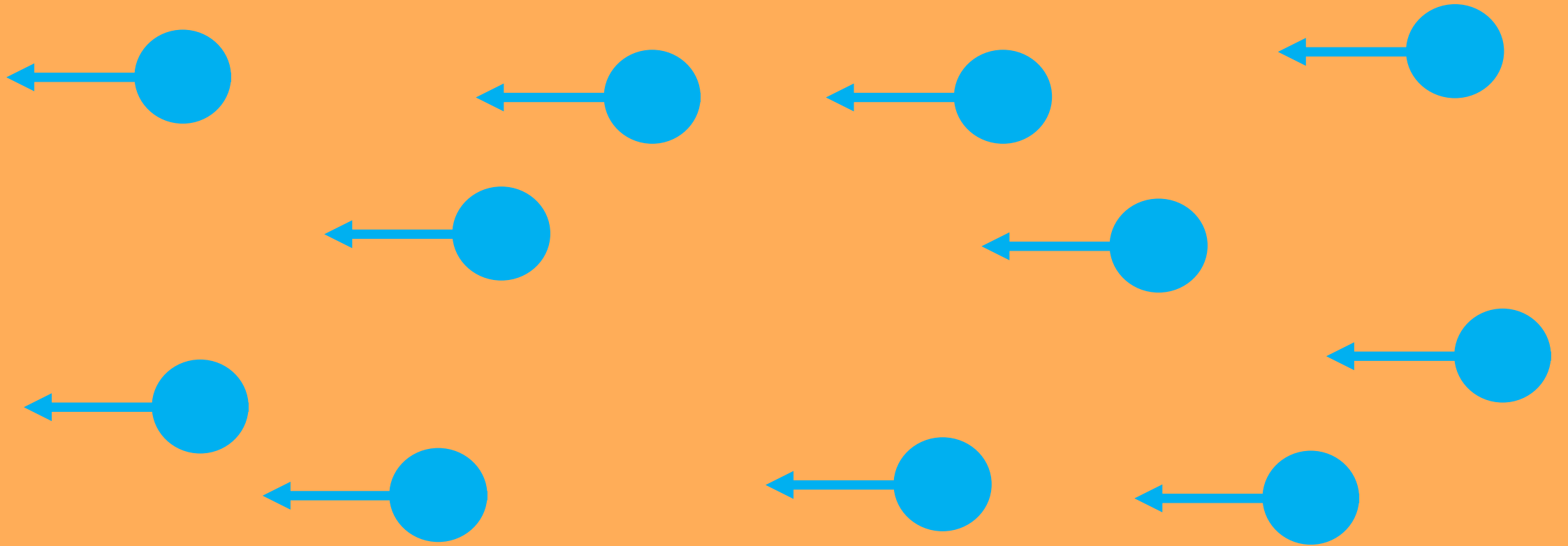
Measuring Light



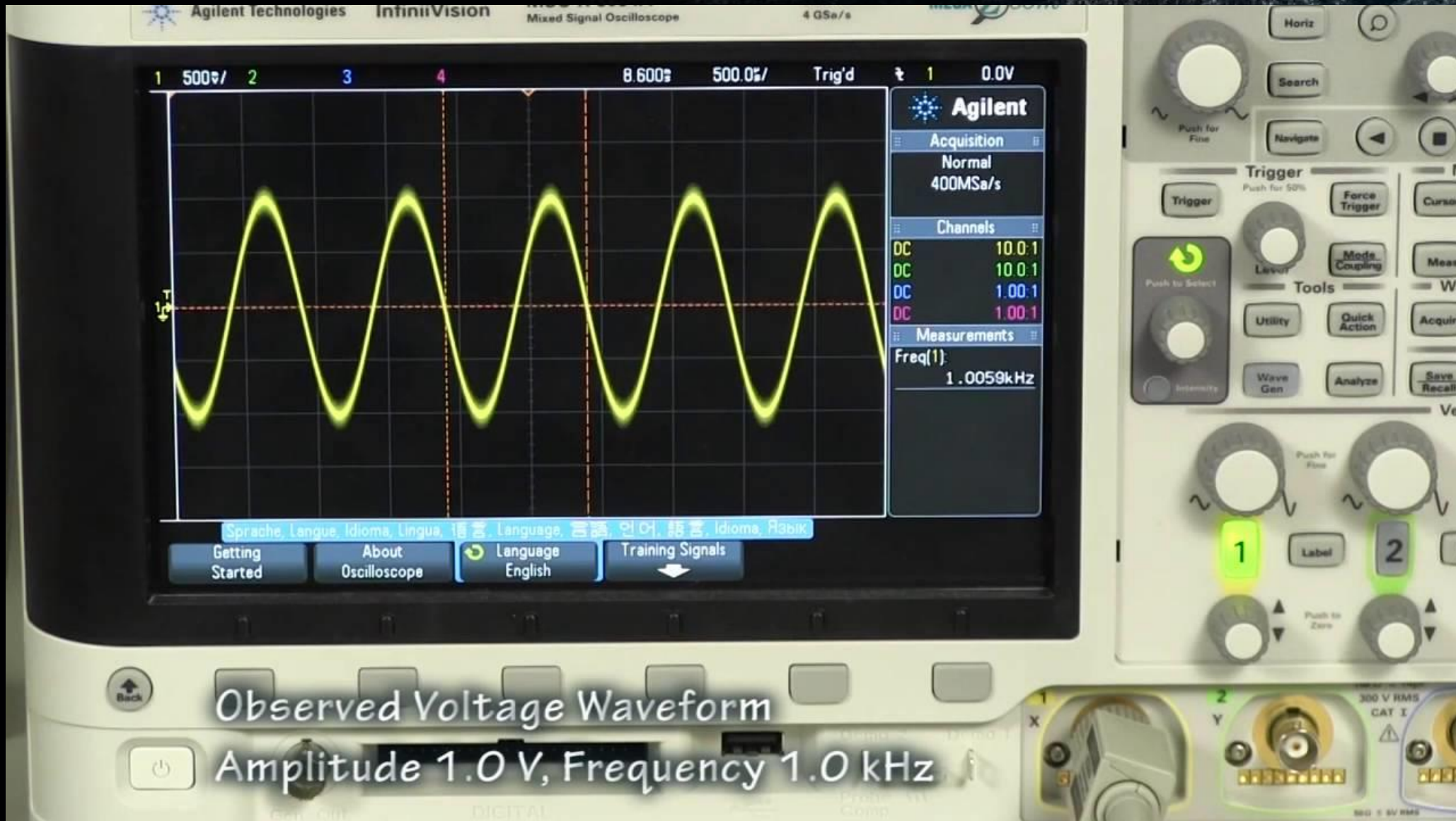
Measuring Light



Measuring Light



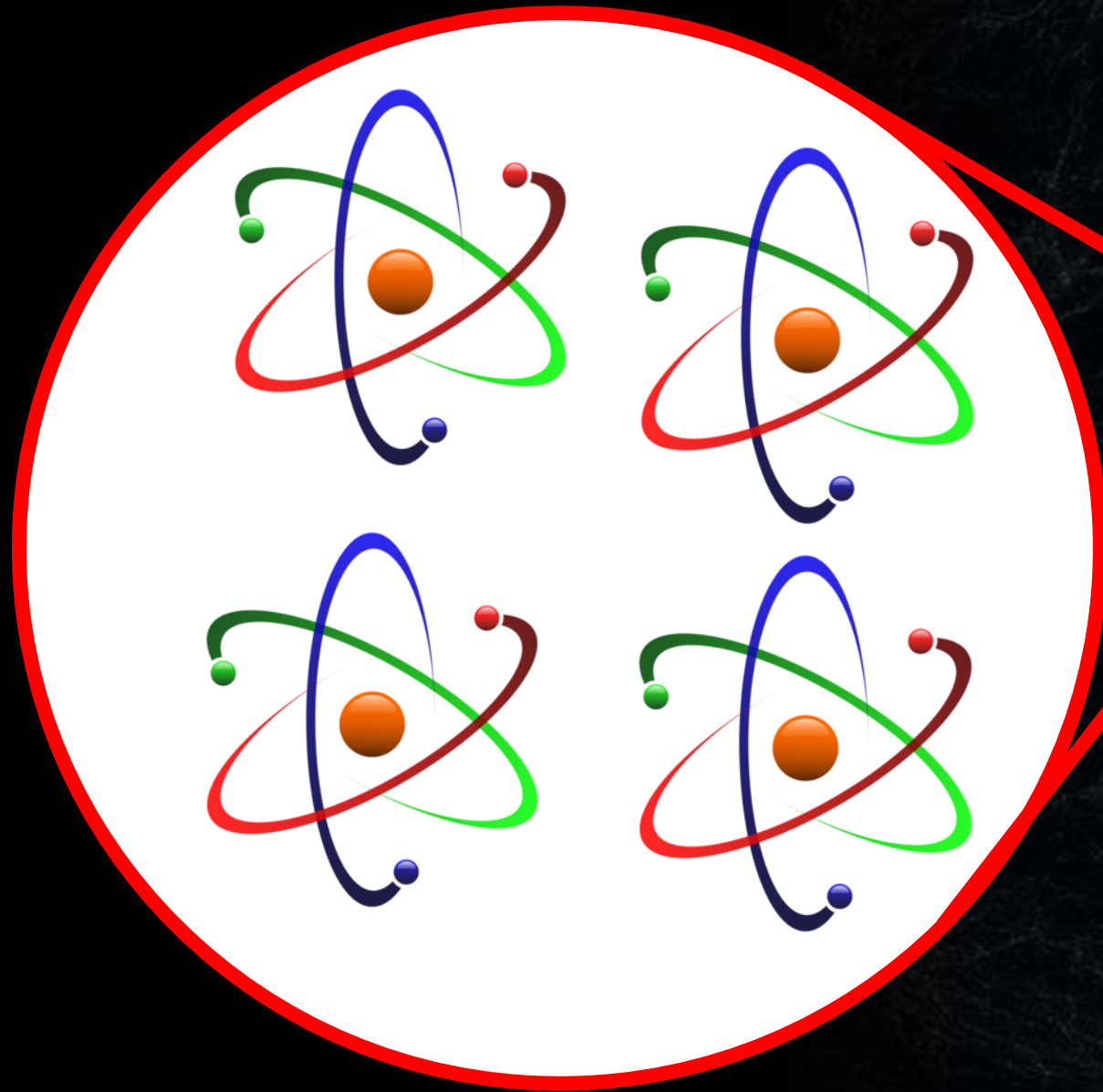
Measuring Light



Everything is an Antenna!



Measuring Light

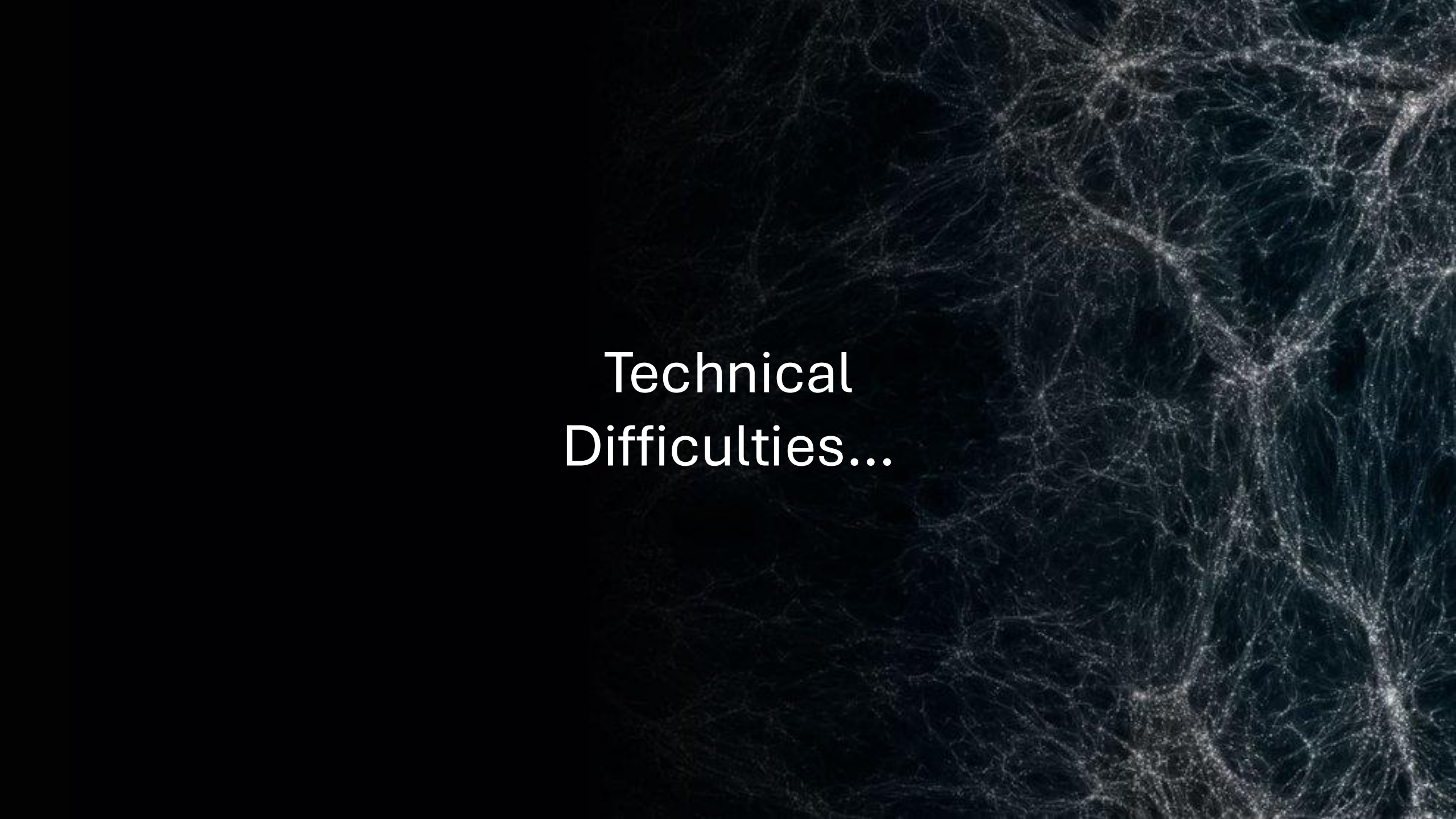


Activity



Everything is an Antenna!



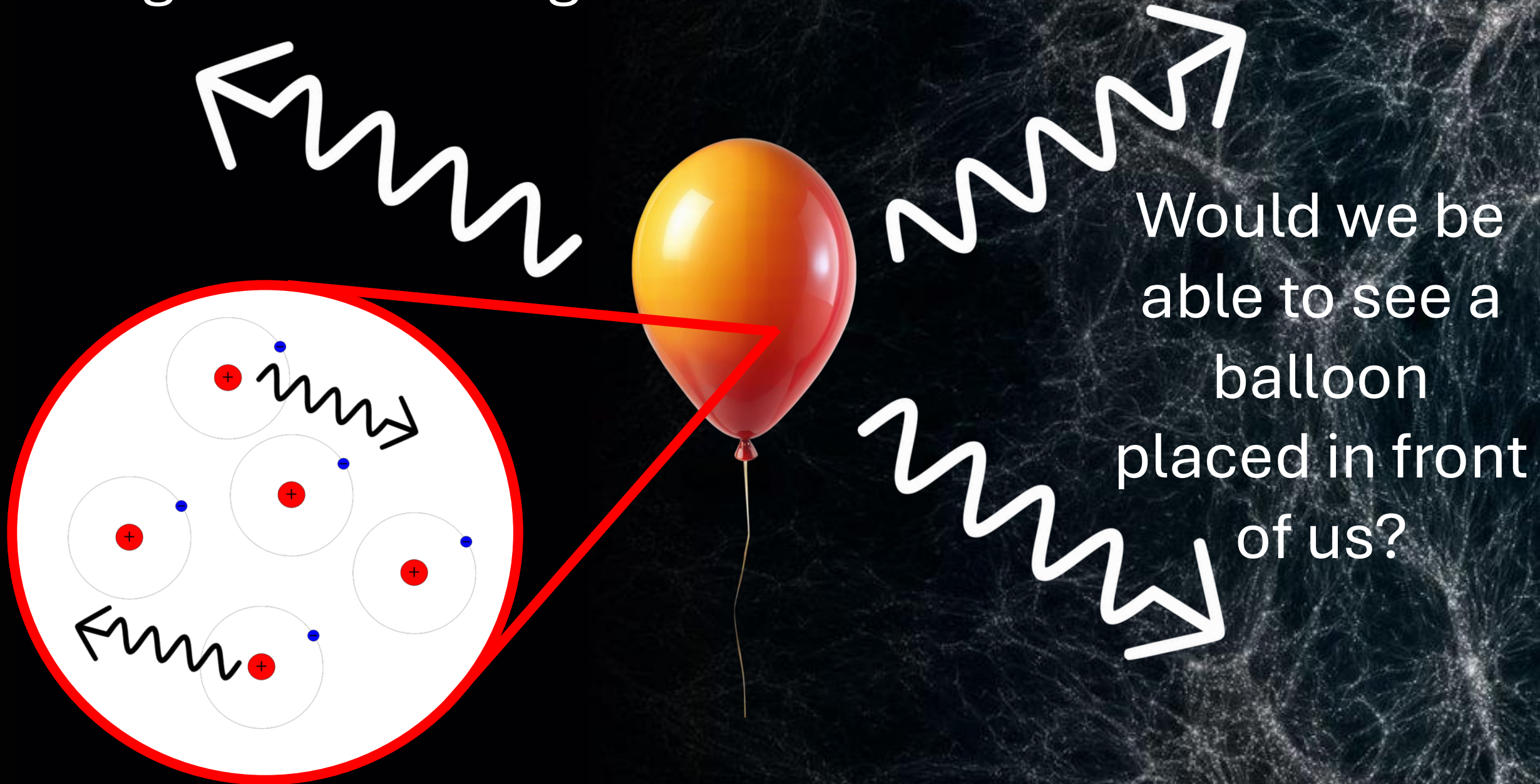


Technical
Difficulties...

Strength of 21cm Signal



Strength of 21cm Signal



Strength of 21cm Signal

~200km!



Can we see the signal in Montreal?

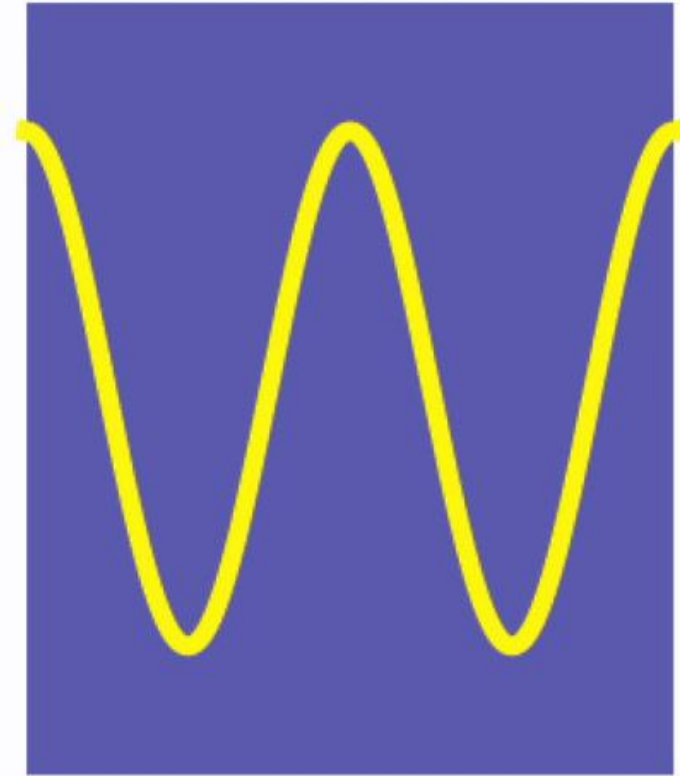


Can we see the signal in Montreal?

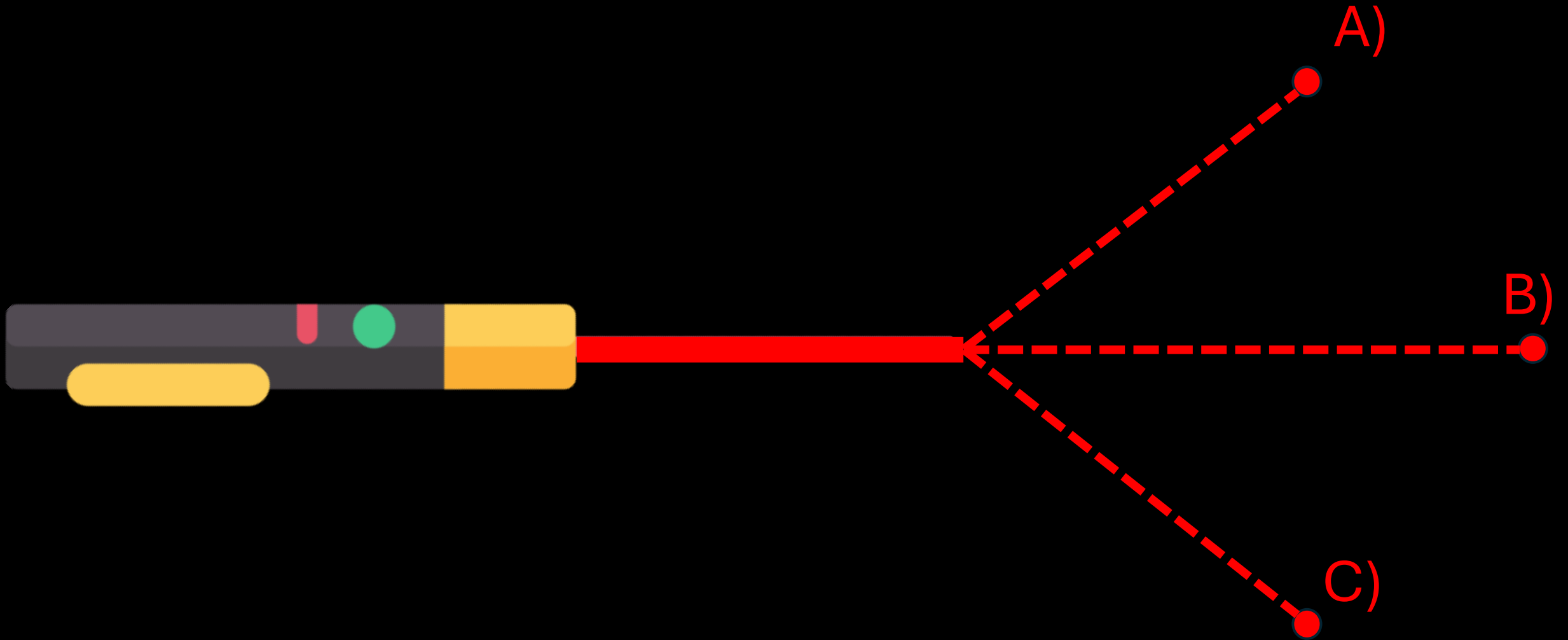
21 cm = 1420 MHz

Expansion of the universe
has stretched signal to
10-50 MHz

Terrestrial
communication:
1-100 MHz



Interactions with the Ionosphere



Interactions with the Ionosphere

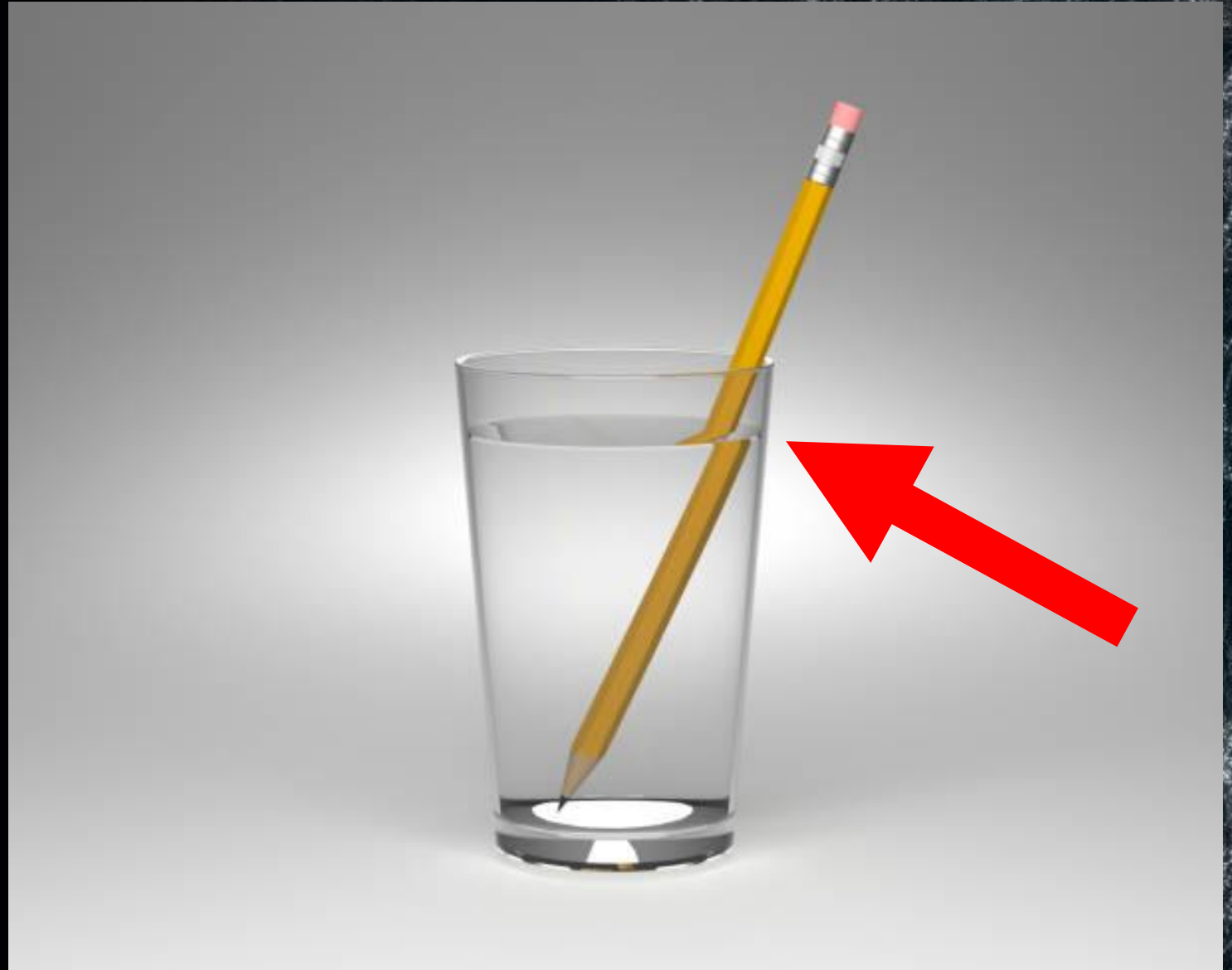
What do you
see?



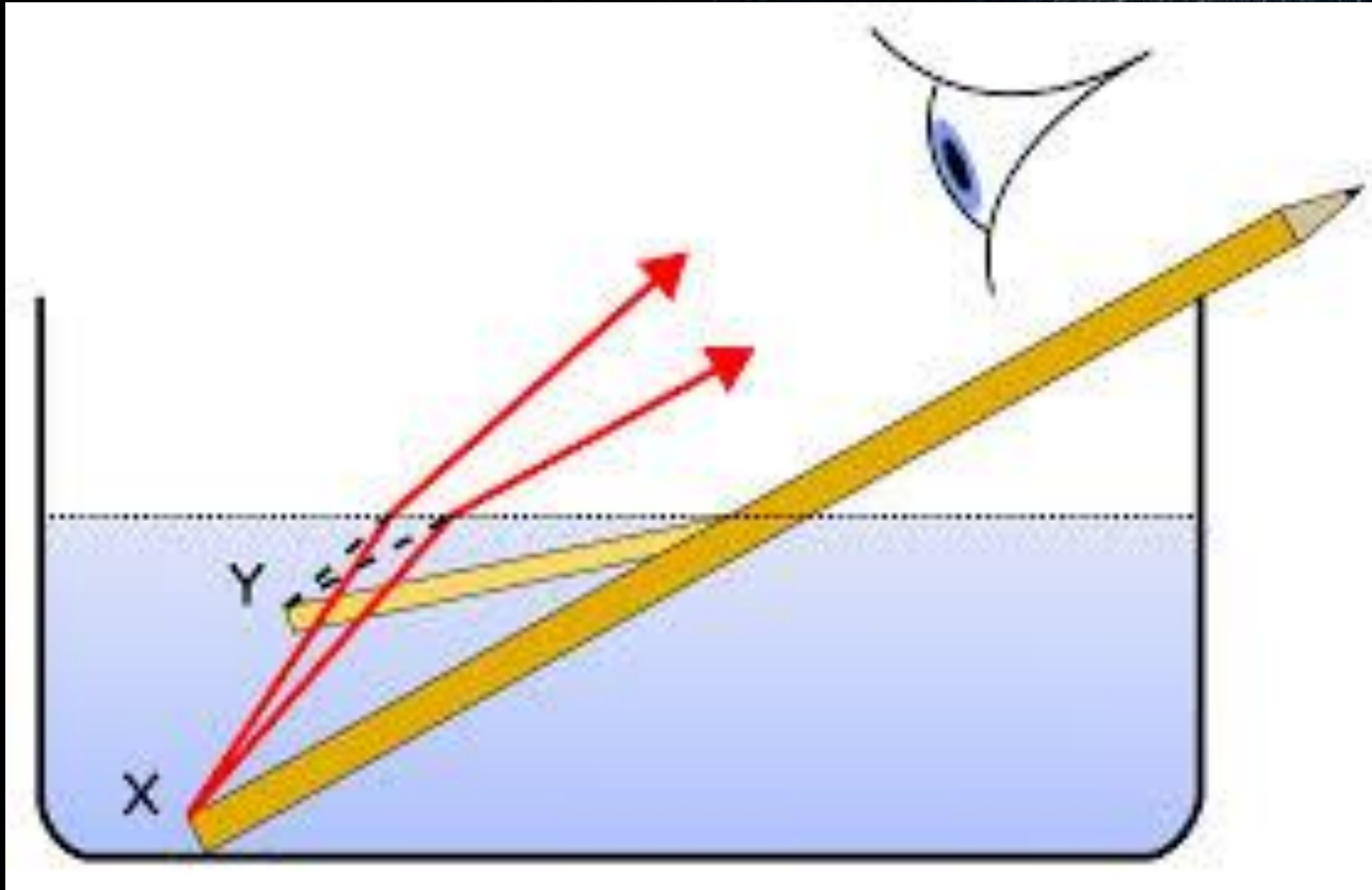
Interactions with the Ionosphere

Why does this happen?

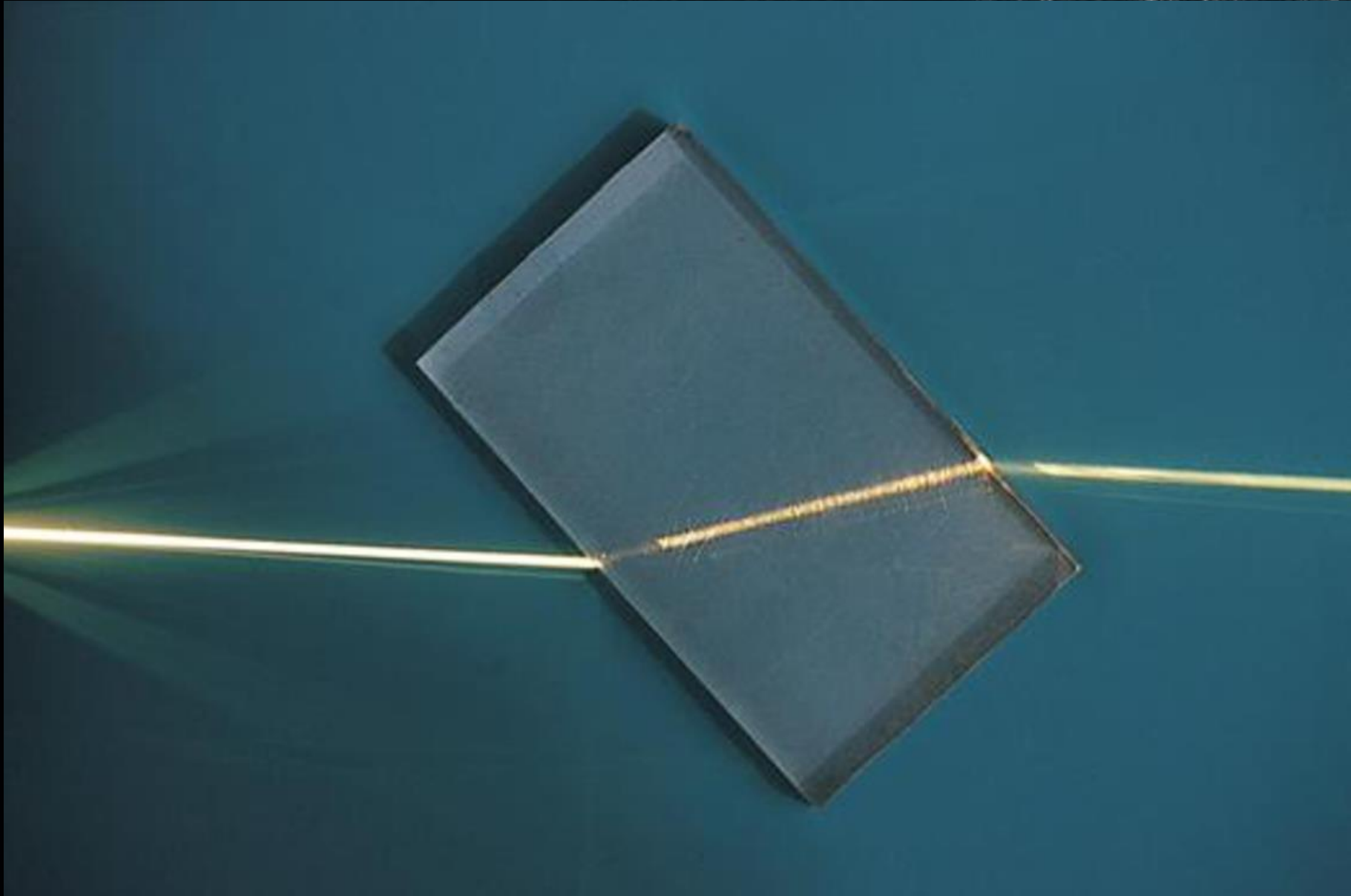
What path does the light take?



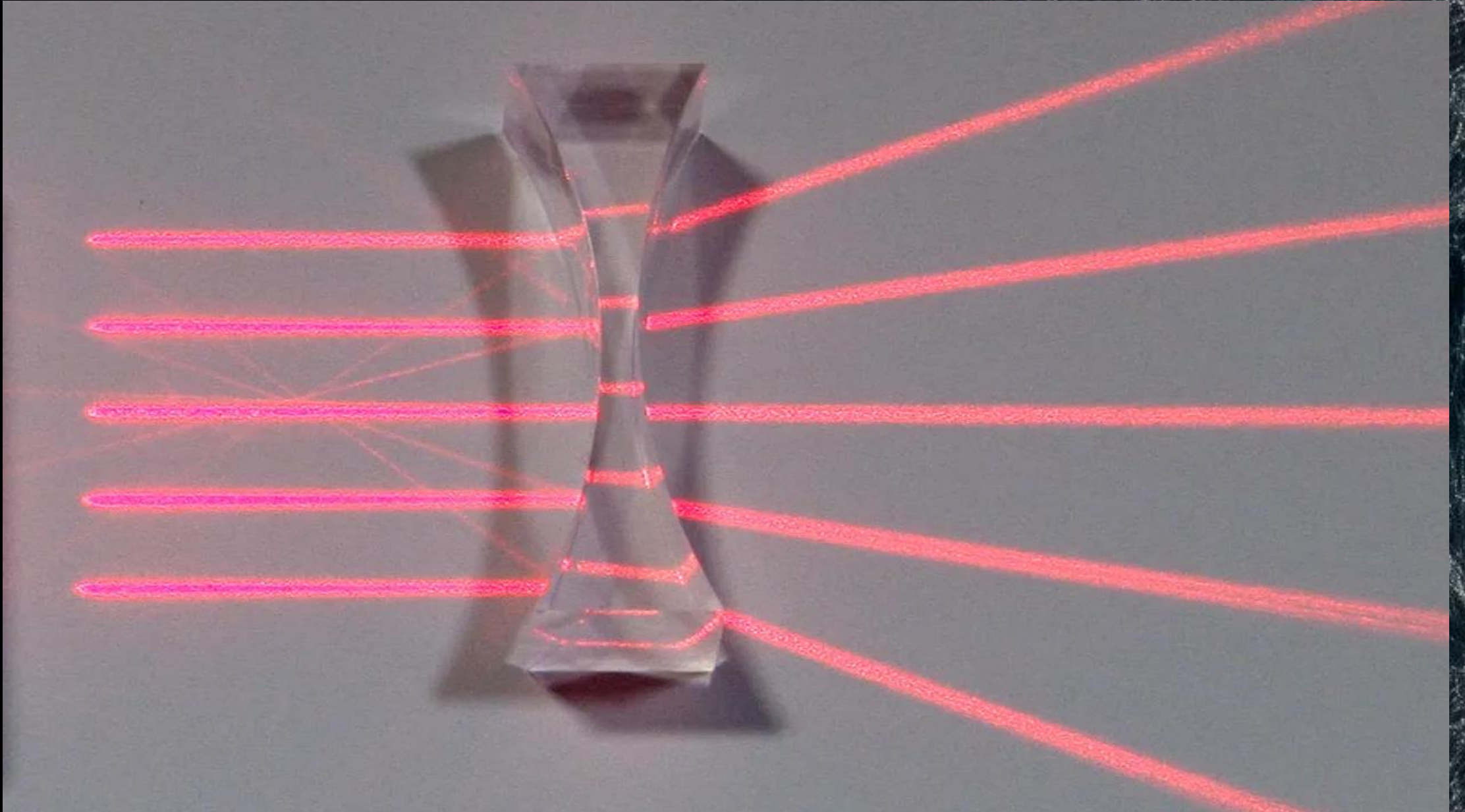
Interactions with the Ionosphere



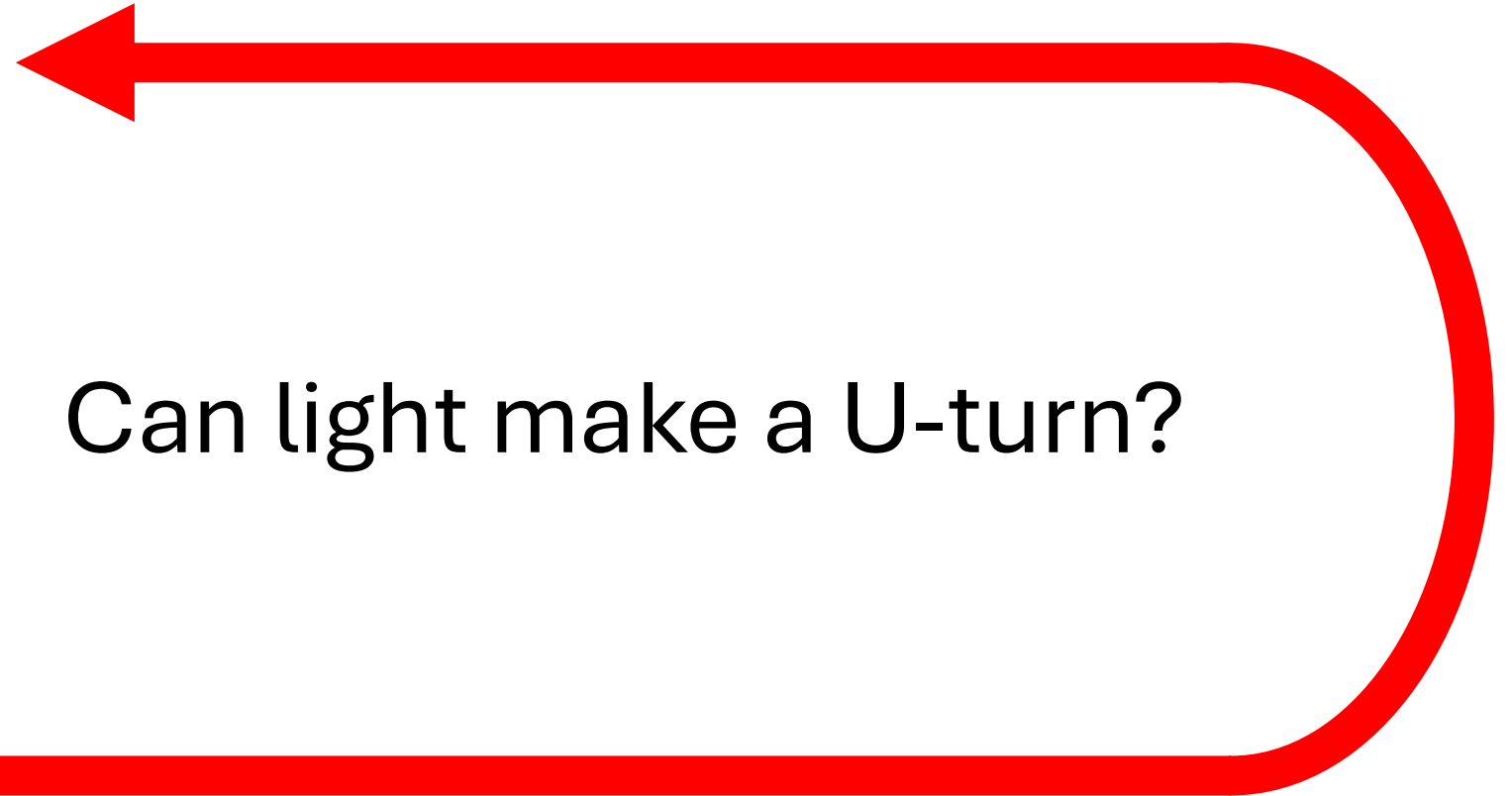
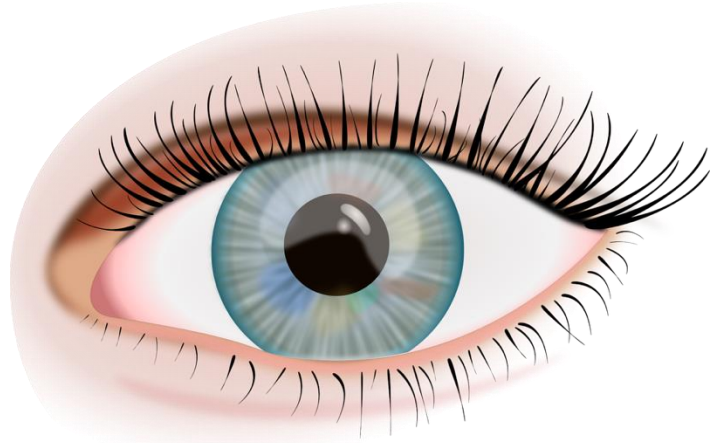
Interactions with the Ionosphere



Interactions with the Ionosphere



Interactions with the Ionosphere



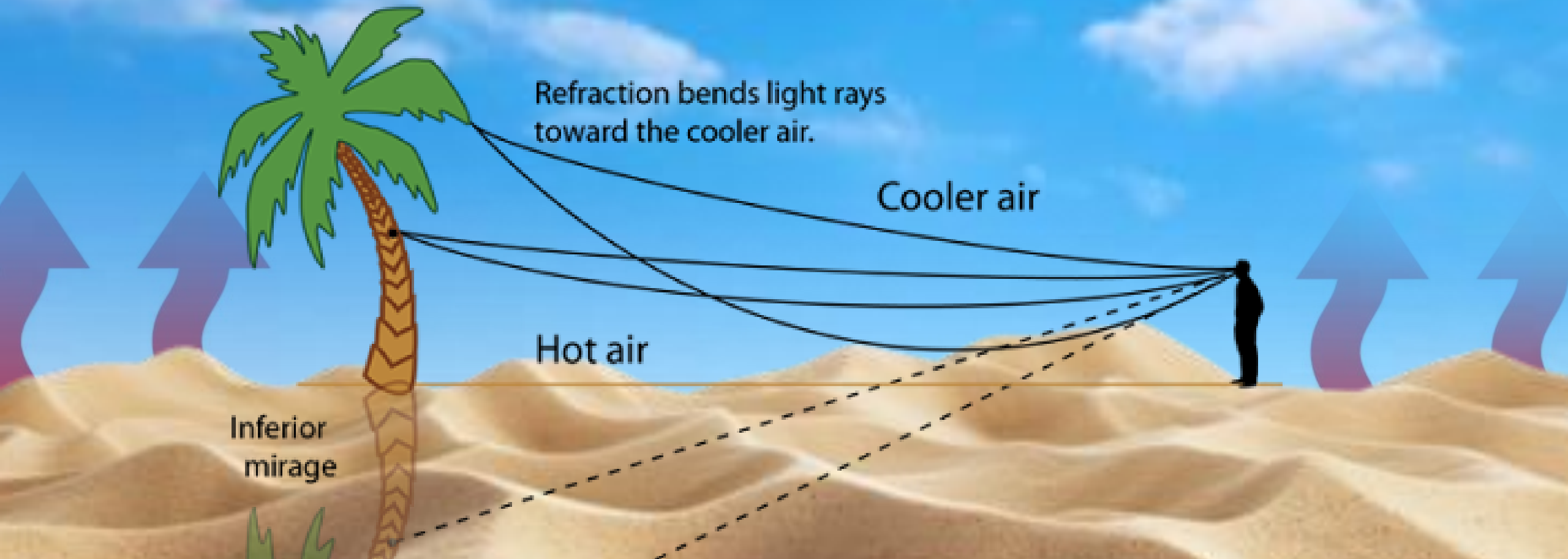
Can light make a U-turn?

Interactions with the Ionosphere

What do you
see?



Interactions with the Ionosphere



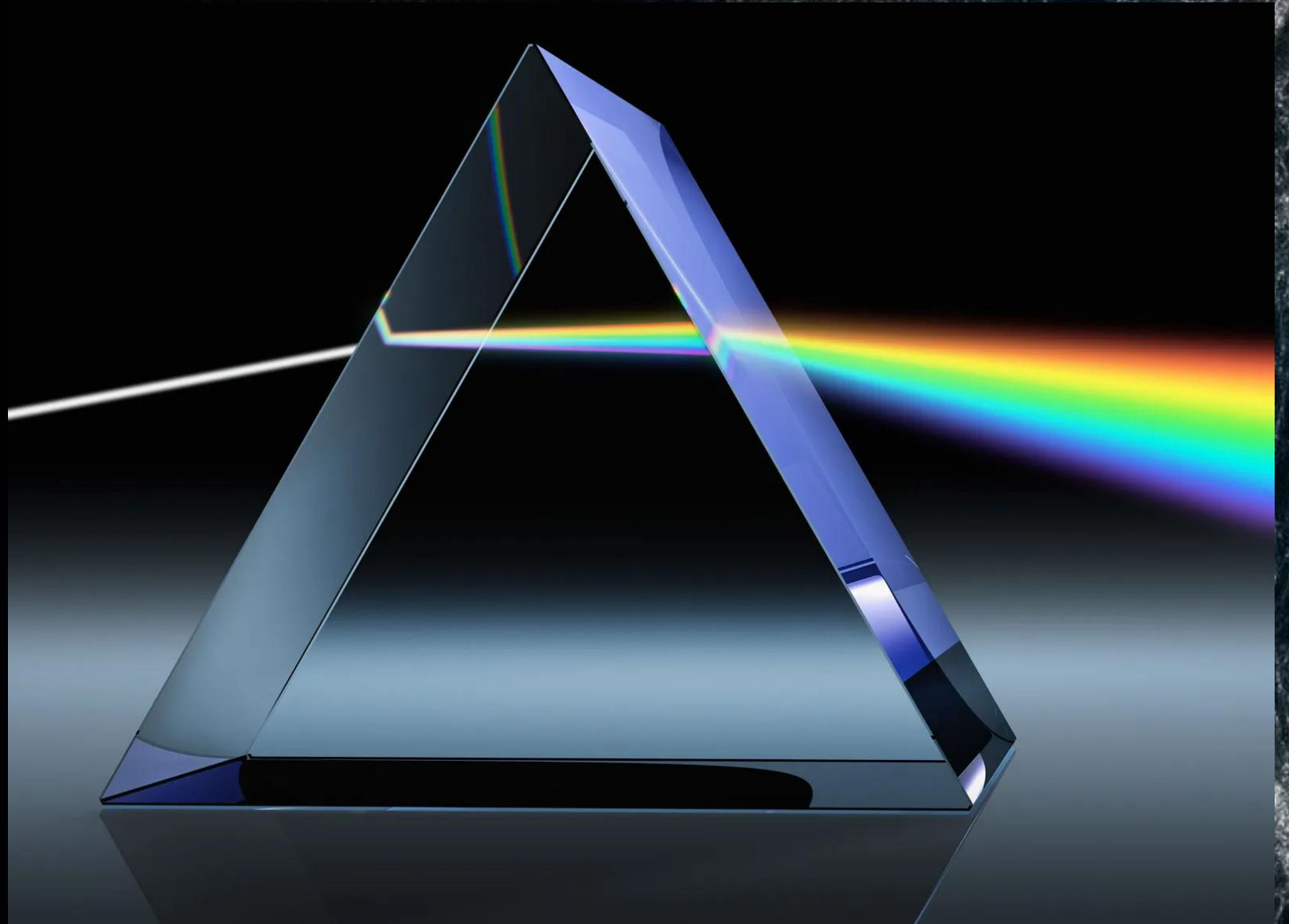
Interactions with the Ionosphere

Is this the
same at all
wavelengths?

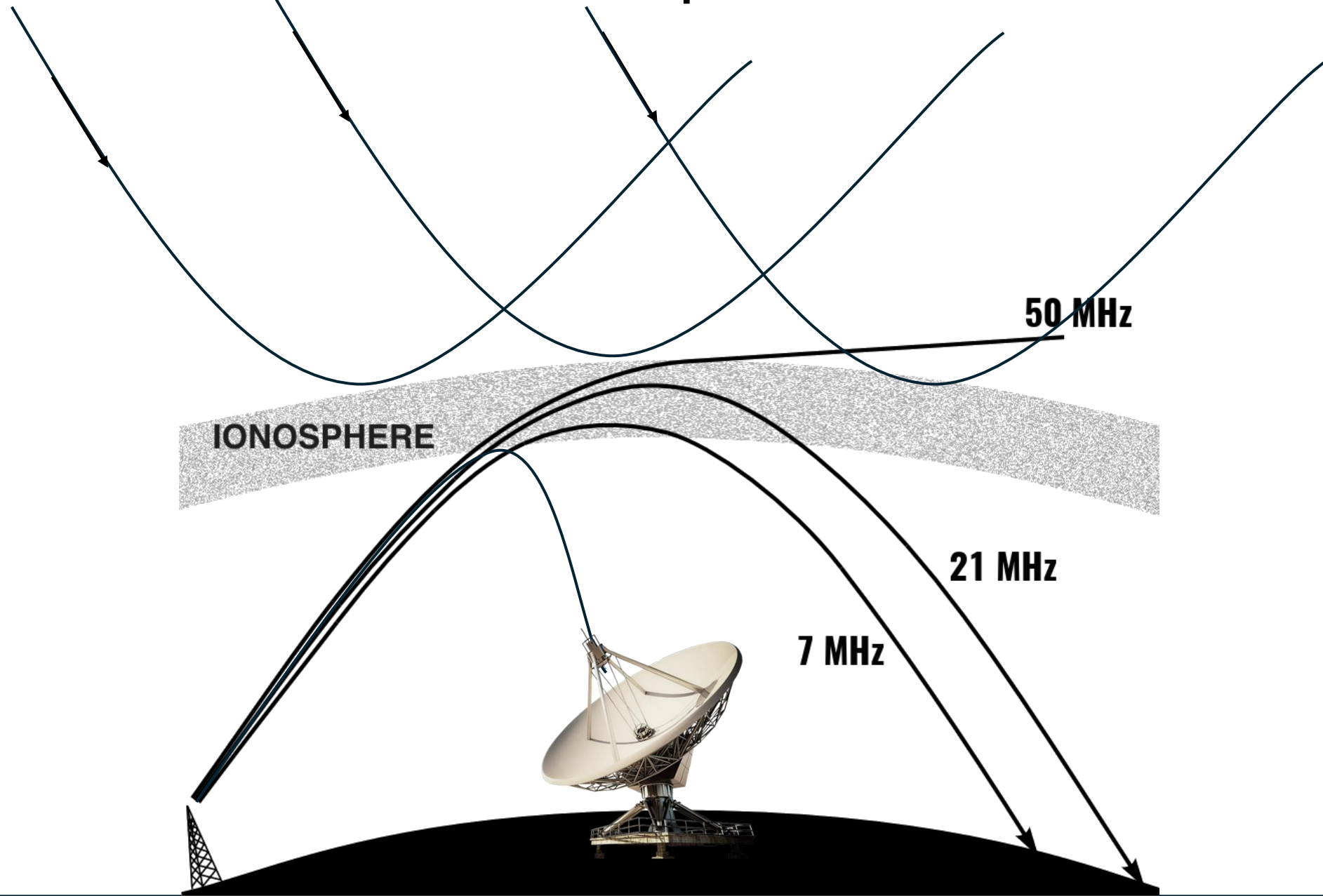


Interactions with the Ionosphere

Is this the
same at all
wavelengths?



Interactions with the Ionosphere



A buildup of problems!

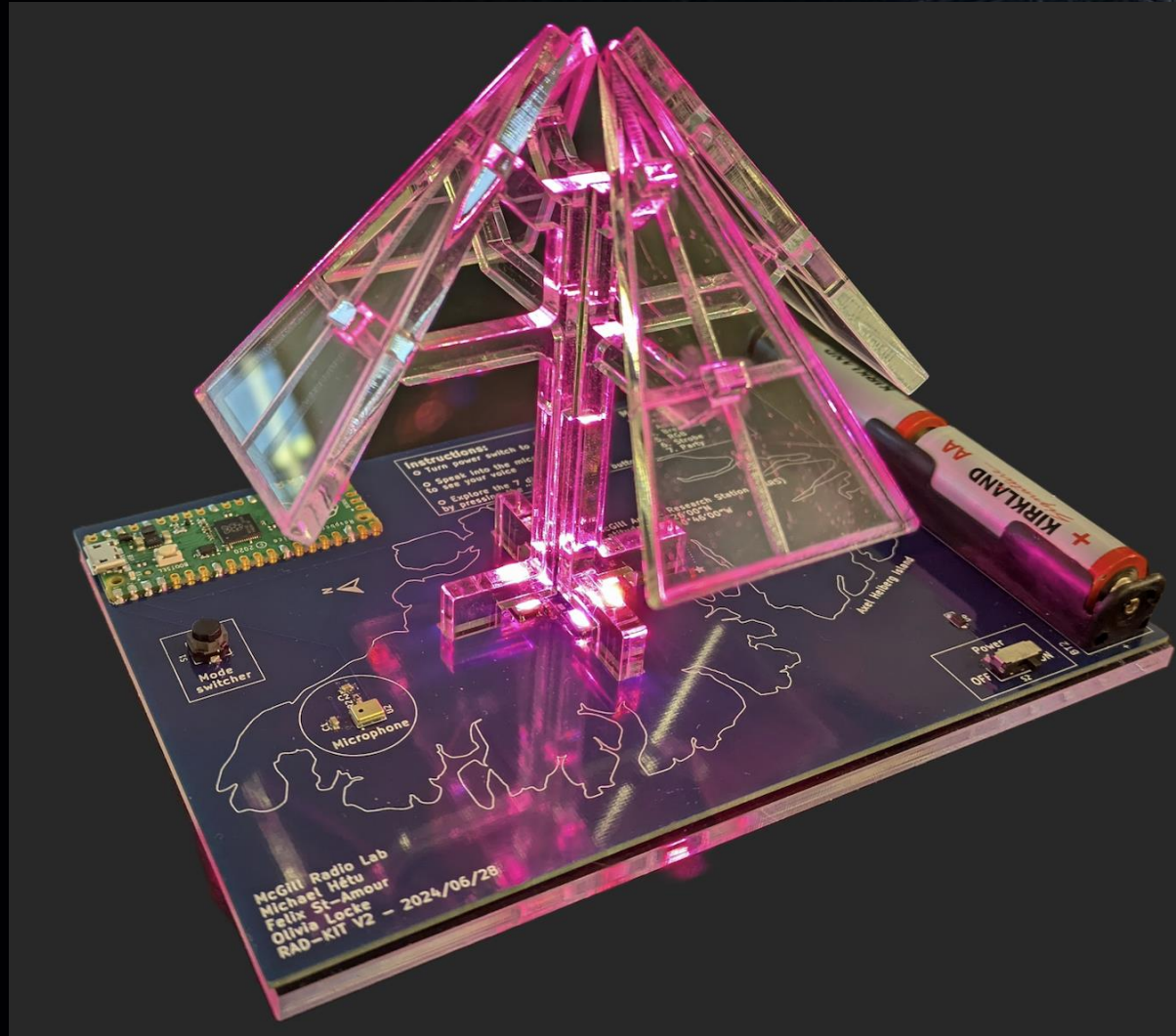


Lots of noise
coming from Earth

Weakening of
signal (from
ionosphere)

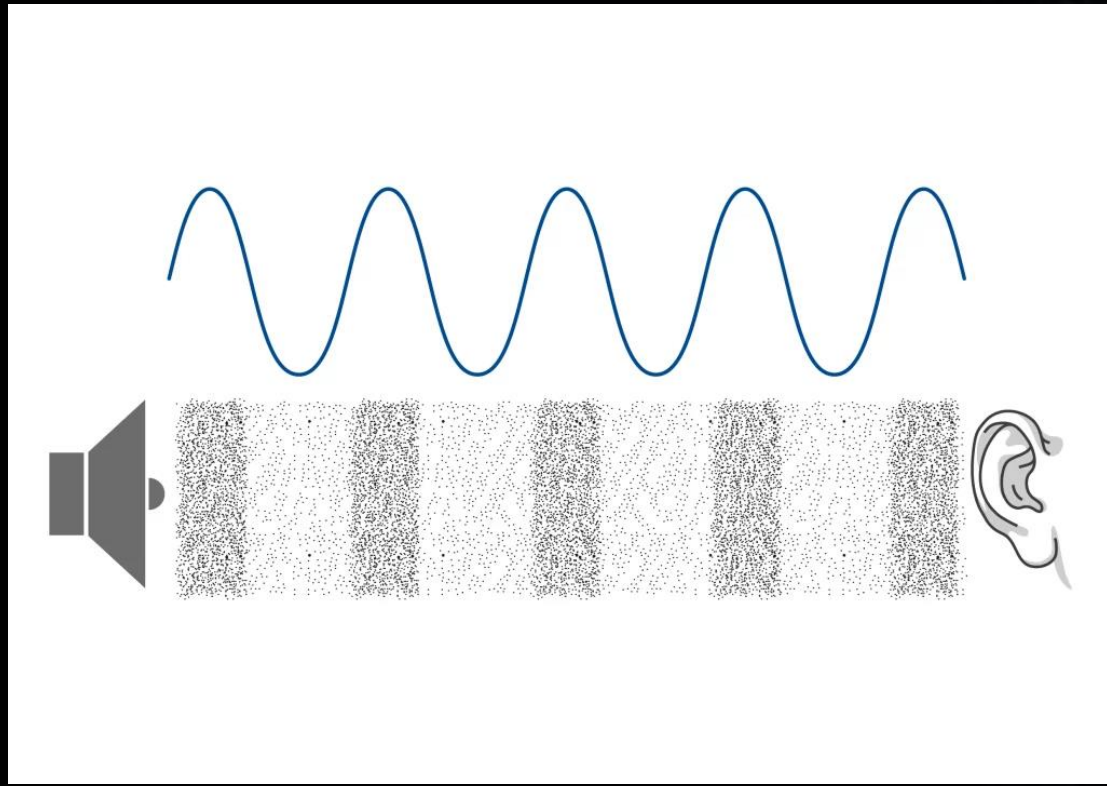


Your turn to explore!

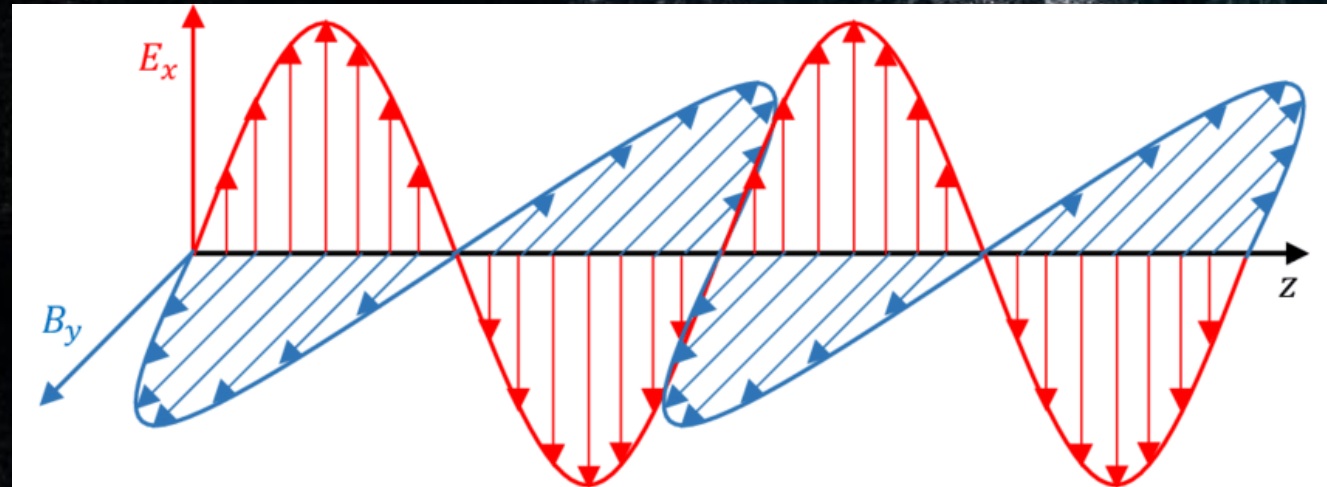


ALBATROS: model vs. reality

Sound wave

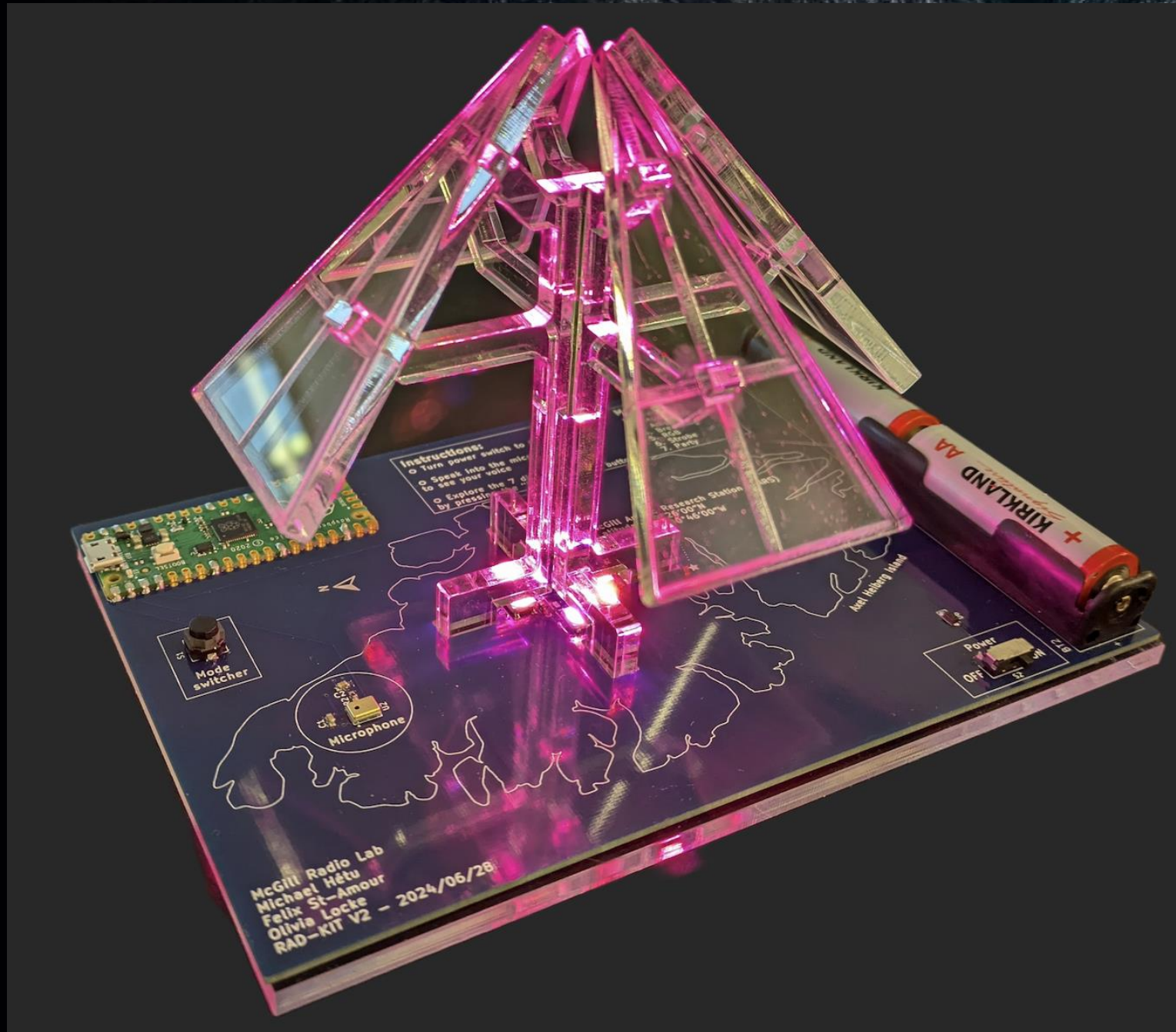


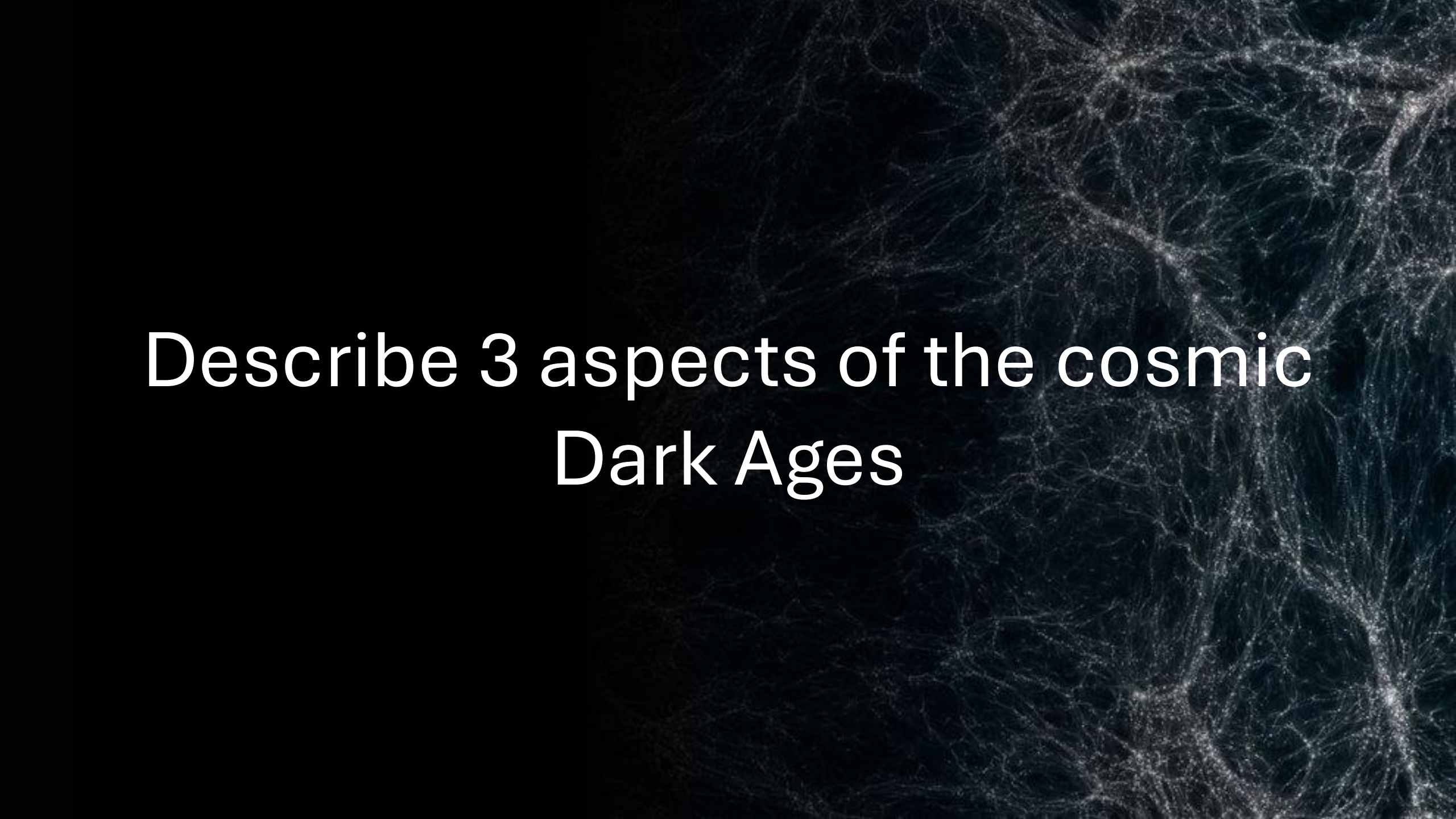
Light wave



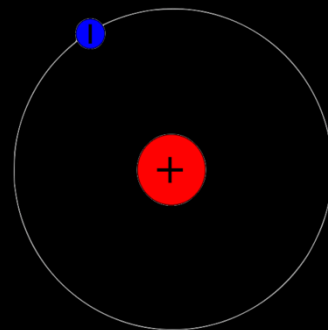
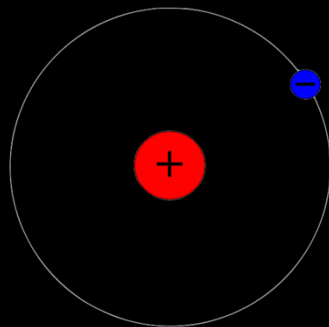
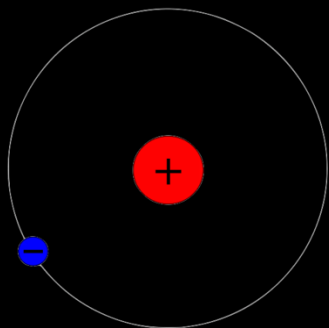
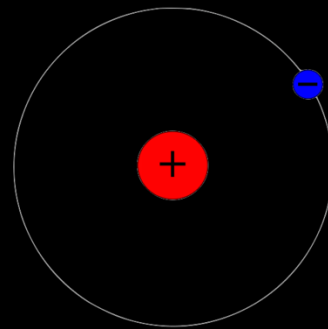
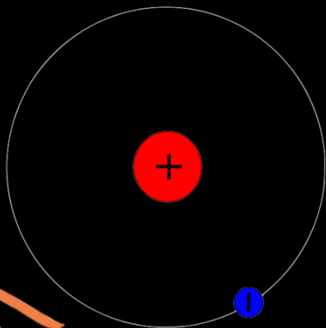
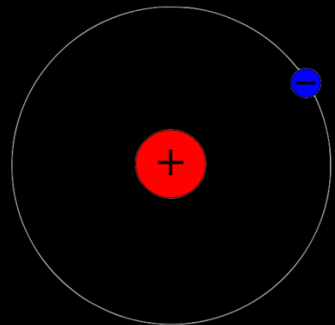
MODE #1:
Frequency
(colours)

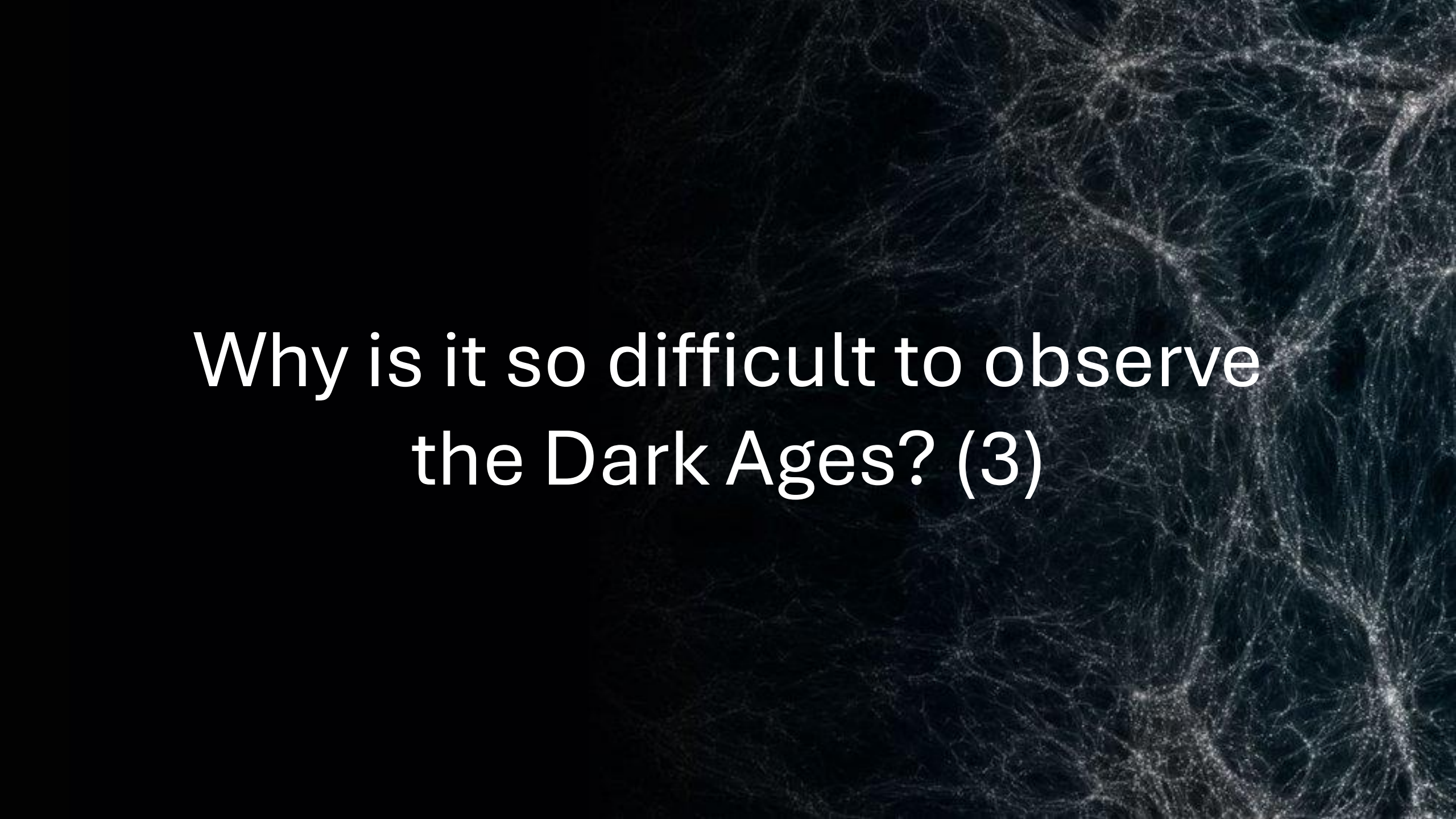
MODE #3:
Amplitude



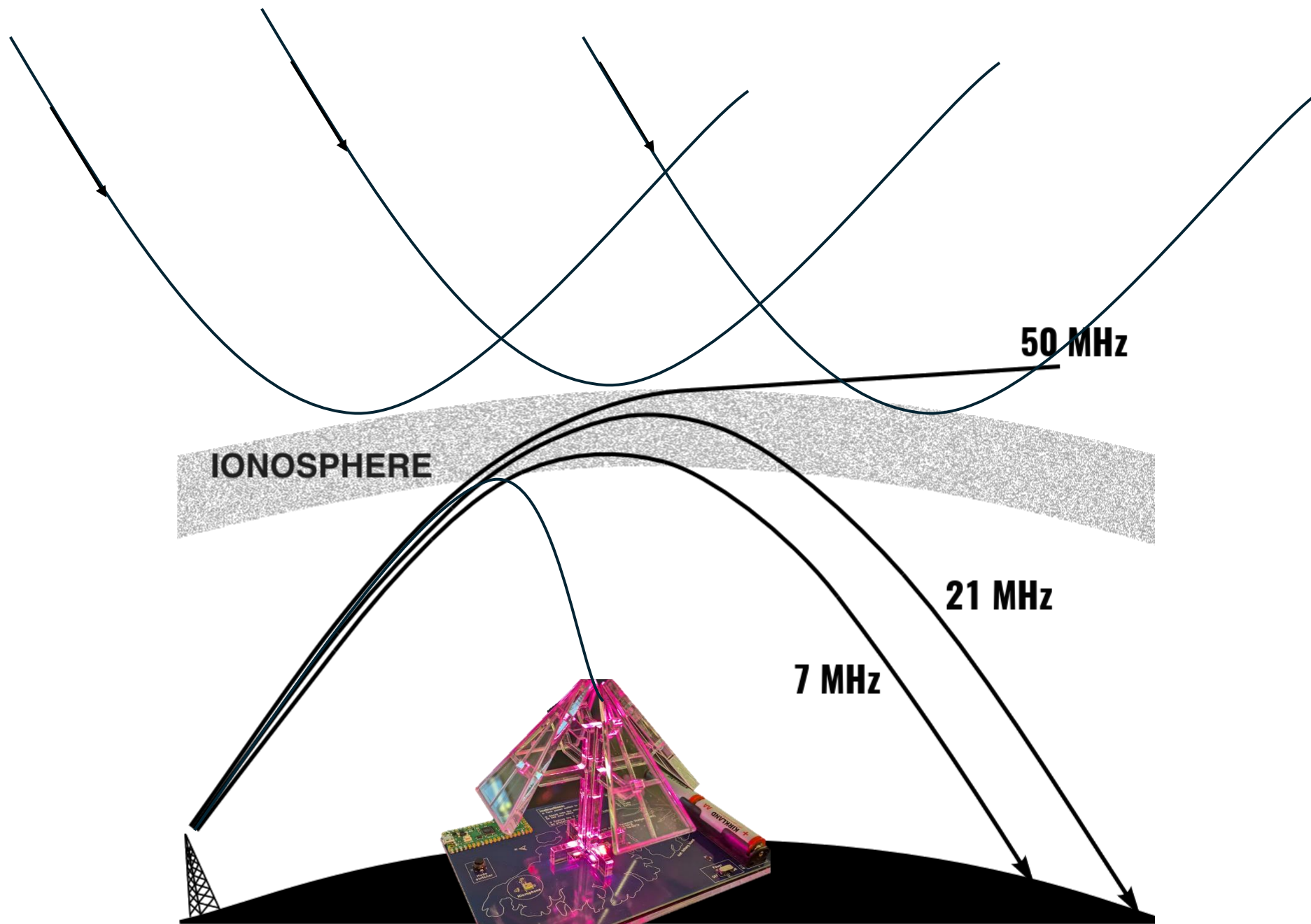
The background of the slide is a dark, almost black, field filled with a complex, web-like pattern of fine, light-colored lines and small dots. This pattern represents the cosmic web, a large-scale structure of the universe consisting of filaments of galaxies and dark matter. The lines are most prominent on the right side of the image, where they form a dense, branching network. The overall effect is one of deep space and cosmic structure.

Describe 3 aspects of the cosmic
Dark Ages



The background of the slide is a dark, deep blue or black space filled with a complex, glowing network of fine, white and light blue lines. These lines represent the cosmic web, showing the large-scale structure of the universe with filaments and clusters of matter. The lines are most prominent on the right side of the image, where they form a dense, branching structure that resembles a spiderweb or a complex web of threads. The overall effect is one of vastness and complexity, typical of astronomical imagery.

Why is it so difficult to observe
the Dark Ages? (3)



The Dark Age of the Cosmos

Thank You!

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